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Nitta et al.

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(54) **IMAGE FORMING APPARATUS HAVING
DETACHABLE TONER CARTRIDGES**

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G03G 15/00 (2006.01)
G03G 15/08 (2006.01)
G03G 21/16 (2006.01)

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(2013.01); **G03G 15/0868** (2013.01);
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CPC G03G 15/0713; G03G 15/0184; G03G
15/0868; G03G 15/0875; G03G 15/5008;
(Continued)

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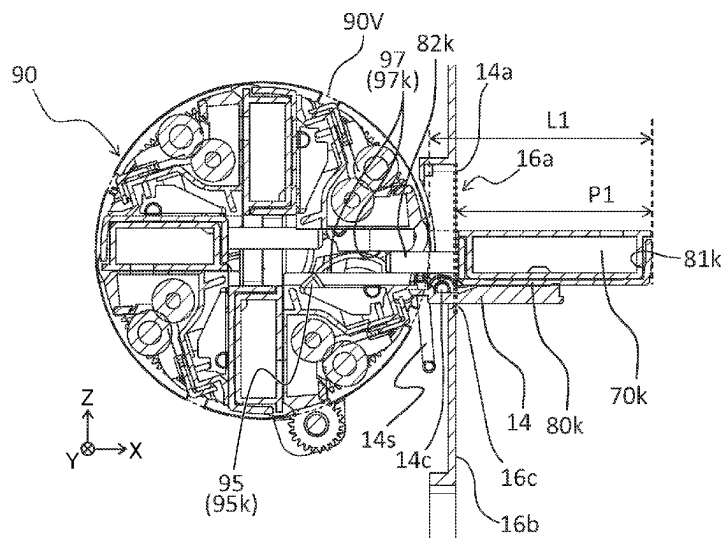
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(57) **ABSTRACT**

An image forming apparatus includes a rotatable rotary including a developing roller and a storage frame including a storage portion to store toner, a body frame with an opening, a toner cartridge which is attachable to and detachable from the rotary through the opening, which is movable to an attached position and a retracted position, a moving device to move the toner cartridge from the attached position to the retracted position, and an opening/closing member provided to be movable between a closed position and an open position. The opening/closing member moves from the closed position to the open position in an interconnected manner with movement of the toner cartridge from the attached position to the retracted position by the moving device.

28 Claims, 21 Drawing Sheets



(52) **U.S. Cl.**

CPC **G03G 15/751** (2013.01); **G03G 21/1633**
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G03G 2215/025 (2013.01); *G03G 2215/066*
 (2013.01)

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CPC G03G 15/751; G03G 21/1633; G03G
 21/1647; G03G 21/1676; G03G
 2215/00544; G03G 2215/025; G03G
 2215/066; G03G 2221/163; G03G
 2221/1684; G03G 15/0173

See application file for complete search history.

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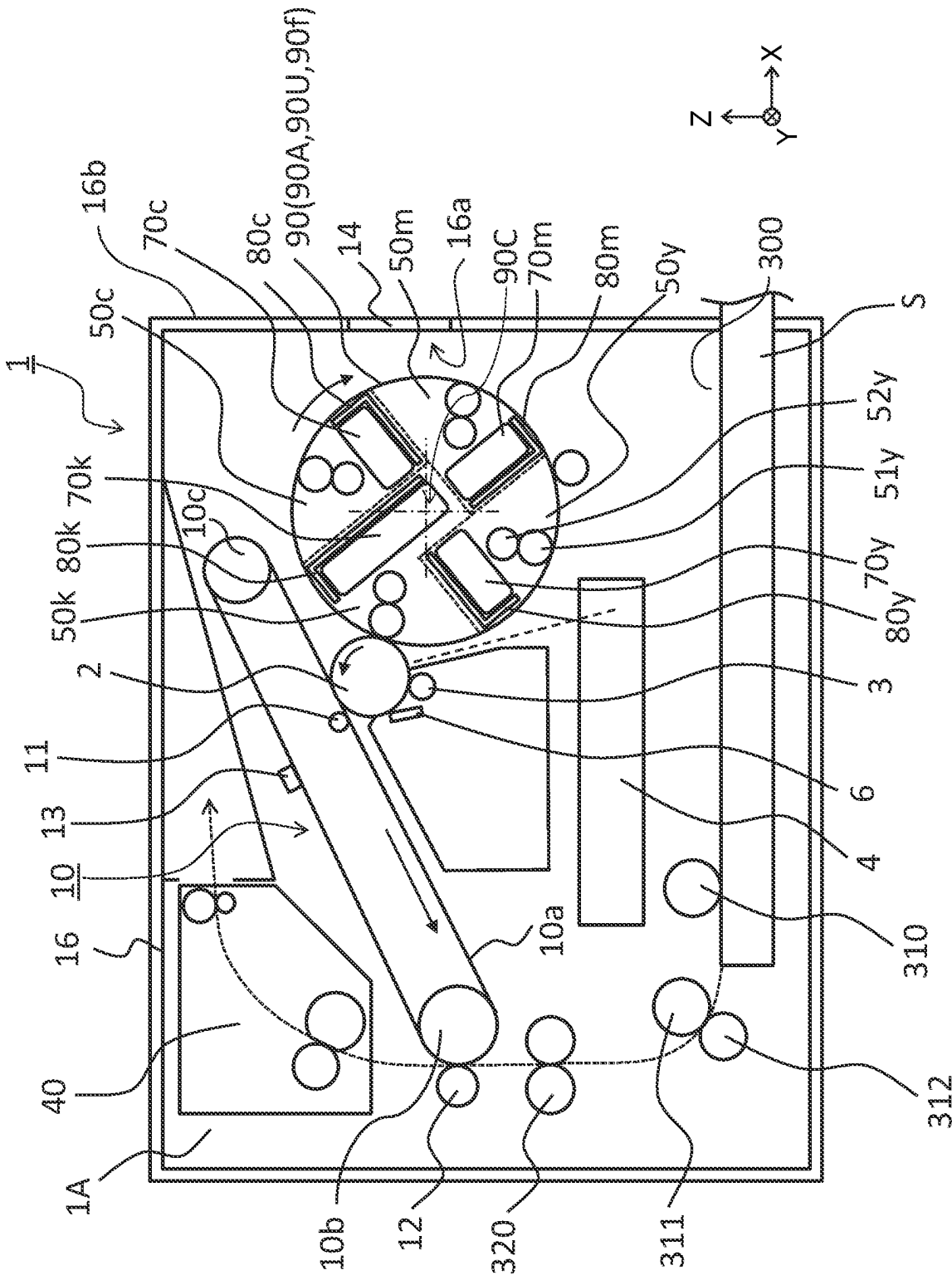


FIG.2

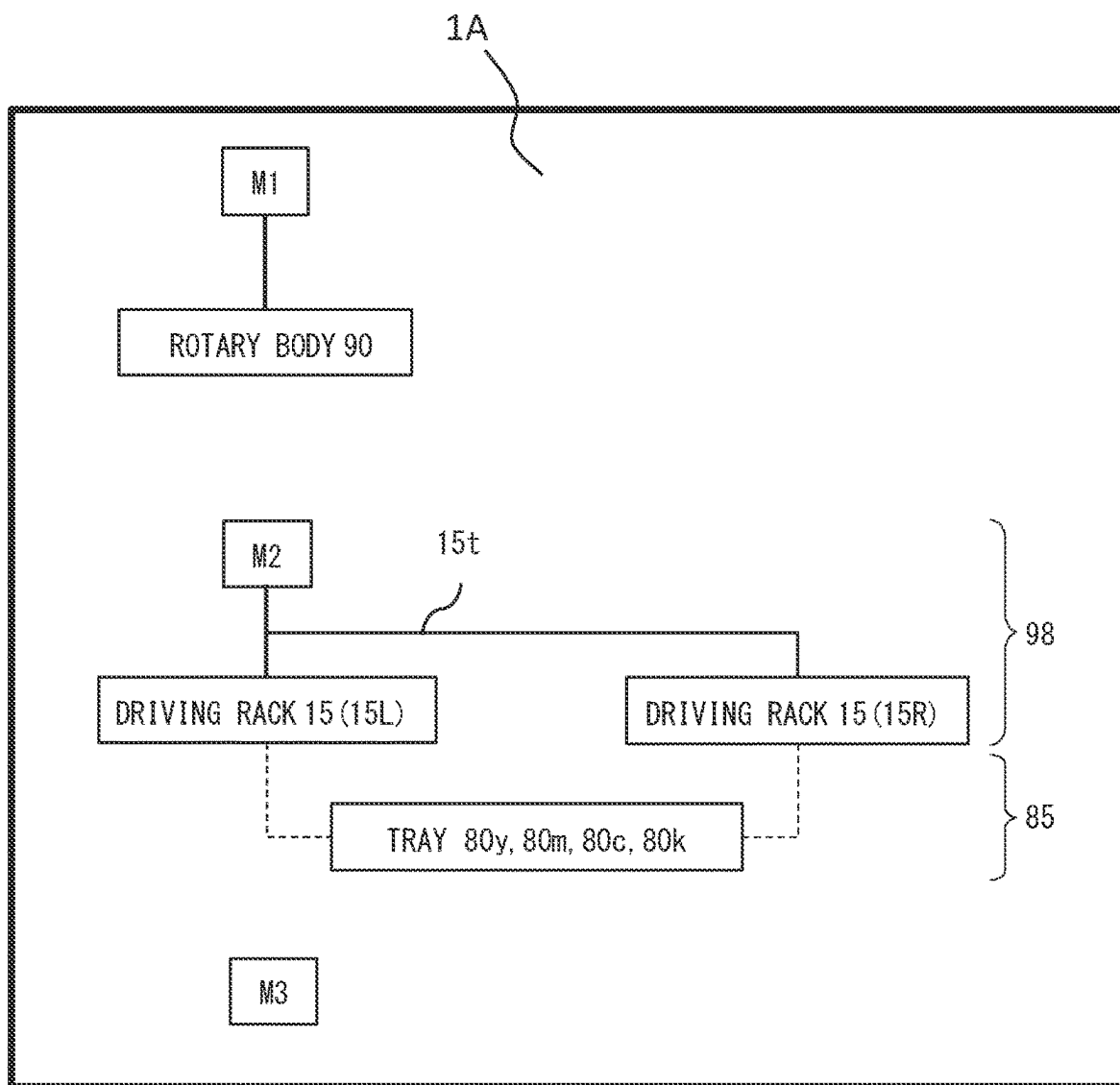


FIG.3

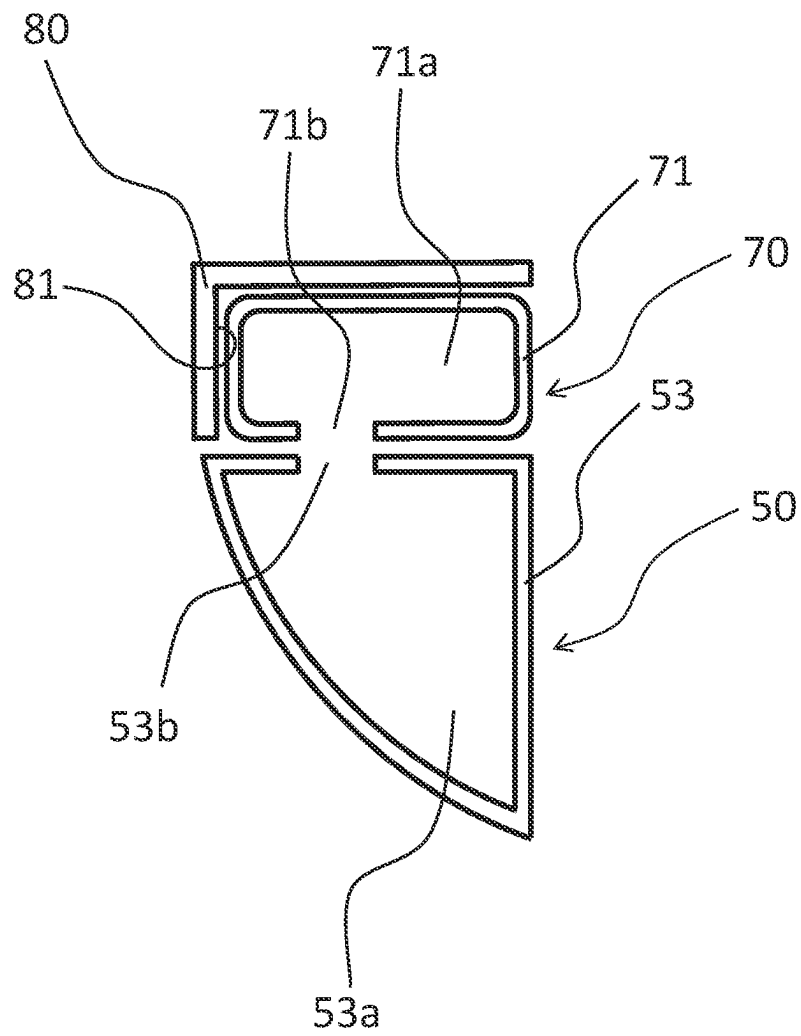


FIG. 4A

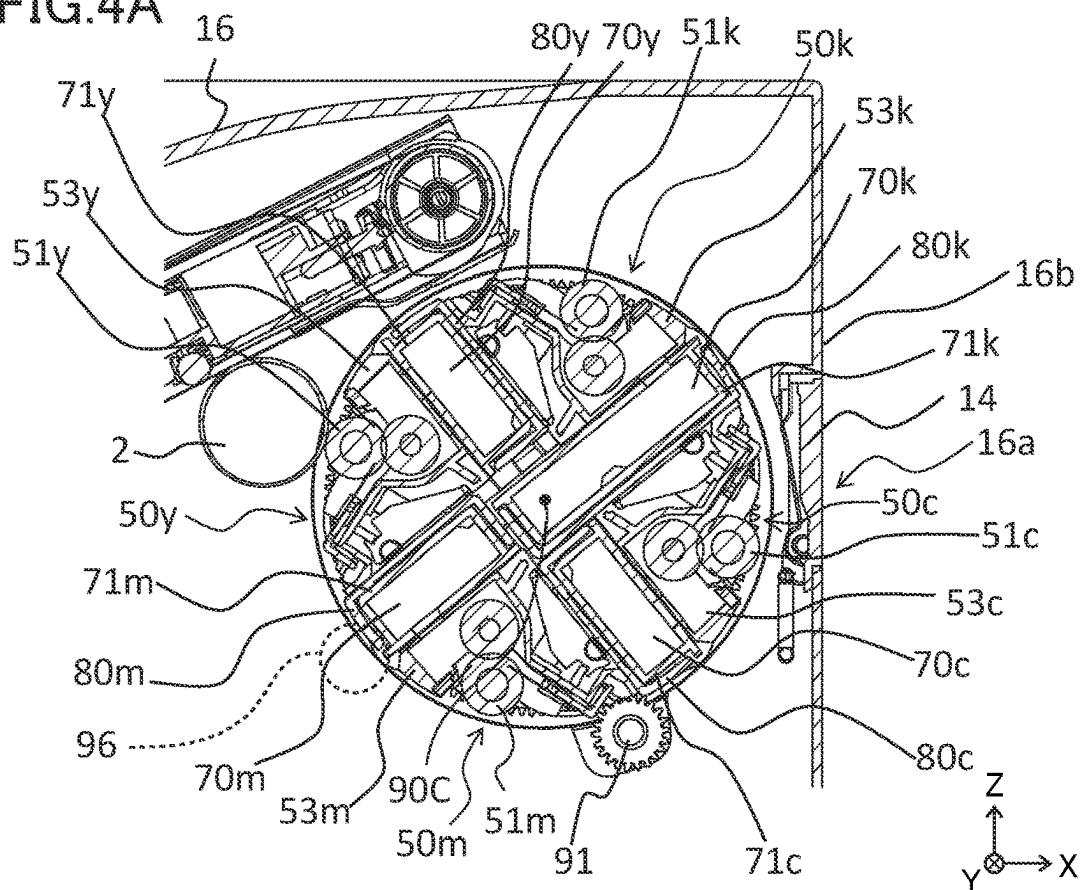


FIG. 4B

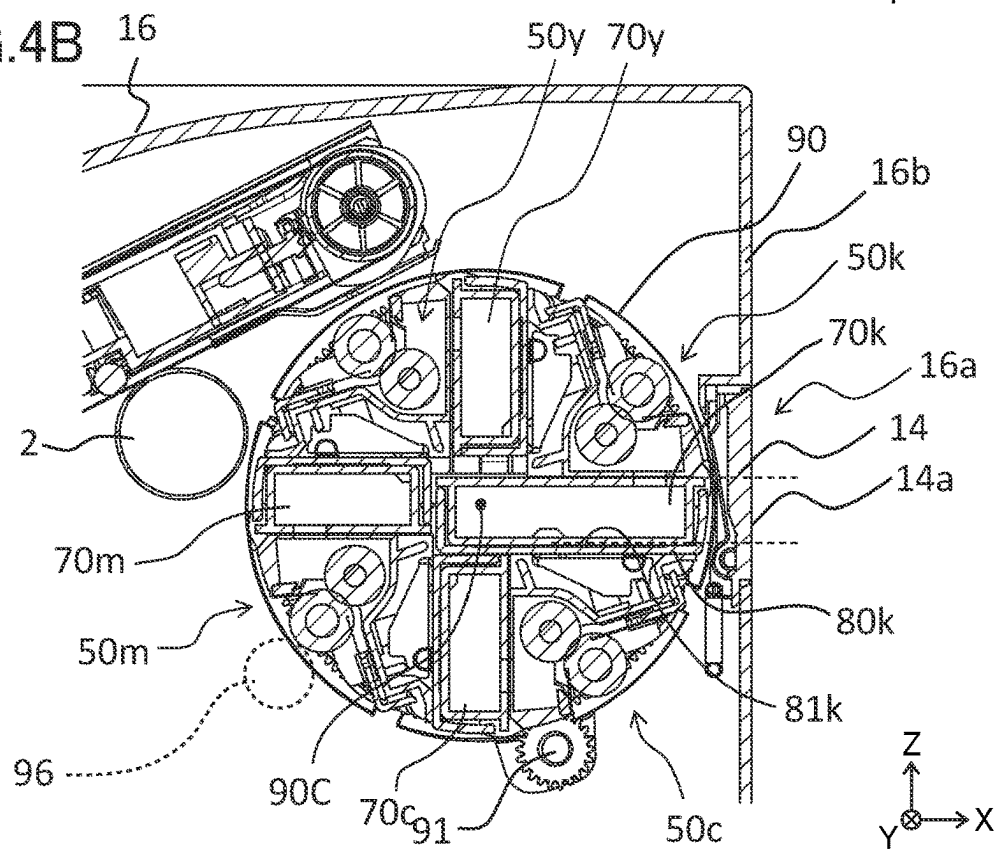


FIG. 5

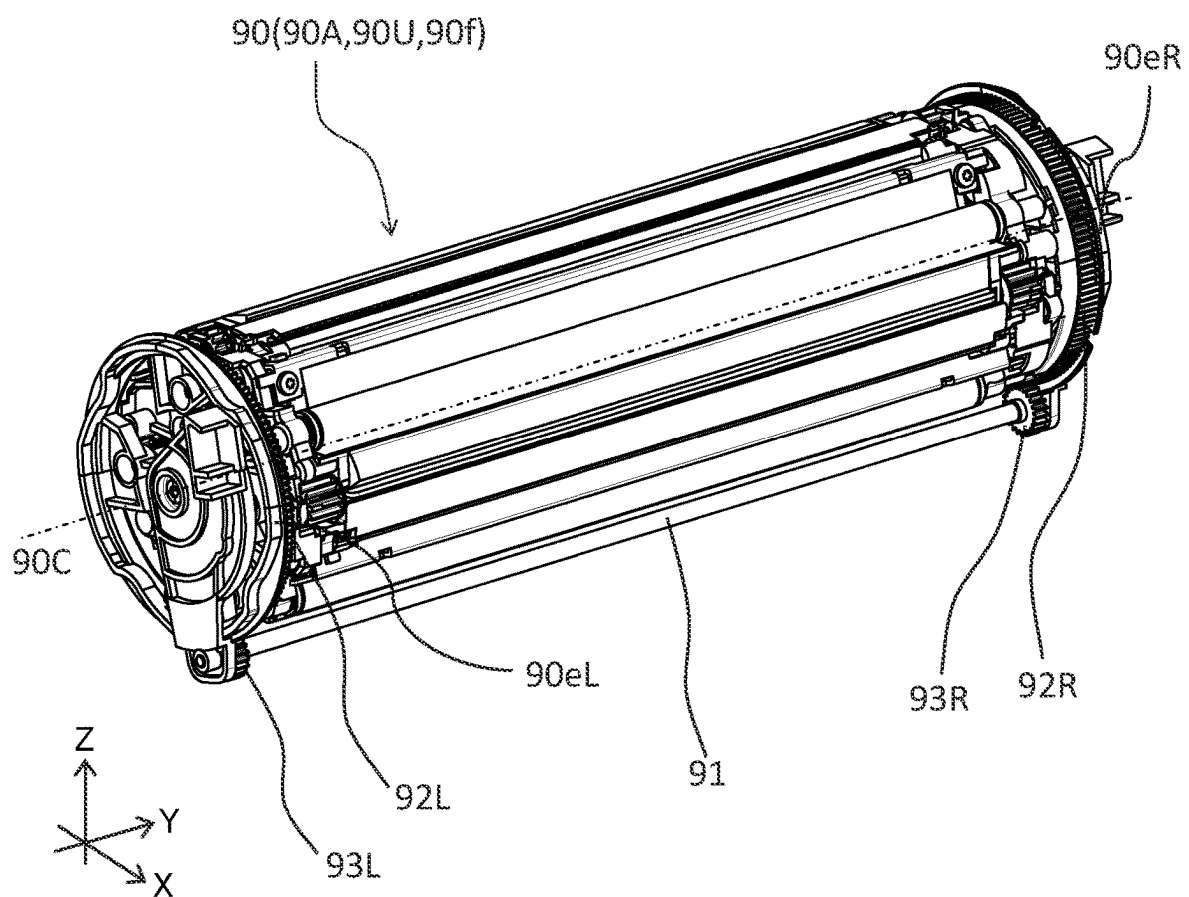


FIG.6A

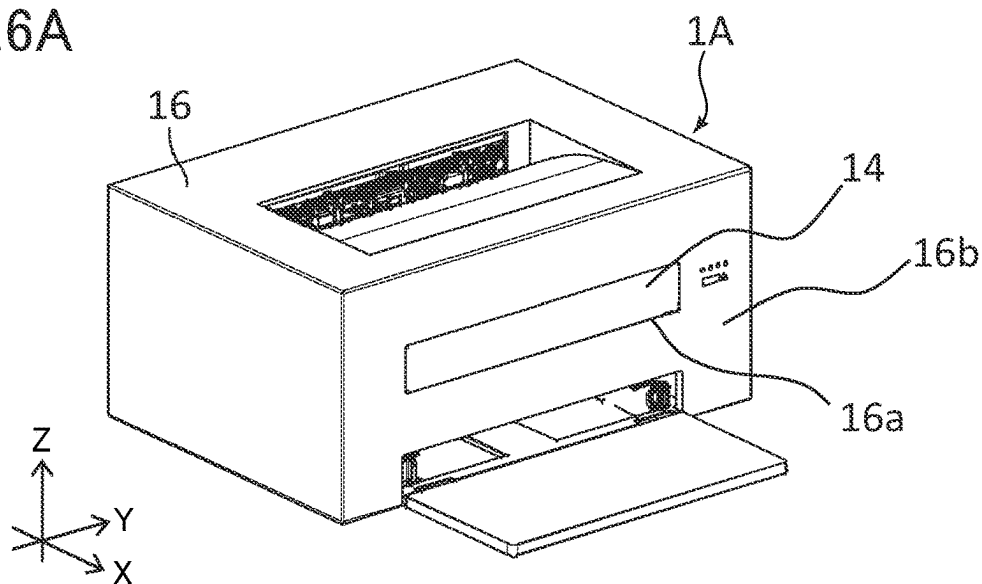


FIG.6B

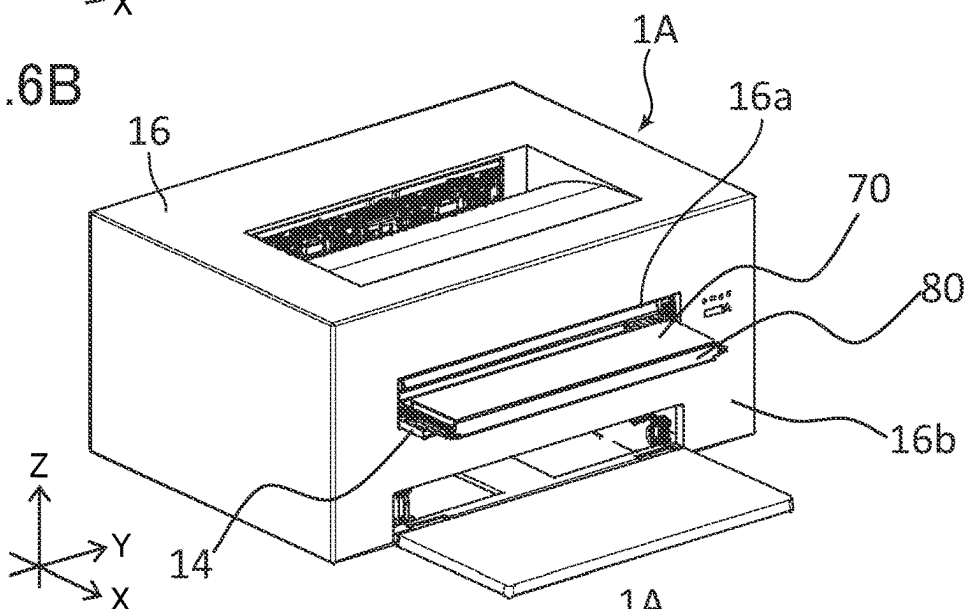


FIG.6C

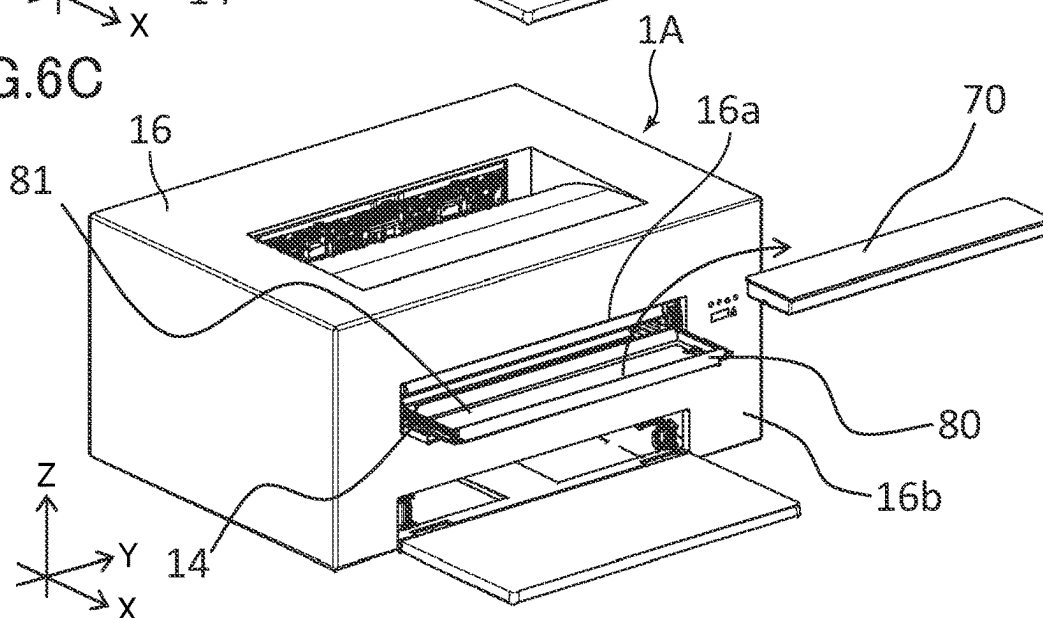


FIG. 7A

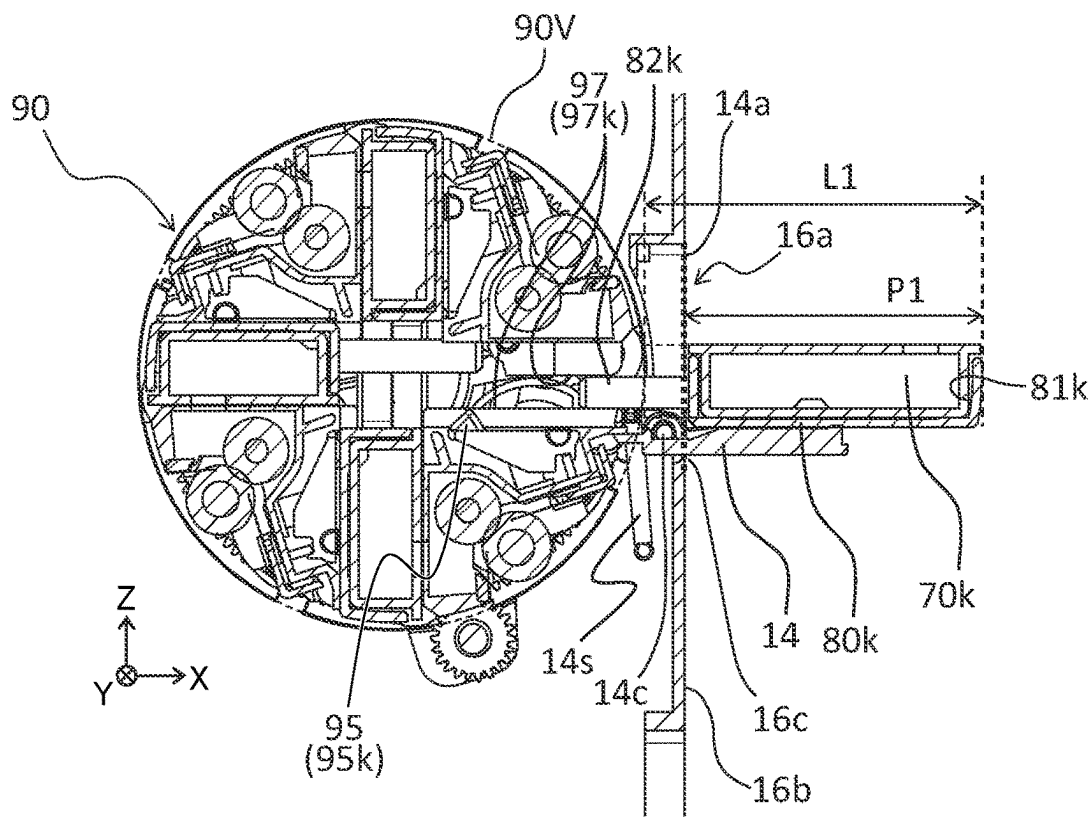


FIG. 7B

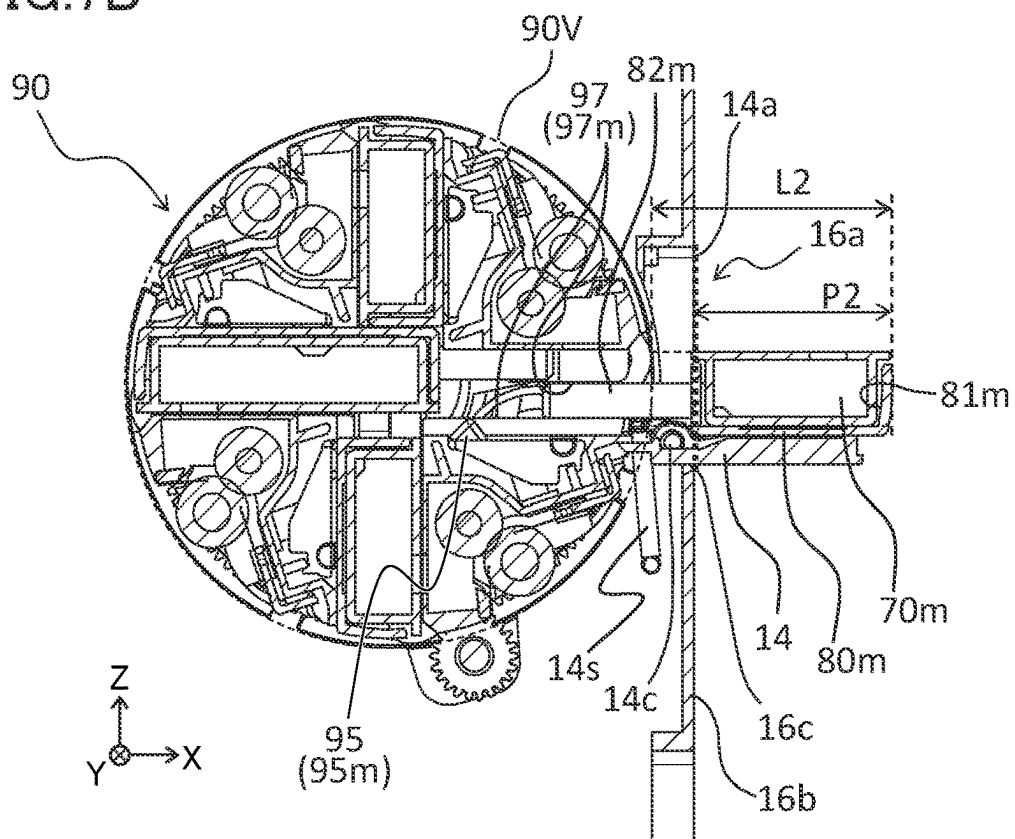


FIG.8

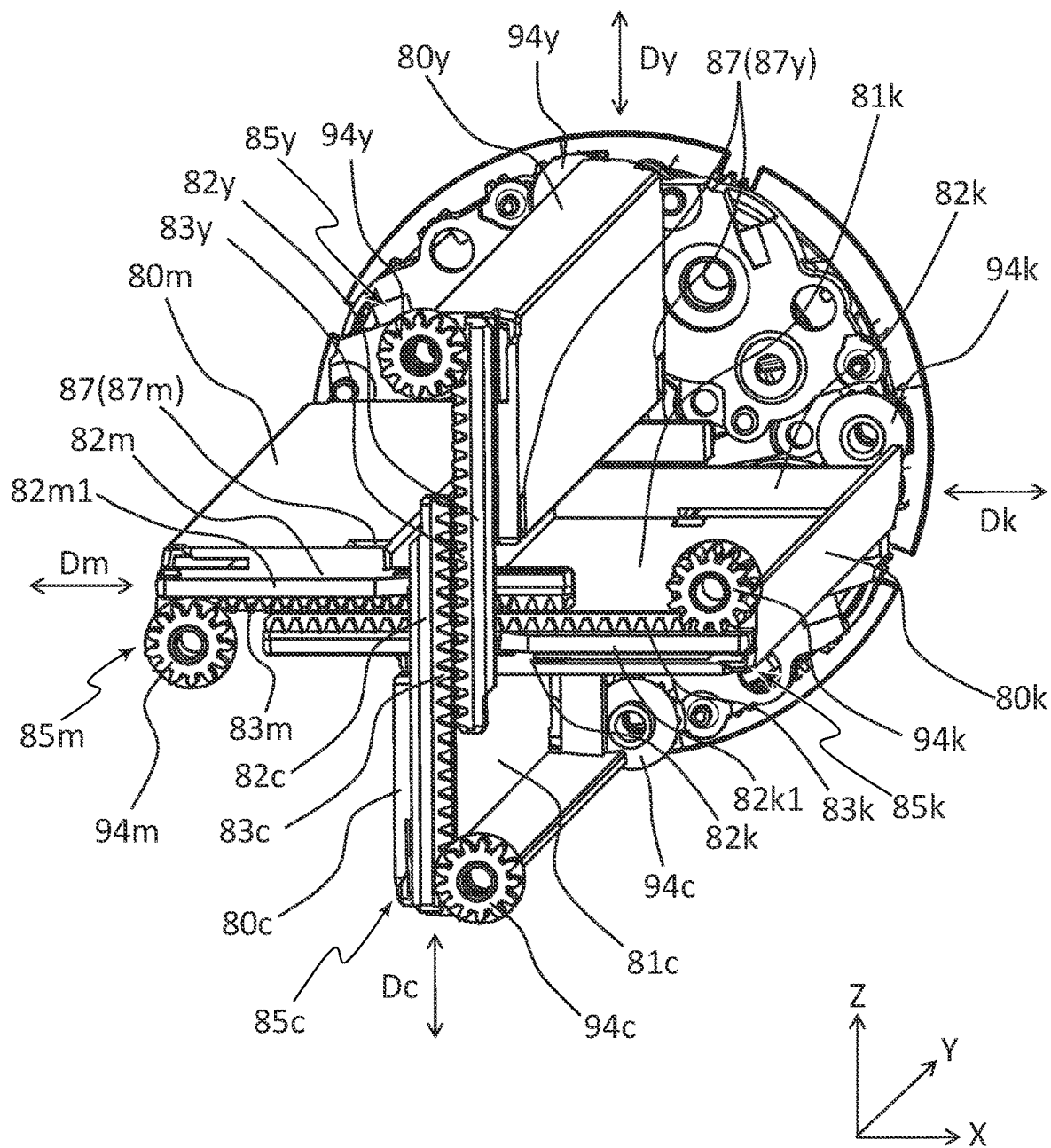


FIG. 9

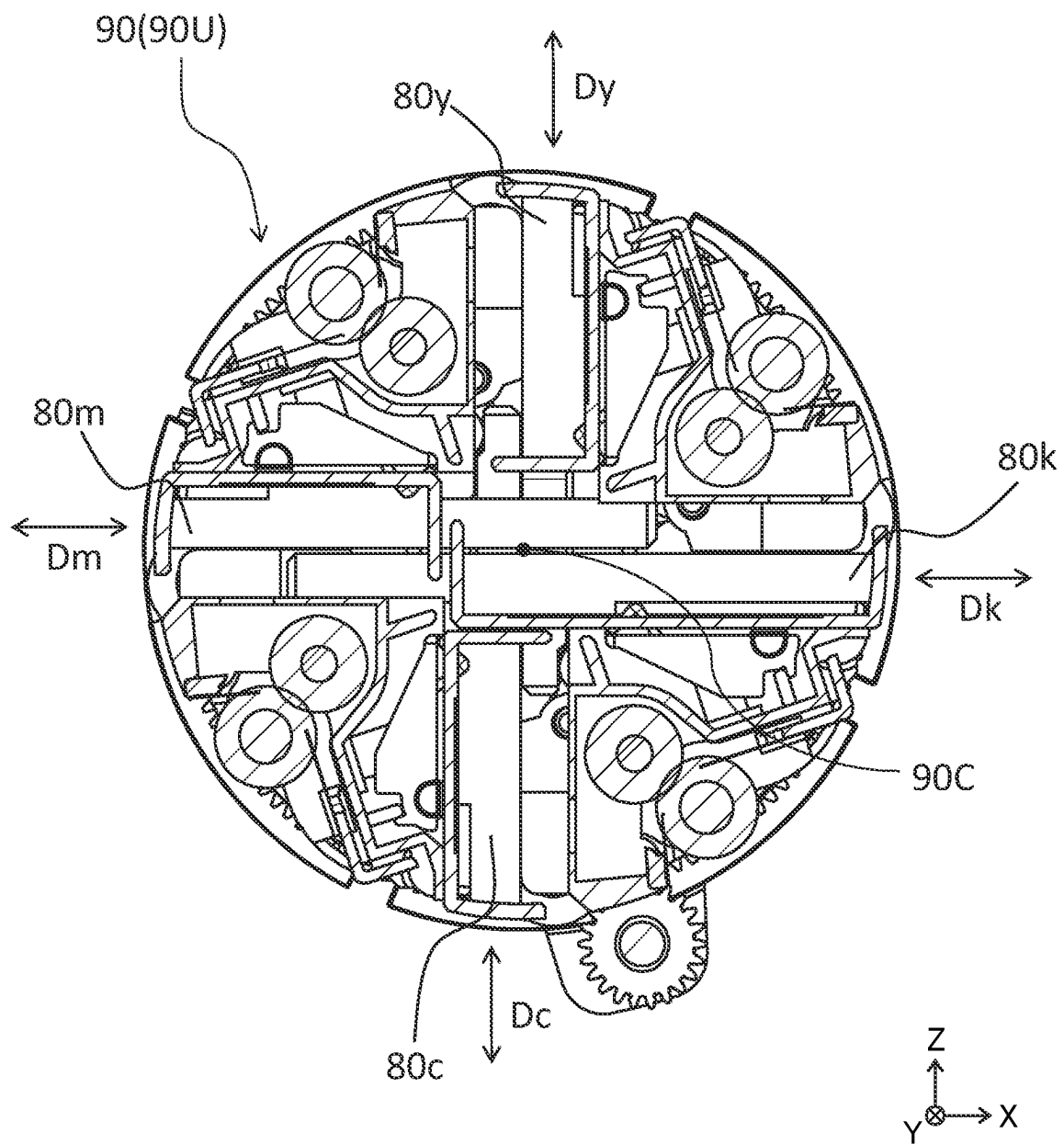


FIG.10

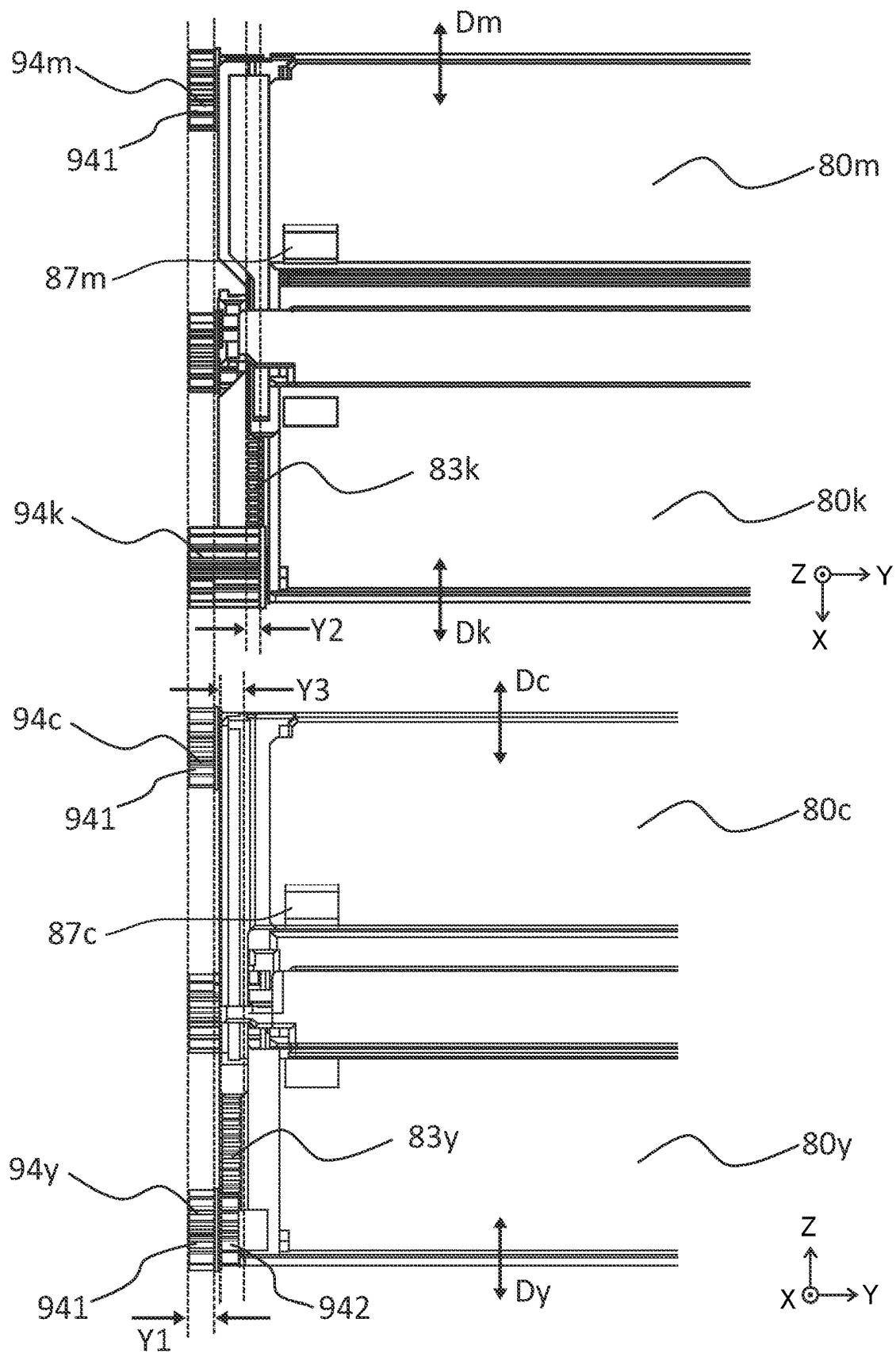


FIG.11A

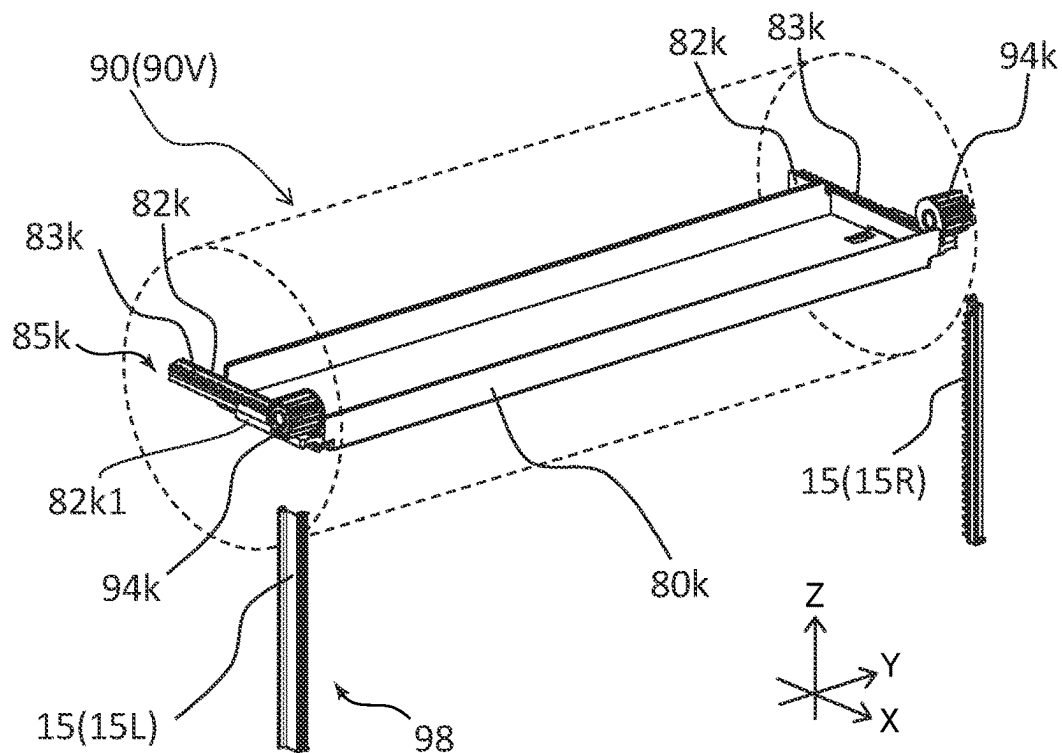


FIG.11B

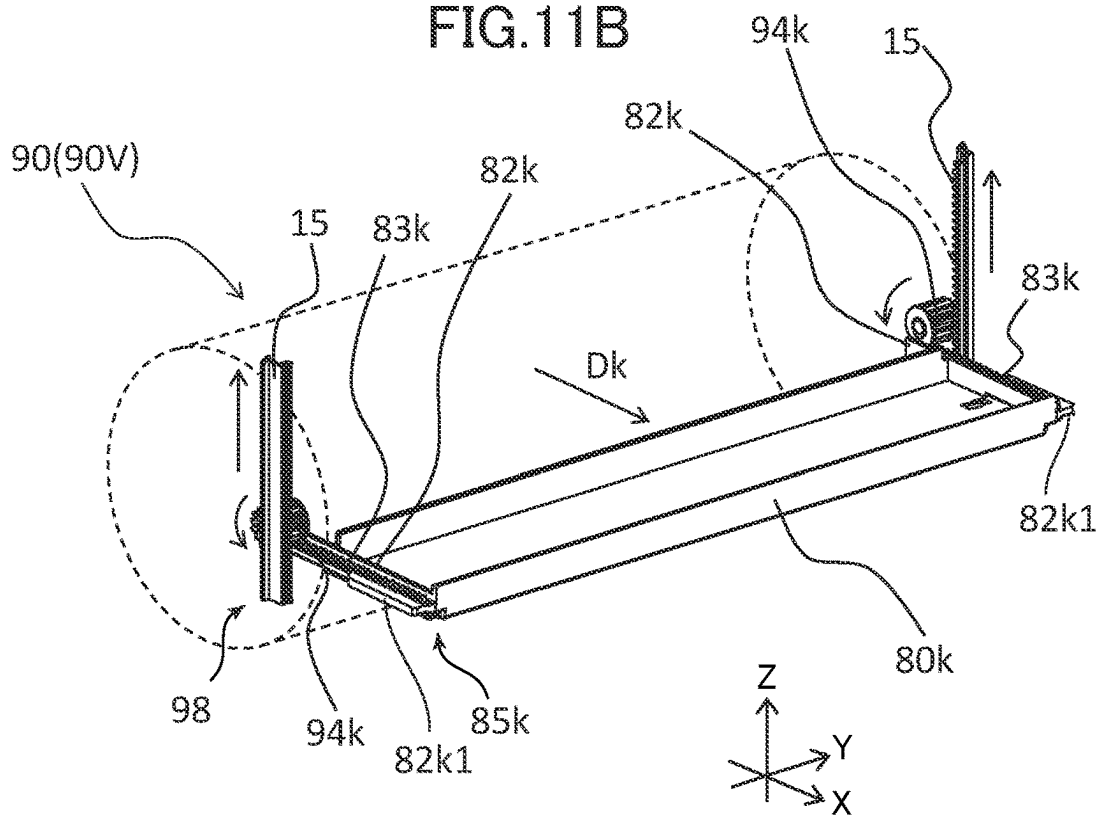


FIG. 12A

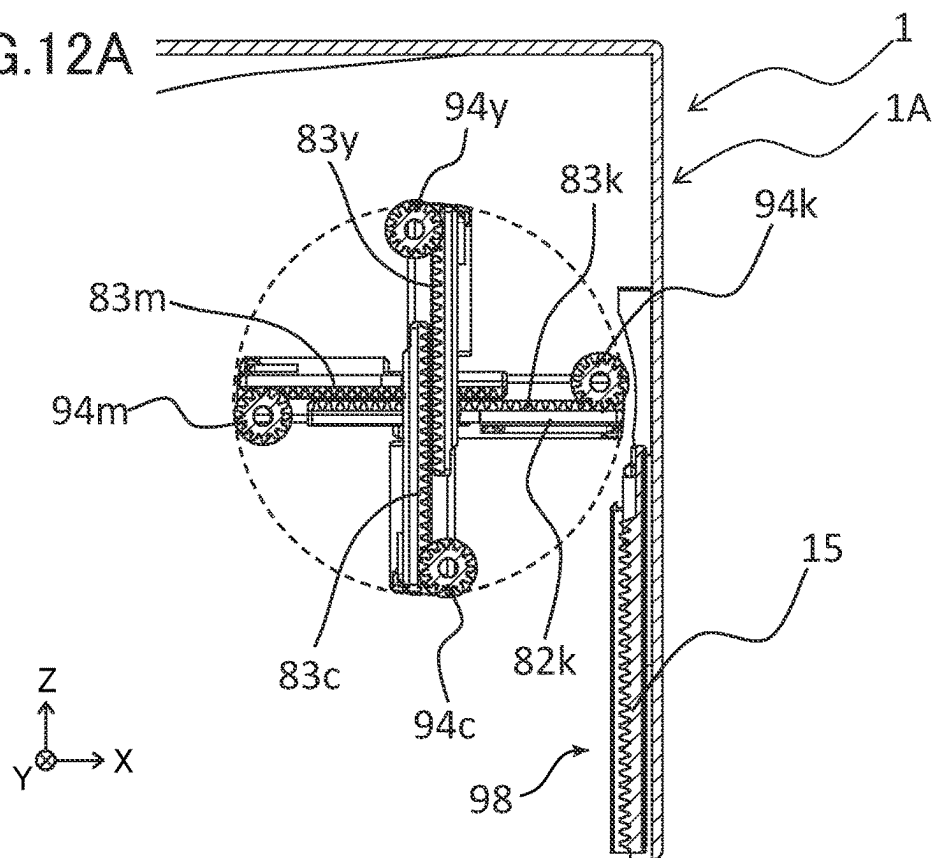


FIG. 12B

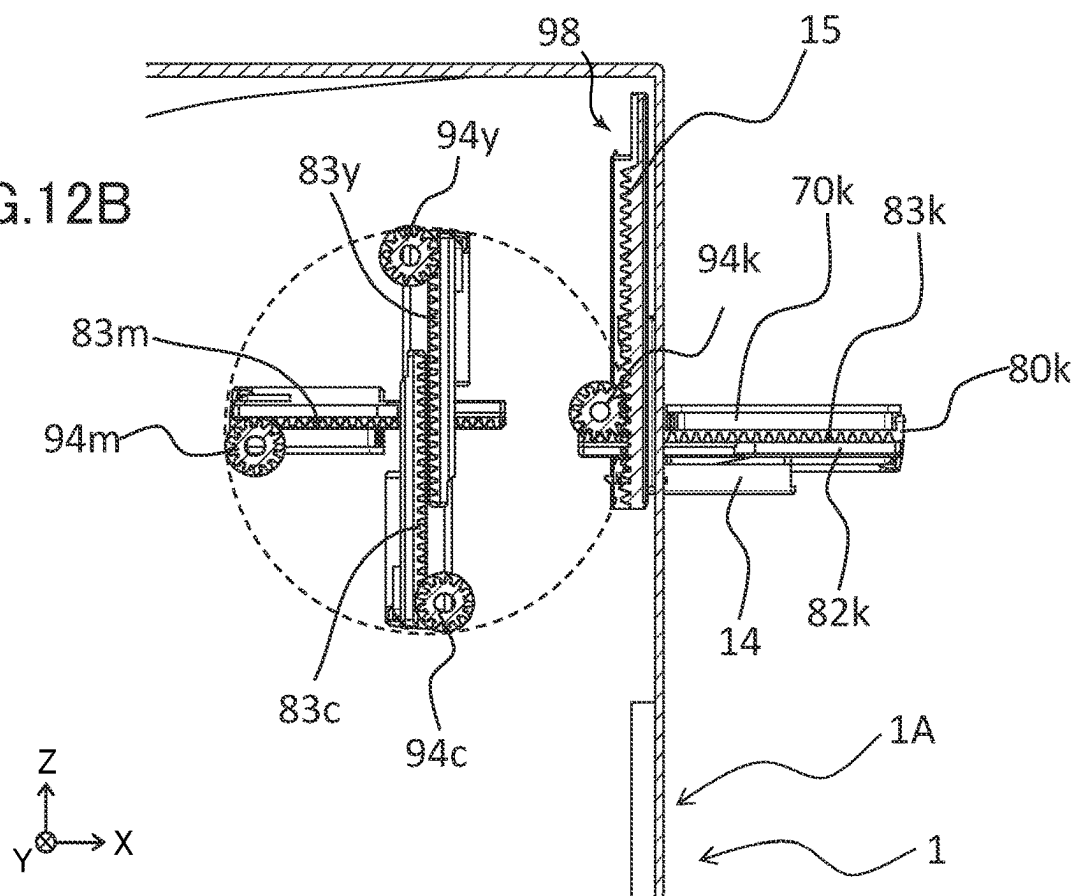


FIG. 13

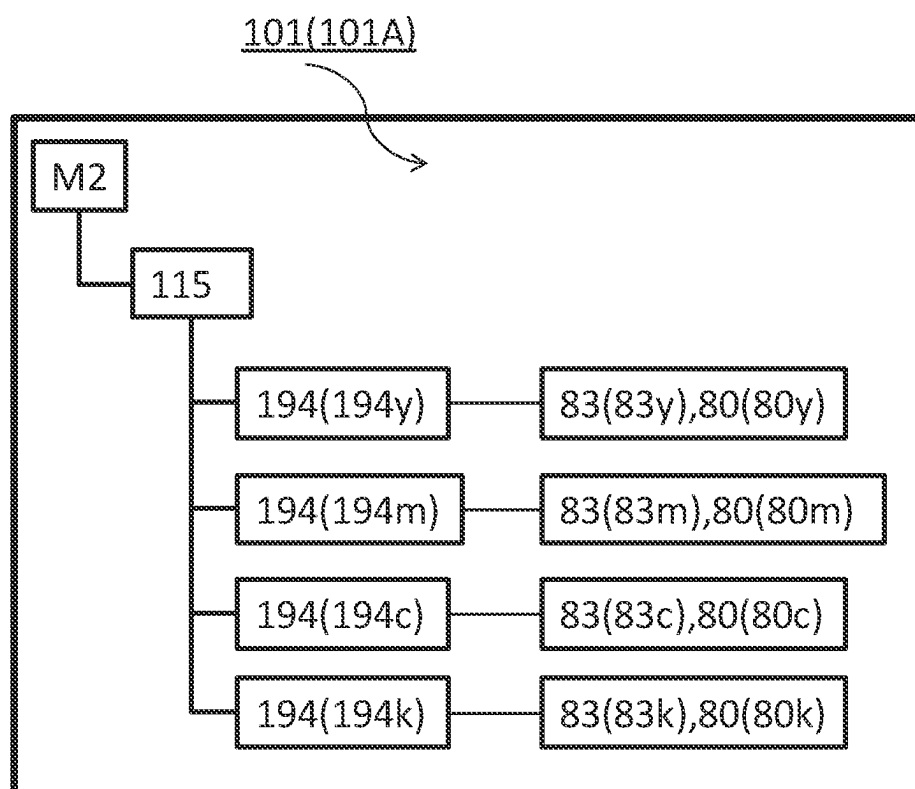


FIG.14A

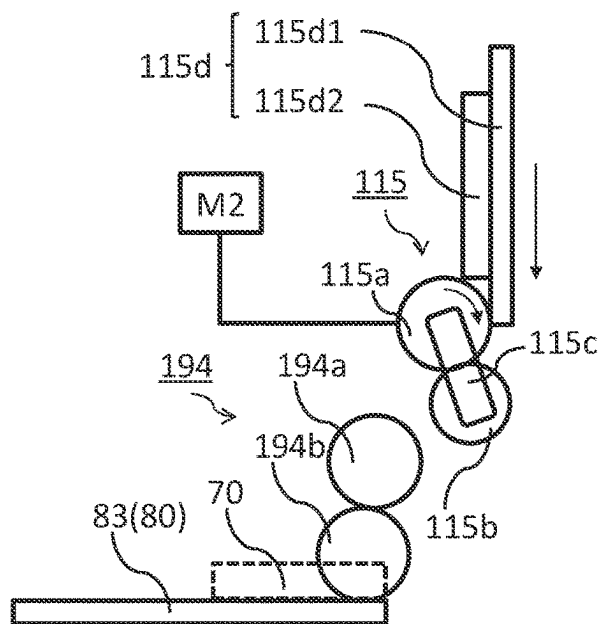


FIG.14B

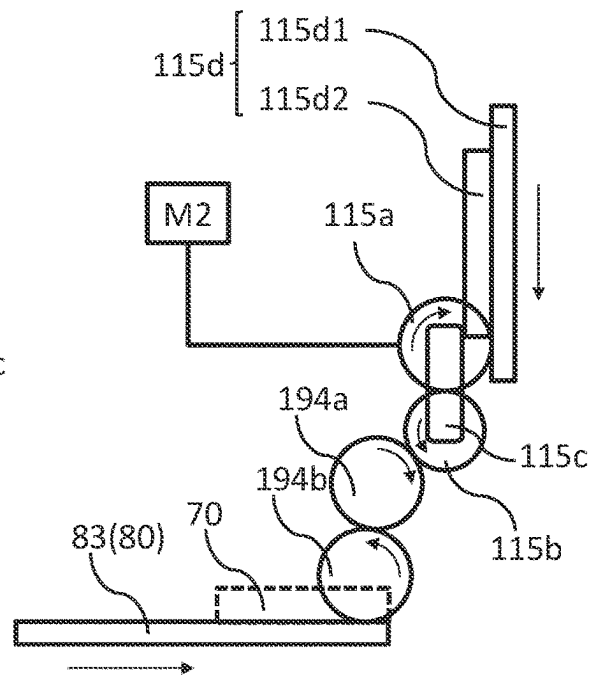


FIG.14C

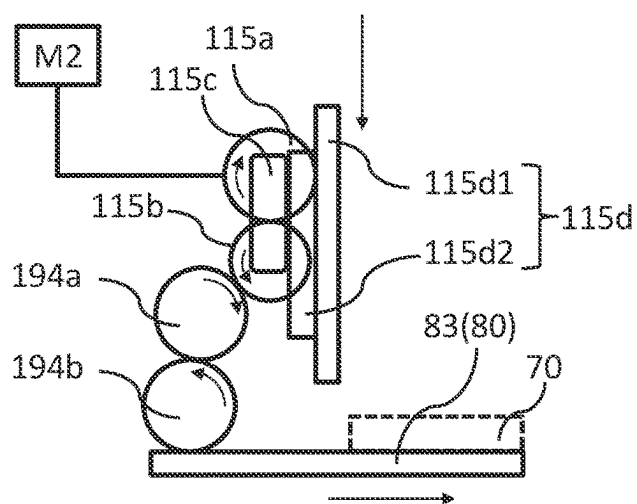


FIG. 15

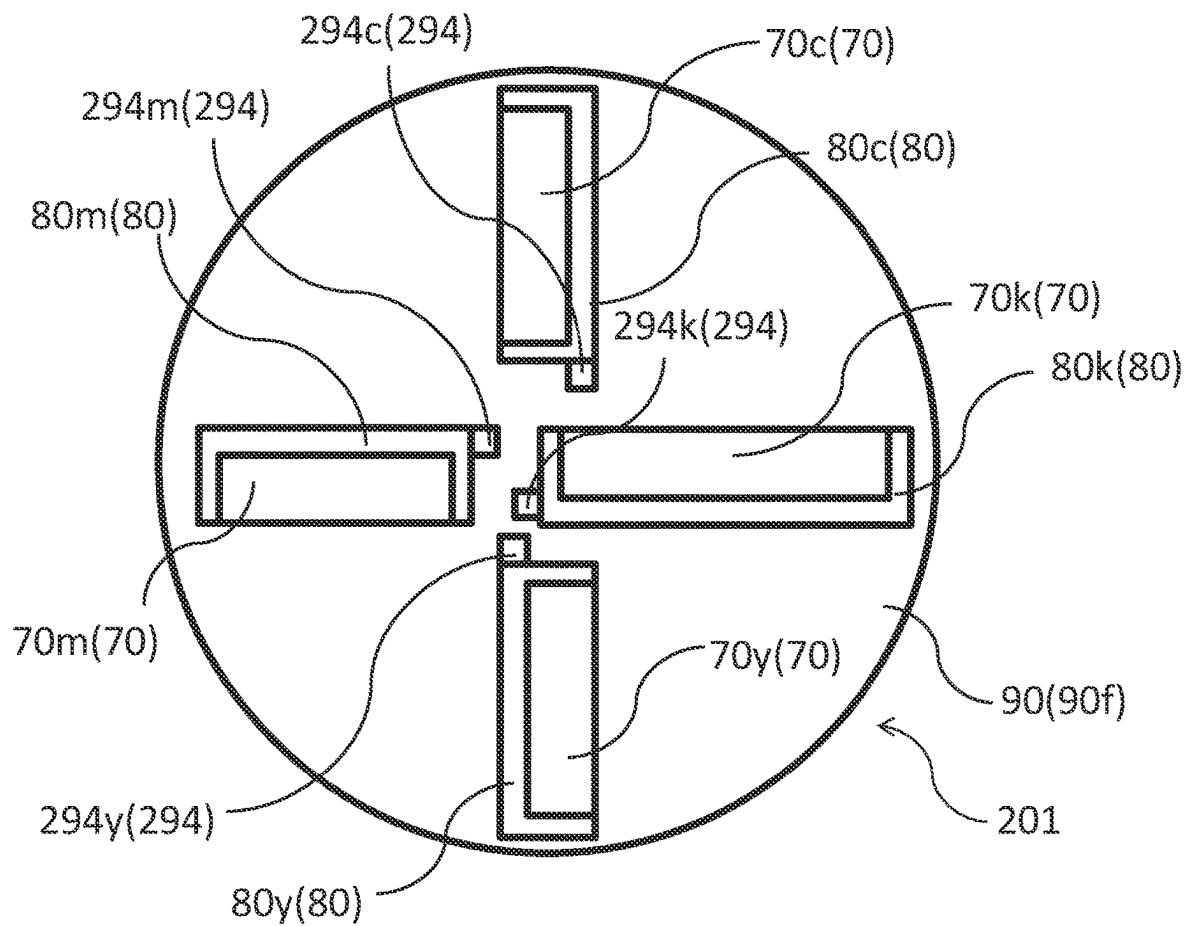


FIG.16A

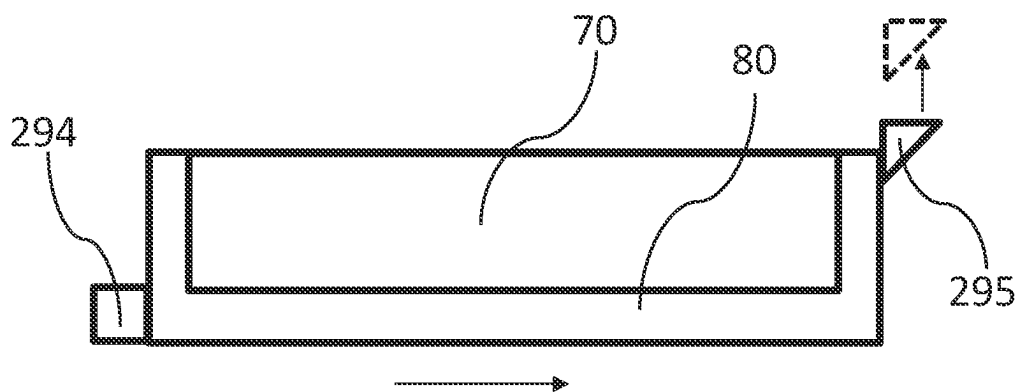


FIG.16B

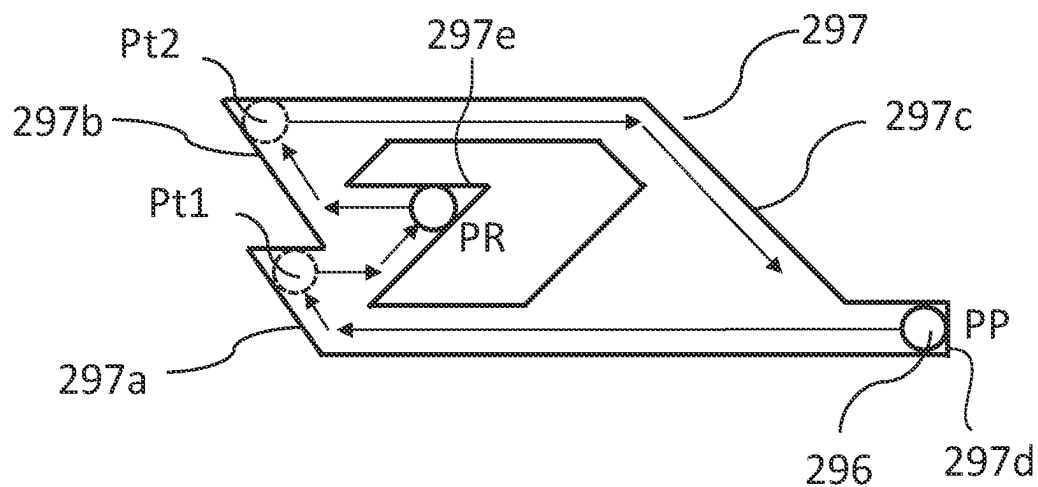


FIG.16C

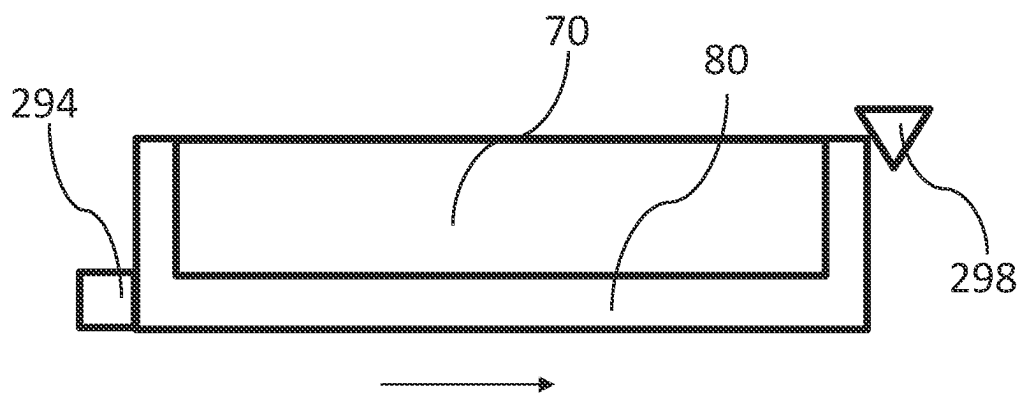


FIG.17A

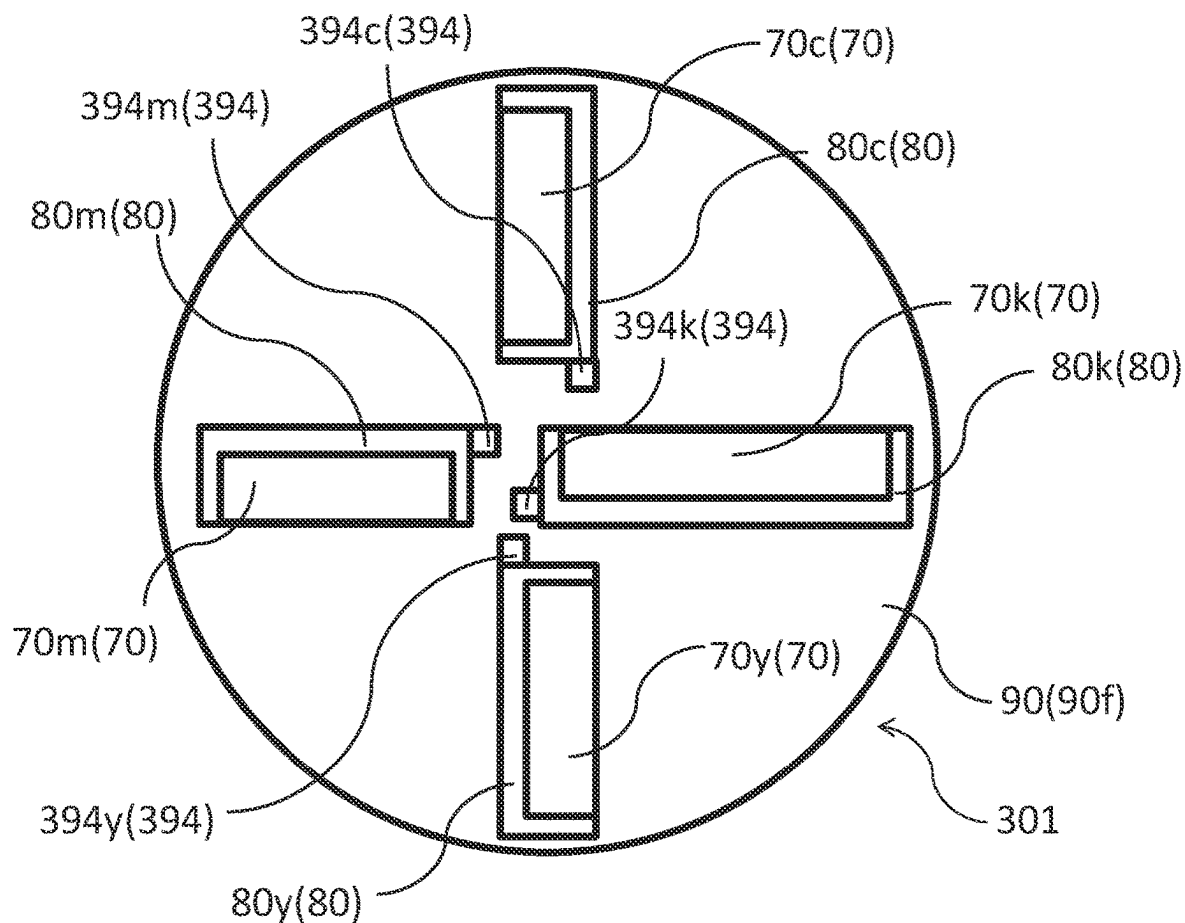
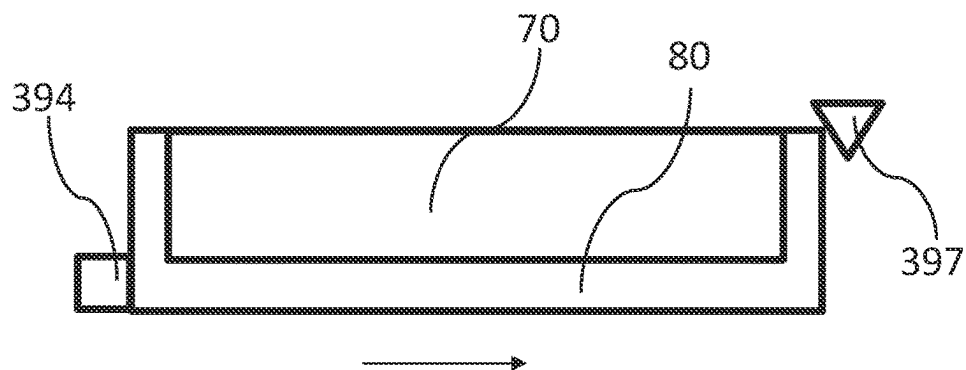


FIG.17B



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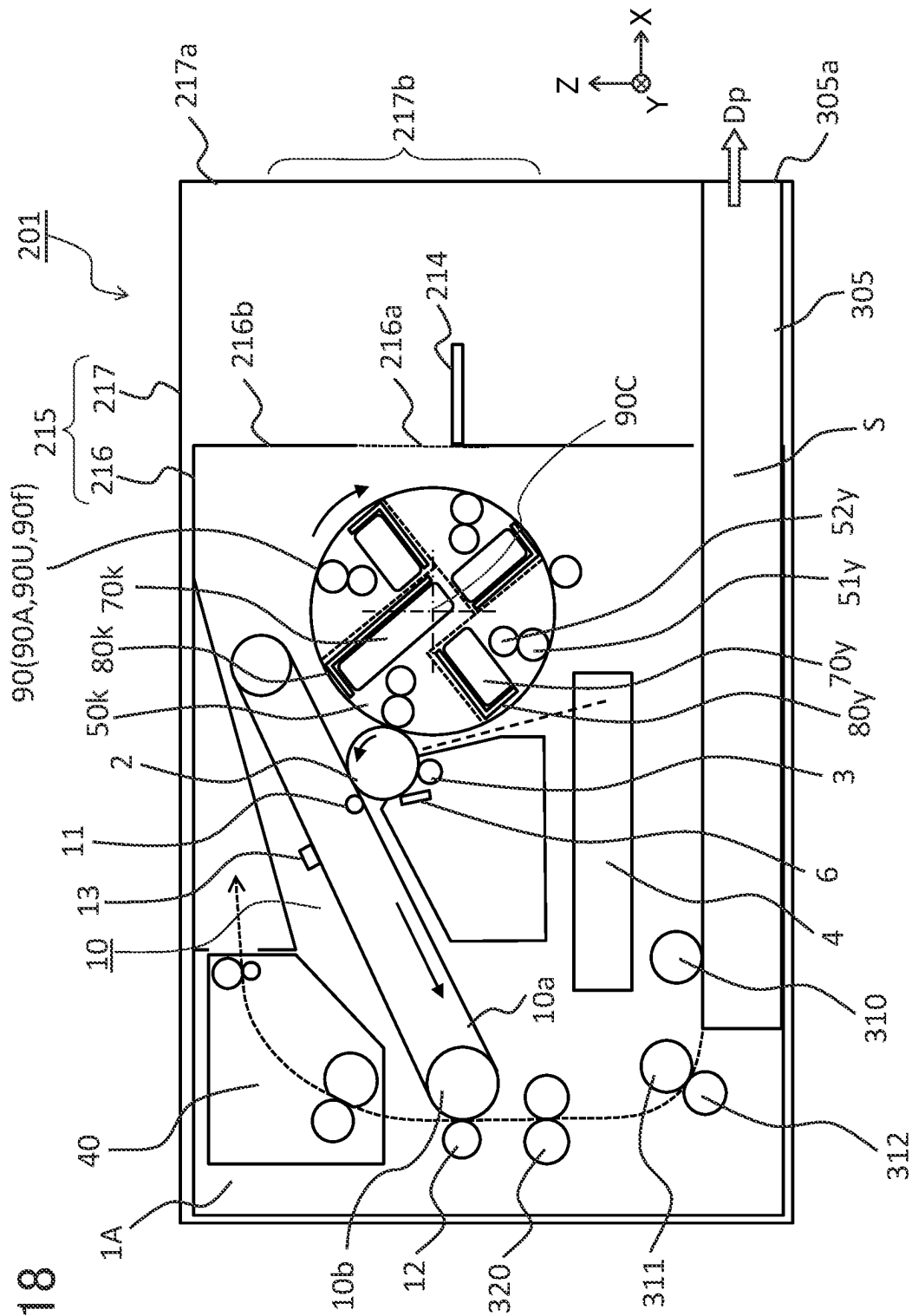


FIG. 19A

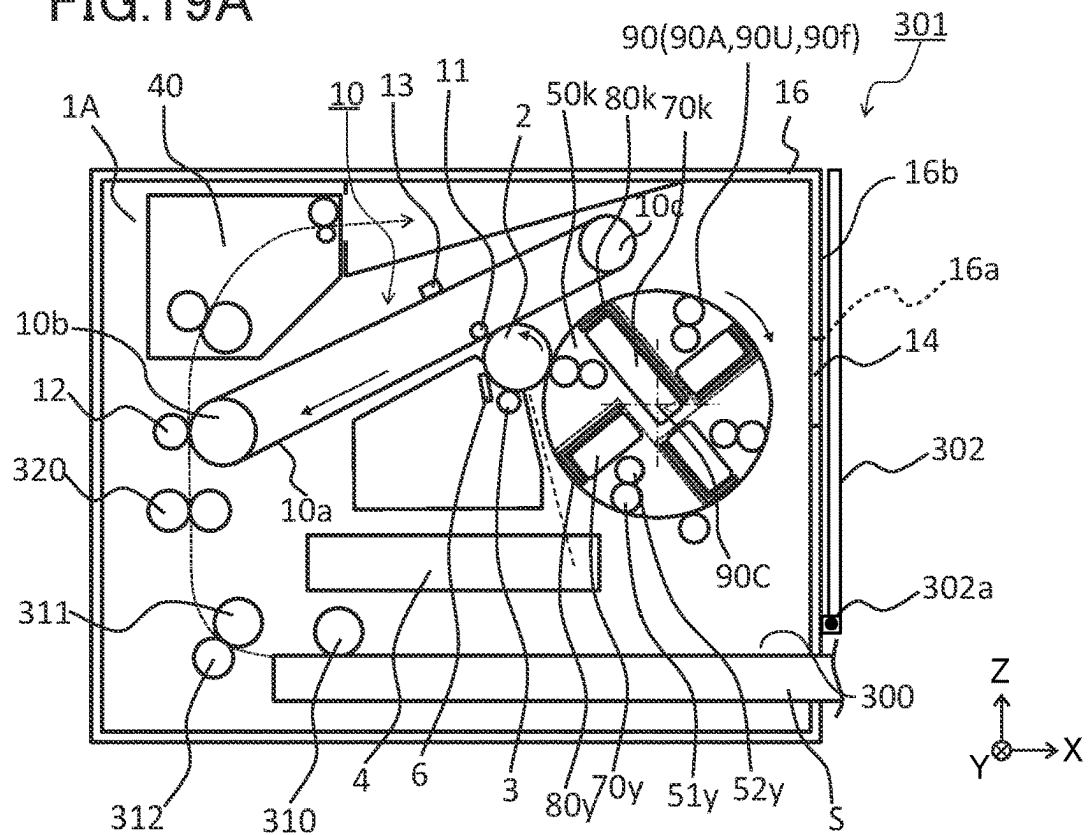


FIG. 19B

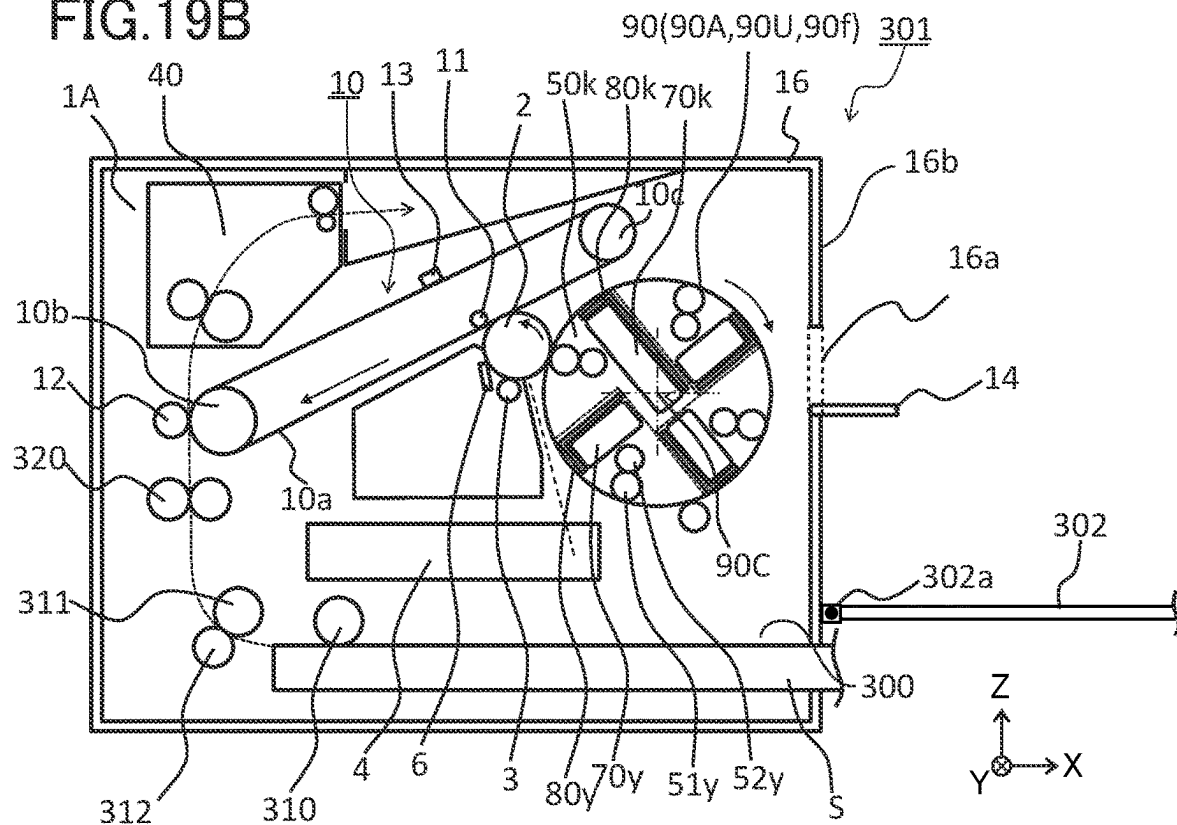


FIG.20A

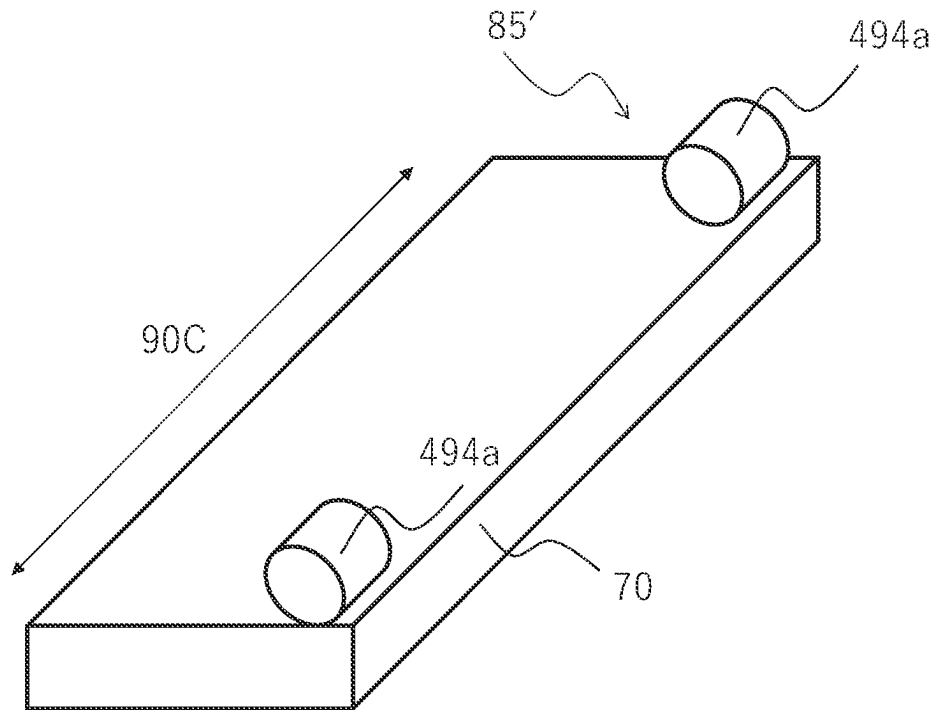


FIG.20B

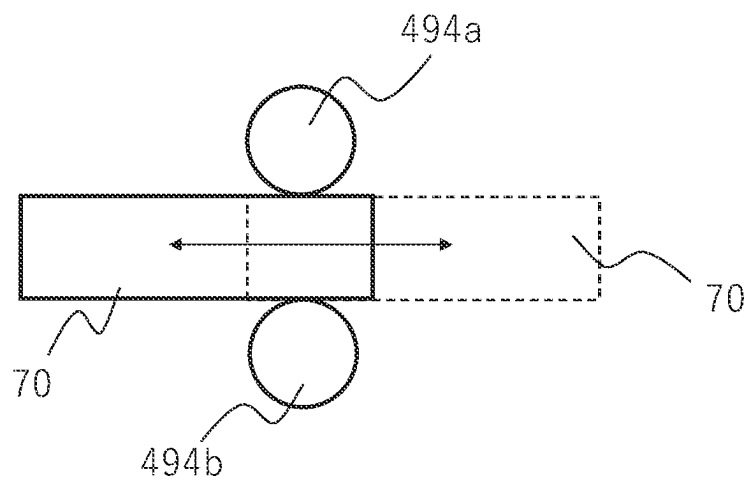
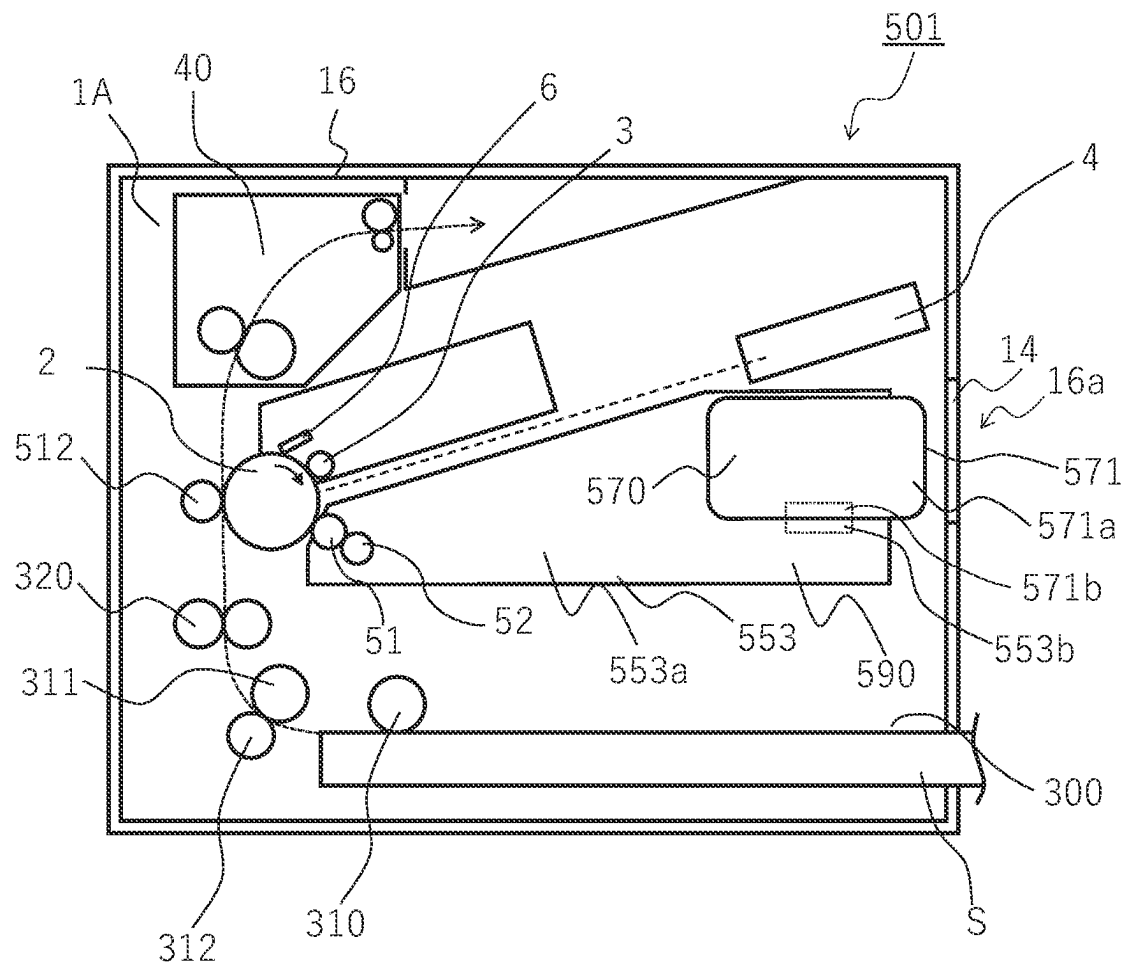


FIG. 21



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**IMAGE FORMING APPARATUS HAVING
DETACHABLE TONER CARTRIDGES****BACKGROUND OF THE INVENTION****Field of the Invention**

The present invention relates to an image forming apparatus that forms an image on a recording material.

Description of the Related Art

In an image forming apparatus of an electrophotographic system, a rotary developing system that forms a color image by rotating a rotary including a plurality of developing rollers is known. Japanese Patent Application Laid-Open Nos. 2007-183305 and 2008-096852 each disclose an image forming apparatus including a rotary including a plurality of developing rollers, and a plurality of toner cartridges (toner storage containers) each attachable to and detachable from the rotary.

SUMMARY OF THE INVENTION

According to one aspect of the invention, an image forming apparatus includes a rotary configured to rotate and including a developing roller and a storage frame including a storage portion configured to store toner to be supplied to the developing roller, a body frame with an opening, the body frame being configured to accommodate the rotary, a toner cartridge which is configured to store the toner, which is attachable to and detachable from the rotary through the opening, which is movable to an attached position and a retracted position where the toner cartridge is retracted from the attached position, with respect to the storage frame, and which is configured to supply the toner to the storage portion in a state in which the toner cartridge is at the attached position, a moving device configured to move the toner cartridge from the attached position to the retracted position, and an opening/closing member provided to be, with respect to the body frame, movable between a closed position where the opening/closing member covers the opening and an open position where the opening/closing member opens the opening to allow attachment and detachment of the toner cartridge to and from the rotary through the opening, wherein the opening/closing member is configured to move from the closed position to the open position in an interconnected manner with movement of the toner cartridge from the attached position to the retracted position by the moving device.

According to another aspect of the invention, an image forming apparatus includes a rotary including a first developing roller, a second developing roller, a first storage frame, and a second storage frame, the first storage frame including a first storage portion configured to store first toner to be supplied to the first developing roller, the second storage frame including a second storage portion configured to store second toner to be supplied to the second developing roller, a body frame configured to accommodate the rotary and having an opening, a first toner cartridge which is configured to store the first toner, which is attachable to and detachable from the rotary through the opening, which is movable to a first attached position and a first retracted position where the first toner cartridge is retracted from the first attached position, with respect to the first storage frame, and which is configured to supply the first toner to the first storage portion in a state in which the first toner cartridge is at the first

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attached position, a second toner cartridge which is configured to store the second toner, which is attachable to and detachable from the rotary through the opening, which is movable to a second attached position and a second retracted position where the second toner cartridge is retracted from the second attached position, with respect to the second storage frame, and which is configured to supply the second toner to the second storage portion in a state in which the second toner cartridge is at the second attached position, a first moving device configured to move the first toner cartridge from the first attached position to the first retracted position, and a second moving device configured to move the second toner cartridge from the second attached position to the second retracted position, and an opening/closing member provided to be, with respect to the body frame, movable between a closed position where the opening/closing member covers the opening and an open position where the opening/closing member opens the opening to allow attachment and detachment of the first toner cartridge and the second toner cartridge to and from the rotary through the opening, wherein the opening/closing member is configured to move from the closed position to the open position in an interconnected manner with movement of the first toner cartridge from the first attached position to the first retracted position by the first moving device, and wherein the opening/closing member is configured to move from the closed position to the open position in an interconnected manner with movement of the second toner cartridge from the second attached position to the second retracted position by the moving device.

According to another aspect of the invention, an image forming apparatus includes a developing device including a developing roller and a storage frame including a storage portion configured to store toner to be supplied to the developing roller, a body frame with an opening, the body frame being configured to accommodate the developing device, a toner cartridge which is configured to store the toner, which is attachable to and detachable from the developing device through the opening, which is movable between an attached position and a retracted position where the toner cartridge is retracted from the attached position, with respect to the storage frame, and which is configured to supply the toner to the storage portion in a state in which the toner cartridge is at the attached position, a moving device including a support member configured to support the toner cartridge, the moving device being configured to move the toner cartridge from the attached position to the retracted position by movement of the support member, and an opening/closing member provided to be, with respect to the body frame, movable between a closed position where the opening/closing member covers the opening and an open position where the opening/closing member opens the opening to allow attachment and detachment of the toner cartridge to and from the developing device through the opening, wherein the opening/closing member is configured to move from the closed position to the open position in an interconnected manner with movement of the toner cartridge from the attached position to the retracted position by the movement of the support member.

Further features of the present invention will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic view of an image forming apparatus according to a first embodiment.

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FIG. 2 is a configuration diagram of the image forming apparatus according to the first embodiment.

FIG. 3 is a schematic view of a developing unit, a toner cartridge, and a tray according to the first embodiment.

FIGS. 4A and 4B are each a section view of the image forming apparatus according to the first embodiment.

FIG. 5 is a perspective view of a rotary body according to the first embodiment.

FIGS. 6A to 6C are each a perspective view of the image forming apparatus according to the first embodiment.

FIGS. 7A and 7B are each a section view of the image forming apparatus according to the first embodiment.

FIG. 8 is an explanatory diagram of the rotary body according to the first embodiment.

FIG. 9 is an explanatory diagram of the rotary body according to the first embodiment.

FIG. 10 is an explanatory diagram of the rotary body according to the first embodiment.

FIGS. 11A and 11B are each an explanatory diagram of elements related to the movement of the tray according to the first embodiment.

FIGS. 12A and 12B are each an explanatory diagram of elements related to the movement of the tray according to the first embodiment.

FIG. 13 is a configuration diagram of an image forming apparatus according to a second embodiment.

FIGS. 14A to 14C are each an explanatory diagram of elements related to the movement of a tray according to the second embodiment.

FIG. 15 is an explanatory diagram of elements related to the movement of a tray according to a third embodiment.

FIGS. 16A to 16C are each an explanatory diagram of elements related to the movement of the tray according to the third embodiment.

FIGS. 17A and 17B are each an explanatory diagram of elements related to the movement of a tray according to a fourth embodiment.

FIG. 18 is a schematic view of an image forming apparatus according to a fifth embodiment.

FIGS. 19A and 19B is a schematic view of an image forming apparatus according to a sixth embodiment.

FIGS. 20A and 20B are each an explanatory diagram illustrating a moving device according to a modification example.

FIG. 21 is a schematic view of an image forming apparatus according to a fifth embodiment.

DESCRIPTION OF THE EMBODIMENTS

Embodiments of the present disclosure will be described below with reference to drawings.

First Embodiment

An image forming apparatus 1 according to a first embodiment will be described with reference to FIGS. 1 to 12B. In the description below and each drawing, the vertical direction in the case where the image forming apparatus 1 is disposed on a horizontal surface will be referred to as a Z direction. A direction that intersects with the Z direction and that is the direction of a rotational axis 90C of a rotary body 90 (rotational axis direction of a rotary) that will be described later will be referred to as a Y direction. A direction intersecting with both the Z direction and the Y direction will be referred to as an X direction. The X direction and the Y direction are preferably horizontal directions. In addition, the X direction, the Y direction, and the Z

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direction are preferably orthogonal to each other. In addition, the sides pointed by arrows X, Y, and Z illustrated in each diagram will be respectively referred to as +X, +Y, and +Z sides, and sides opposite thereto will be respectively referred to as -X, -Y, and -Z sides.

Overall Configuration of Image Forming Apparatus

First, an overall configuration of the image forming apparatus 1 will be described. The image forming apparatus 1 is a laser beam printer that forms an image on a sheet S by using an electrophotographic method. Specifically, the image forming apparatus 1 is a color laser beam printer including four developing units 50y, 50m, 50c, and 50k. As the sheet S serving as a recording material (recording medium), various sheet materials of different sizes and different materials can be used. Examples of the various sheet materials include paper sheets such as plain paper sheets and cardboards, plastic films, cloths, surface-treated sheet materials such as coated paper sheets, and sheet materials of irregular shapes such as envelopes and index paper sheets.

A schematic configuration of the image forming apparatus 1 and an image forming operation will be described with reference to FIGS. 1, 2, and 3. FIG. 1 is a schematic diagram illustrating a sectional configuration of the image forming apparatus 1. FIG. 2 is a diagram for describing drive sources of the image forming apparatus 1. FIG. 3 is a concept diagram illustrating elements for supplying toner from a toner cartridge 70 to a developing unit 50.

As illustrated in FIG. 1, the image forming apparatus 1 includes an image forming apparatus main body (hereinafter referred to as an apparatus body 1A), and toner cartridges 70y, 70m, 70c, and 70k that are attachable to and detachable from the apparatus body 1A. The apparatus body 1A of the present embodiment is a part of the image forming apparatus 1 excluding the toner cartridges 70y, 70m, 70c, and 70k.

The apparatus body 1A of the image forming apparatus 1 includes a photosensitive member 2 for use in an electrophotographic system. The photosensitive member has a drum shape (cylindrical shape) and is hereinafter referred to as a photosensitive drum 2. The photosensitive member 2 serves as an image bearing member that bears an electrostatic latent image. A charging roller 3, a scanner 4 serving as an exposing device, and a cleaning unit 6 are disposed around the photosensitive drum 2.

The charging roller 3 is an example of a charging means or a charging unit for uniformly charging the photosensitive drum 2. The scanner 4 is an example of an exposing means or an exposing unit that exposes the photosensitive drum 2 by irradiating the photosensitive drum 2 with laser light in accordance with image information. By irradiating the photosensitive drum 2 with the laser light after charging, an electrostatic latent image is formed on each surface of the photosensitive drum 2. The cleaning unit 6 is an example of a cleaning means or a cleaning portion that removes toner remaining on the surface of the photosensitive drum 2.

Further, the apparatus body 1A includes a sheet storage portion 300, a pickup roller 310, a feed roller 311, a separation roller 312, a conveyance roller pair 320, a secondary transfer roller 12, a fixing device 40, and an intermediate transfer unit 10. The pickup roller 310 is an example of a feeding means or a feeding unit that feeds the sheet S. The feed roller 311 and the separation roller 312 are an example of a separation conveyance unit that conveys sheets S while separating the sheets S from each other by frictional force. The secondary transfer roller 12 is an example of a transfer means or a transfer unit that transfers an image from an intermediate transfer belt 10a onto the sheet S.

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The intermediate transfer unit 10 includes an intermediate transfer belt 10a, a belt driving roller 10b, a tension roller 10c, a cleaning device 13, and a primary transfer roller 11. The intermediate transfer belt 10a is an example of an intermediate transfer member that bears an image transferred from the photosensitive drum 2 through primary transfer and conveys the image so as to transfer the image onto the sheet S through secondary transfer. The intermediate transfer belt 10a is stretched over the belt driving roller 10b and the tension roller 10c. The belt driving roller 10b is a driving member that is rotationally driven by a drive source to convey the intermediate transfer belt 10a.

In addition, the apparatus body 1A includes a rotary body 90, which serves as a rotary, rotator, or developing device, including the developing units 50y, 50m, 50c, and 50k. As will be described later, trays 80y, 80m, 80c, and 80k, which serve as support members, are attached to the rotary body 90 in the present embodiment. Toner cartridges 70y, 70m, 70c, and 70k are detachably attached to the trays 80y, 80m, 80c, and 80k.

In the description below, a plurality of members having similar functions can be distinguished by numbers given thereto. For example, one of the toner cartridges 70y, 70m, 70c, and 70k may be referred to as a first toner cartridge, one of the remaining three may be referred to as a second toner cartridge, one of the remaining two may be referred to as a third toner cartridge, and the last one may be referred to as a fourth toner cartridge. Similarly, one of the trays 80y, 80m, 80c, and 80k may be referred to as a first tray, one of the remaining three may be referred to as a second tray, one of the remaining two may be referred to as a third tray, and the last one may be referred to as a fourth tray. That is, one of the trays 80y to 80k is an example of a first support member, another one of the trays 80y to 80k is an example of a second support member, yet another one of the trays 80y to 80k is an example of a third support member, and the last one of the trays 80y to 80k is an example of a fourth support member. This numbering is merely used for the sake of convenience of description, and can be interchanged appropriately in principle.

The developing units 50y, 50m, 50c, and 50k serving as first to fourth developing units are examples of developing means or developing portions that each develop (visualize) an electrostatic latent image formed on the photosensitive drum 2 into a toner image by using toner of a corresponding color. The developing units 50y, 50m, 50c, and 50k each develop the electrostatic latent image formed on the photosensitive drum 2 by using corresponding one of yellow toner, magenta toner, cyan toner, and black toner. The developing units 50y, 50m, 50c, and 50k may be arranged in an order different from the order illustrated in FIG. 1.

The developing unit 50y includes a developing roller 51y, a supply roller 52y, and a developing blade. The developing roller 51y is a developer bearing member that rotates while bearing toner serving as developer or developing agent, and supplies the toner to the photosensitive drum 2. The supply roller 52y is a supply member that is disposed in contact with the developing roller 51y and supplies toner to the developing roller 51y. The developing blade is a regulation member that regulates the thickness of a toner layer borne on the developing roller 51y. The other developing units 50m, 50c, and 50k respectively include developing rollers 51m, 51c, and 51k, supply rollers, and developing blades that are configured in a similar manner.

The toner cartridges 70y, 70m, 70c, and 70k corresponding to the developing units 50y, 50m, 50c, and 50k are attached to the rotary body 90. The toner cartridges 70y,

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70m, 70c, and 70k respectively store therein yellow toner, magenta toner, cyan toner, and black toner to be supplied to the developing units 50y, 50m, 50c, and 50k. One of the toners of four colors may be referred to as first toner, one of the toners of remaining three colors may be referred to as second toner, one of the toners of remaining two colors may be referred to as third toner, and the toner of the last remaining color may be referred to as fourth toner. For example, the black toner may be referred to as an example of first toner, and the magenta toner may be referred to as an example of second toner. This numbering is merely used for the sake of convenience of description, and can be interchanged appropriately in principle.

Here, the rotary body 90 includes a rotary frame 90f supporting the developing units 50y, 50m, 50c, and 50k. The developing units 50y, 50m, 50c, and 50k are supported by the rotary frame 90f that is a rotary support member that is rotatable.

In addition, the trays 80y, 80m, 80c, and 80k are attached to the rotary body 90. The rotary body 90 and the trays 80y, 80m, 80c, and 80k as a combination can be referred to as a rotary unit 90U. In other words, the rotary unit 90U includes the rotary body 90 and the trays 80y, 80m, 80c, and 80k.

The toner cartridges 70y to 70k are detachably held by the trays 80y to 80k. As will be described later, the trays 80y to 80k are supported so as to be slidable to the outside of the rotary body 90. The rotary unit 90U and the toner cartridges 70y, 70m, 70c, and 70k as combined can be referred to as a rotary assembly 90A. In other words, the rotary assembly 90A includes the rotary unit 90U and toner cartridges 70y, 70m, 70c, and 70k.

As will be described later, the rotary body 90 is rotatable about a rotational axis (rotational center) 90C. The rotational axis 90C coincides with a rotational axis of the rotary frame 90f, that of the rotary unit 90U, and that of the rotary assembly 90A. In addition, the rotational axis 90C is substantially parallel to the rotational axis (rotational center) of the photosensitive drum 2.

The rotary body 90 rotates about the rotational axis 90C, and thus can take developing postures in each of which any one of the developing rollers 51y, 51m, 51c, and 51k faces the photosensitive drum 2. A posture in which the developing roller 51y faces the photosensitive drum 2 will be referred to as a yellow developing posture. A posture in which the developing roller 51m faces the photosensitive drum 2 will be referred to as a magenta developing posture. A posture in which the developing roller 51c faces the photosensitive drum 2 will be referred to as a cyan developing posture. A posture in which the developing roller 51k faces the photosensitive drum 2 will be referred to as a black developing posture. That is, the rotary body 90 can rotate about the rotational axis 90C such that the positions of the developing rollers 51y, 51m, 51c, and 51k change with respect to the photosensitive drum 2. The black developing posture is an example of a first developing posture in which the first developing roller (e.g., developing roller 51k) faces the photosensitive drum 2. The other developing postures are examples of a second developing posture in which a second developing roller (e.g., one of the developing rollers 51y to 51c) faces the photosensitive drum 2. The yellow/magenta/cyan/black developing postures can be referred to as first to fourth developing postures. This numbering is merely used for the sake of convenience of description, and can be interchanged appropriately in principle.

As illustrated in FIG. 2, the apparatus body 1A includes motors M1, M2, and M3 serving as drive sources. As will be described later, the motor M1 supplies a driving force for

rotating the rotary body **90** about the rotational axis **90C**. In other words, the motor **M1** rotates the rotary assembly **90A** and the rotary unit **90U** about the rotational axis **90C**.

In addition, the apparatus body **1A** includes a driving device **98** including the motor **M2** and a transmission device. The transmission device includes driving racks **15L** and **15R** serving as driving gears and a transmission portion **15t** that will be described later. The driving force of the motor **M2** is transmitted to the driving racks **15L** and **15R** by the transmission portion **15t**. In other words, the motor **M2** is configured to drive the driving racks **15L** and **15R**, and moves the trays **80y**, **80m**, **80c**, and **80k** with respect to the rotary body **90** via the driving racks **15L** and **15R**.

The motor **M3** drives members that are not driven by the motors **M1** and **M2**. For example, the motor **M3** drives the photosensitive drum **2**, the developing units **50y**, **50m**, **50c**, and **50k**, the pickup roller **310**, the feed roller **311**, the conveyance roller pair **320**, the secondary transfer roller **12**, the belt driving roller **10b**, and the fixing device **40**.

To be noted, the members driven by the motors **M1**, **M2**, and **M3** can be appropriately changed. In addition, the roles of two or three of the motors **M1**, **M2**, and **M3** can be concentrated in one motor. In addition, a drive source other than the motors **M1**, **M2**, and **M3** may be added.

Here, the suffices *y*, *m*, *c*, and *k* given to the developing units **50y**, **50m**, **50c**, and **50k**, the toner cartridges **70y**, **70m**, **70c**, and **70k**, the trays **80y**, **80m**, **80c**, and **80k**, and the like indicate the colors of toner. The developing units **50y**, **50m**, **50c**, and **50k** basically have the same configuration and function. The toner cartridges **70y**, **70m**, **70c**, and **70k** basically have the same configuration and function. In addition, the trays **80y**, **80m**, **80c**, and **80k** basically have the same configuration and function. Therefore, in the case where these do not need to be distinguished, the suffices *y*, *m*, *c*, and *k* will be omitted, and arbitrarily selected one of the four units, four cartridges, and four trays will be described.

As illustrated in FIG. 3, the toner cartridge **70** includes a toner frame **71**. The toner frame **71** includes a toner storage portion **71a** that stores toner, and a discharge opening **71b** communicating with the toner storage portion **71a**.

The developing unit **50** includes a developing frame **53** serving as a storage frame. The developing frame **53** includes a developing-side storage portion **53a** and an inlet opening **53b** communicating with the developing-side storage portion (toner supply chamber) **53a**. To be noted, as described above, although the developing unit **50** includes the developing roller **51**, the supply roller **52**, and the like, illustration of these members is omitted in FIG. 3.

The developing roller **51k** included in the developing unit **50k** is an example of a first developing roller. The developing roller **51m** included in the developing unit **50m** is an example of a second developing roller. A developing frame **53k** of the developing unit **50k** including the developing-side storage portion **53a** illustrated in FIG. 4A is an example of a first storage frame including a first storage portion. A developing frame **53m** of the developing unit **50m** including the developing-side storage portion **53a** illustrated in FIG. 4A is an example of a second storage frame including a second storage portion. The rotary body **90** is an example of a rotary that is rotatable and includes a first developing roller, a second developing roller, a first storage frame including a first storage portion, and a second storage frame including a second storage portion. In the present embodiment, the rotary body **90** includes first to fourth developing rollers and first to fourth storage frames.

As will be described later, the toner cartridge **70** is movable to an attached position and a retracted position where the toner cartridge **70** is retracted from the attached position, with respect to the developing frame **53**. In a state in which the toner cartridge **70** is at the attached position with respect to the developing frame **53**, the discharge opening **71b** faces the inlet opening **53b**. That is, the toner storage portion **71a** of the toner cartridge **70** and the developing-side storage portion **53a** of the developing unit **50** communicate with each other via the discharge opening **71b** and the inlet opening **53b**. When toner is supplied from the toner cartridge **70** to the developing unit **50**, at least part of the inlet opening **53b** is positioned below at least part of the discharge opening **71b**.

Then, toner stored in the toner storage portion **71a** is discharged through the discharge opening **71b**, and the toner discharged through the discharge opening **71b** is stored in the developing-side storage portion **53a** through the inlet opening **53b**. The toner stored in the developing-side storage portion **53a** is supplied to the developing roller **51** by the supply roller **52**. The toner stored in the toner storage portion **71a** is supplied to the developing roller **51** through such a path.

The toner cartridge **70** preferably includes an unillustrated sealing member (first sealing member) that covers the discharge opening **71b**. In addition, the developing unit **50** preferably includes an unillustrated sealing member (second sealing member) that covers the inlet opening **53b**.

In a state in which the toner cartridge **70** is not attached to the developing unit **50**, the discharge opening **71b** and the inlet opening **53b** are preferably each covered by a sealing member such that leakage of toner through the discharge opening **71b** and the inlet opening **53b** is suppressed.

Image Forming Operation

An image forming operation in the present embodiment will be described. First, the photosensitive drum **2** is rotated in an arrow direction (counterclockwise direction) in FIG. 1 in synchronization with the rotation of the intermediate transfer belt **10a**. Further, the surface of the photosensitive drum **2** is uniformly charged by the charging roller **3**.

In the case of forming a color image on the sheet **S**, the rotary body **90** rotates in an arrow direction (clockwise direction) in FIG. 1 while supporting the developing units **50y**, **50m**, **50c**, and **50k**. Then, an electrophotographic process is repeatedly performed while moving the developing rollers **51y**, **51m**, **51c**, and **51k** to a developing position one by one.

First, the scanner **4** emits laser light based on image data corresponding to a yellow image, and thus forms an electrostatic latent image corresponding to the yellow image on the surface of the photosensitive drum **2**. In parallel with the formation of this electrostatic latent image, the motor **M1** rotates the rotary body **90**, and the rotary body **90** takes the yellow developing posture. When the rotary body **90** is in the yellow developing posture, the developing roller **51y** is in the developing position, and the electrostatic latent image formed on the photosensitive drum **2** is developed with yellow toner.

Here, in the present embodiment, the developing rollers **51y**, **51m**, **51c**, and **51k** are each an elastic roller formed by covering a metal shaft with rubber. At the developing position, the developing rollers **51y**, **51m**, **51c**, and **51k** each develop the electrostatic latent image in a state of being in contact with the photosensitive drum **2**. That is, a contact development system is employed for the image forming apparatus **1** of the present embodiment. However, at the developing position, each of the developing rollers **51y**,

51m, **51c**, and **51k** may develop the electrostatic latent image with a gap between the developing roller and the photosensitive drum **2**. That is, a non-contact development system may be employed for the image forming apparatus **1**.

After the yellow toner image is developed, the yellow toner image on the photosensitive drum **2** is transferred onto the intermediate transfer belt **10a** through primary transfer by the primary transfer roller **11** disposed on the inner peripheral side of the intermediate transfer belt **10a**.

After this, toner images of respective colors are formed by rotating the rotary body **90** and thus sequentially moving the developing rollers **51m**, **51c**, and **51k** to the developing position. That is, after the yellow toner image is formed on the intermediate transfer belt **10a**, the rotary body **90** takes a magenta developing posture, and a magenta toner image is formed on the intermediate transfer belt **10a**. After the magenta toner image is formed on the intermediate transfer belt **10a**, the rotary body **90** takes a cyan developing posture, and a cyan toner image is formed on the intermediate transfer belt **10a**. After the cyan toner image is formed on the intermediate transfer belt **10a**, the rotary body **90** takes a black developing posture, and a black toner image is formed on the intermediate transfer belt **10a**. After the black toner image is formed on the intermediate transfer belt **10a**, the rotary body **90** rotates about the rotational axis **90C** in an arrow direction (clockwise direction) illustrated in FIG. **1**, and returns to the yellow developing posture. To be noted, the color of the image to be formed first on the intermediate transfer belt **10a** can be arbitrarily selected, and for example, the black toner image may be formed first.

Then, primary transfer is repeated so as to superimpose the toner images of four colors on each other on the intermediate transfer belt **10a**, and thus a color image is formed on the intermediate transfer belt **10a**. To be noted, before the color image is formed on the intermediate transfer belt **10a**, the secondary transfer roller **12** and the cleaning device **13** are not in contact with the intermediate transfer belt **10a**.

Meanwhile, the sheet **S** is fed by the pickup roller **310** from the sheet storage portion **300** provided in a lower portion of the apparatus body **1A**. The sheet **S** is conveyed to the conveyance roller pair **320** in a state in which one sheet **S** is separated from a stack of sheets **S** by the feed roller **311** and the separation roller **312**. The conveyance roller pair **320** delivers out the fed sheet **S** to a transfer portion (secondary transfer portion) that is a nip portion between the intermediate transfer belt **10a** and the secondary transfer roller **12**. The color image on the intermediate transfer belt **10a** is transferred onto the surface of the conveyed sheet **S** through secondary transfer.

The sheet **S** onto which a color image has been transferred is conveyed to the fixing device **40**. In the fixing device **40**, the sheet **S** is heated and pressurized, and thus the image is fixed to the sheet **S**. The sheet **S** having passed the fixing device **40** is discharged to the outside of the image forming apparatus **1** as a product.

In contrast, in the case of forming a black-and-white image (monochrome image) on the sheet **S**, the rotary body **90** takes the black developing posture. In this state, an electrostatic latent image is formed on the surface of the photosensitive drum **2** by charging and exposing the photosensitive drum **2**, and then the electrostatic latent image is developed with black toner by the developing roller **51k** positioned at the developing position. The black toner image is transferred onto the intermediate transfer belt **10a** through primary transfer, and then the toner image is transferred onto

the sheet **S** through secondary transfer. Steps after this are similar to the case of a color image.

Rotary Configuration

The configuration of the rotary body **90** will be described with reference to FIGS. **1**, **4A**, **4B**, and **5**. FIGS. **4A** and **4B** are each a section view of the rotary body **90** of the image forming apparatus **1** and the surroundings thereof. To be noted, FIGS. **4A** and **4B** are each a section view taken along a virtual plane orthogonal to the rotational axis **90C** of the rotary body **90**. FIG. **5** is a perspective view of the rotary body **90**.

As has been described, the toner cartridges **70y** to **70k** are attachable to and detachable from the rotary body **90**. In the case where toner in the toner cartridges **70y** to **70k** has run out, the user can replenish the image forming apparatus **1** with toner by replacing the toner cartridges **70y** to **70k**.

As illustrated in FIG. **1**, the apparatus body **1A** includes a frame **16** accommodating the rotary body **90**. The frame **16** is a body frame of the image forming apparatus **1** of the present embodiment. The frame **16** is a casing or skeleton of the apparatus body **1A** constituted by a frame and exterior members, and has an approximately rectangular parallelepiped shape.

The frame **16** has an opening **16a**. More specifically, the frame **16** has a side surface **16b** extending in a direction intersecting with the horizontal direction. The side surface **16b** constitutes at least part of the exterior surface of the apparatus body **1A** on the +X side. The opening **16a** is provided in the side surface **16b**. The side surface **16b** is a side surface disposed on the downstream side of the discharge port in a discharge direction in which the sheet **S** on which an image has been formed is discharged from the apparatus body **1A** through the discharge port. From the side surface **16b** side of the image forming apparatus **1**, the user can access the sheet storage portion **300** to replenish the sheet storage portion **300** with sheets **S**, and can obtain the sheet **S** discharged through the discharge port. Therefore, the side surface **16b** can be referred to as the front surface of the apparatus body **1A**.

The toner cartridges **70y**, **70m**, **70c**, and **70k** are attachable to and detachable from the rotary body **90** through the opening **16a**. That is, the toner cartridge **70k** can be referred to as an example of a first toner cartridge that stores toner to be supplied to the first developing roller (developing roller **51k**) and that is attachable to and detachable from the rotary (rotary body **90**) through the opening **16a** of the frame **16** of the apparatus body **1A**. The toner cartridge **70m** can be referred to as an example of a second toner cartridge that stores toner to be supplied to the second developing roller (developing roller **51m**) and that is attachable to and detachable from the rotary (rotary body **90**) through the opening **16a** of the frame **16** of the apparatus body **1A**.

In the present embodiment, the toner cartridges **70y**, **70m**, **70c**, and **70k** are attached to and detached from the rotary body **90** through the opening **16a** in the state of being supported by the trays **80y** to **80k**. In other words, the user can attach and detach the toner cartridges **70y** to **70k** to and from the rotary body **90** via the trays **80y** to **80k**.

The opening **16a** is disposed in the side surface **16b** of the frame **16**. In the present embodiment, the side surface **16b** is a surface approximately parallel to the rotational axis **90C** of the rotary body **90**. Therefore, in the case of replacing the toner cartridge **70**, the toner cartridge **70** passes through the opening **16a** in a direction intersecting with the rotational axis **90C** (preferably a direction orthogonal to the rotational axis **90C**).

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The image forming apparatus 1 includes a door 14 that covers the opening 16a of the frame 16. The door 14 is an opening/closing member movable to a closed position illustrated in FIG. 6A where the door 14 covers the opening 16a and an open position illustrated in FIGS. 6B and 6C where the opening 16a is exposed.

As described above, in the present embodiment, the toner cartridge 70 is configured to be attachable to and detachable from the rotary body 90 via the tray 80. Therefore, the toner cartridge 70 can be stably attached to and detached from the rotary body 90.

More specifically, the user can replace the toner cartridge 70 by an operation of attaching and detaching the toner cartridge 70 to and from the tray 80 configured to be movable with respect to the rotary body 90 (that is, with respect to the apparatus body 1A). In the case of a configuration in which the toner cartridge is replaced by the user directly inserting and removing the toner cartridge in and from the apparatus body, the user is required to insert the toner cartridge to a predetermined attached position in the apparatus body. In the present embodiment, the tray 80 is capable of moving such that the toner cartridge 70 moves to the attached position in a state in which the tray 80 is supporting the toner cartridge 70. Therefore, the user can replace the toner cartridge 70 by a simple operation of placing the toner cartridge 70 on the tray 80, and thus the operability is improved.

To be noted, the toner cartridge 70 has a thin elongated shape with the Y direction parallel to the rotational axis 90C of the rotary body 90 as a longitudinal direction. That is, the dimension of the toner cartridge 70 in the longitudinal direction is larger than its height and width in a cross-section orthogonal to the longitudinal direction. In the case of using the toner cartridge 70 having a thin elongated shape as described above, the toner cartridge 70 can be passed through the opening 16a in a short movement distance by providing the opening 16a in the side surface 16b of the frame 16 that is approximately parallel to the longitudinal direction (Y direction) of the toner cartridge 70. The replacement of the toner cartridge 70 becomes easier than, for example, a case where the toner cartridge 70 is inserted or removed through an opening provided in a side surface on one side (+Y side or -Y side) of the frame 16 in the longitudinal direction of the toner cartridge 70.

In the present embodiment, the four toner cartridges 70y, 70m, 70c, and 70k can be replaced via the opening 16a that is a shared opening by rotating the rotary body 90. Therefore, there is no need to provide a plurality of openings respectively for the toner cartridges in the frame 16, and the total area of the opening provided in the frame 16 for toner cartridge replacement can be reduced. Since the total area of the opening in the frame 16 is smaller, the strength of the frame 16 can be improved. In addition, since only one opening 16a is provided as a user interface for attaching and detaching a plurality of toner cartridges 70, the user is not confused about where to attach or detach the toner cartridge 70, and thus the replacement work can be easily performed.

The rotary body 90 rotates about the rotational axis 90C, and thus can take a replacement posture in which detachment of one of the toner cartridges 70y to 70k from the rotary body 90 is allowed. A posture in which the detachment of the toner cartridge 70y is allowed will be referred to as a yellow replacement posture. A posture in which the detachment of the toner cartridge 70m is allowed will be referred to as a magenta replacement posture. A posture in which the detachment of the toner cartridge 70c is allowed will be referred to as a cyan replacement posture. A posture in which

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the detachment of the toner cartridge 70k is allowed will be referred to as a black replacement posture. The black replacement posture is an example of a first replacement posture in which detachment of a first toner cartridge from the rotary body 90 is allowed. The yellow/magenta/cyan replacement postures are examples of a second replacement posture in which detachment of a second toner cartridge from the rotary body 90 is allowed. The yellow/magenta/cyan/black replacement postures can be referred to as first to fourth replacement postures. This numbering is merely used for the sake of convenience of description, and can be interchanged appropriately in principle.

The rotary body 90 rotates about the rotational axis 90C in a counterclockwise direction of FIG. 1, and can sequentially take the yellow/magenta/cyan/black replacement postures. In the present embodiment, the rotary body 90 rotates about the rotational axis 90C in the counterclockwise direction of FIG. 1, and thus the developing posture and the replacement posture can be switched alternately. For example, in FIG. 1, the rotary body 90 is in the black developing posture. By rotating the rotary body 90 in the clockwise direction from this state, the posture of the rotary body 90 can be switched in the order of the cyan replacement posture, the yellow developing posture, the black replacement posture, the magenta developing posture, the yellow replacement posture, the cyan developing posture, and the magenta replacement posture. By rotating the rotary body 90 in the clockwise direction from the magenta replacement posture, the rotary body 90 returns to the black developing posture. That is, the rotary body 90 can rotate more than once (360°) in the clockwise direction.

FIG. 4A illustrates a cross-section of the rotary body 90 in a developing posture (specifically, the yellow developing posture). FIG. 4B illustrates a cross-section of the rotary body 90 in a replacement posture (specifically, the black replacement posture).

As illustrated in FIGS. 4A and 4B, the four trays 80y to 80k are attached to the rotary body 90. The trays 80y to 80k respectively hold the toner cartridges 70y to 70k. In FIGS. 4A and 4B, the trays 80y to 80k are accommodated in the rotary body 90, and this state can be referred to as a state in which the toner cartridges 70y to 70k are attached to the developing units 50y, 50m, 50c, and 50k.

As described above, the toner cartridge 70 is movable to an attached position and a retracted position where the toner cartridge 70 is retracted from the attached position, with respect to the developing frame 53 of the developing unit 50. That is, the first toner cartridge (toner cartridge 70k) is movable to a first attached position and a first retracted position with respect to a first storage frame (developing frame 53k). The second toner cartridge (toner cartridge 70m) is movable to a second attached position and a second retracted position with respect to a second storage frame (developing frame 53m).

In a state in which the toner cartridge 70 is at the attached position with respect to the developing frame 53, the discharge opening 71b and the inlet opening 53b face each other as illustrated in FIG. 3. The toner cartridge 70 is configured to supply toner to the developing-side storage portion 53a through the inlet opening 53b (opening of the storage frame) in this state.

The apparatus body 1A includes a moving device 85 configured to move the toner cartridge 70 from the attached position to the retracted position with respect to the rotary body 90 (more specifically, with respect to the developing frame 53 of the developing unit 50). The moving device 85 will be described below with reference to FIG. 8 and the

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like. In the present embodiment, a plurality of moving devices **85y** to **85k** corresponding to the plurality of toner cartridges **70y** to **70k** are disposed in the rotary body **90**. The trays **80y** to **80k** can be referred to as part of the moving devices **85y** to **85k**.

In the present embodiment, the toner cartridge **70k** storing the black toner is larger in size than the toner cartridges **70y** to **70c** storing yellow toner, magenta toner, and cyan toner, and is capable of storing more toner. In other words, the first toner cartridge is capable of storing a first amount of toner, the second toner cartridge is capable of storing a second amount of toner, and the first amount is larger than the second amount.

Specifically, the length of the black toner cartridge **70k** in a first radial direction with respect to the rotational axis **90C** of the rotary body **90** is larger than the length of the magenta toner cartridge **70m** in a second radial direction. Here, the first radial direction is a rotational radius direction (radial direction of a virtual circle centered on the rotational axis **90C**) of the rotary body **90**, and is a direction in which the toner cartridge **70k** extends with respect to the rotational axis **90C** as viewed in the direction of the rotational axis **90C**. The second radial direction is a rotational radius direction of the rotary body **90**, and is a direction in which the toner cartridge **70m** extends with respect to the rotational axis **90C** as viewed in the direction of the rotational axis **90C**. Similarly, the length of the black toner cartridge **70k** in the first radial direction is larger than the lengths of the other toner cartridges **70y** and **70c** in the radial directions corresponding to the toner cartridges **70y** and **70c**.

Accordingly, the tray **80k** holding the black toner cartridge **70k** is larger in size than the trays **80y** to **80c** holding the other toner cartridges **70y**, **70m**, and **70c**. That is, the four toner cartridges **70y** to **70k** and the trays **80y** to **80k** having different sizes are disposed in the rotary body **90**. In other words, the toner cartridge **70k** serving as an example of a first toner cartridge and the toner cartridge **70y** serving as an example of a second toner cartridge smaller than the first toner cartridge are attachable to and detachable from the rotary body **90**. In accordance with this, the tray **80k** serving as an example of a first support member that supports the first toner cartridge and the tray **80y** serving as an example of a second support member smaller than the first support member are provided in the rotary body **90**. In addition, the toner cartridges **70m** and **70c** serving as examples of a third toner cartridge and a fourth toner cartridge that are smaller in size than the first toner cartridge are attachable to and detachable from the rotary body **90**. In accordance with this, the trays **80m** and **80c** serving as examples of a third support member and a fourth support member that are smaller in size than the first support member are provided in the rotary body **90**.

Here, rotational driving of the rotary body **90** will be described with reference to FIG. 5. As illustrated in FIG. 5, disk gears **92L** and **92R** are formed on respective end portions of the rotary body **90**. In addition, rotary driving gears **93L** and **93R** are formed on respective end portions of a swing shaft **91** in a manner capable of transmitting a driving force. Here, the driving force of the motor **M1** is transmitted to the rotary driving gear **93R** via a drive transmission mechanism. Next, the driving force is transmitted to the disk gears **92L** and **92R** via the rotary driving gears **93L** and **93R**, and thus the rotary body **90** is rotationally driven. The rotary body **90** rotates about the rotational axis **90C** in the clockwise direction in FIG. 1.

In addition, the rotary body **90** is supported so as to be swingable about the swing shaft **91**. The rotary body **90** is

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urged in a counterclockwise direction in FIGS. 4A and 4B about the swing shaft **91** by an unillustrated urging member. This direction can be referred to as a direction in which the developing rollers **51y** to **51k** each approach the photosensitive drum **2**. As a result of this, one of the developing rollers **51y** to **51k** is in contact with the photosensitive drum **2** in a state in which the rotary body **90** is in a developing posture.

Meanwhile, as illustrated in FIG. 5, rotary cams **90eL** and **90eR** are provided on respective end portions of the rotary body **90**. When the rotary body **90** rotates about the rotational axis **90C** in a clockwise direction in FIGS. 4A and 4B, the rotary cams **90eL** and **90eR** come into contact with a roller **96** supported by the frame **16** illustrated in FIGS. 4A and 4B. Then, the rotary body **90** moves in the clockwise direction in FIGS. 4A and 4B about the swing shaft **91**. This direction can be referred to as a direction in which the developing rollers **51y** to **51k** each move away from the photosensitive drum **2**. In addition, this direction can be referred to as a direction in which the rotary body **90** approaches the opening **16a** of the frame **16** and the door **14**.

As a result of this, when the rotary body **90** rotates and switches from the developing posture to the replacement posture, the rotary body **90** swings about the swing shaft **91**. In a state in which the rotary body **90** is in the replacement posture, the developing roller **51** is separated from the photosensitive drum **2**.

As illustrated in FIG. 4B, in the black replacement posture, the toner cartridge **70k** stops at a position where the toner cartridge **70k** faces the opening **16a** and the door **14** provided on the side surface **16b** of the apparatus body **1A**. When the tray **80k** is slid from the attached position for the developing unit **50k** to the outside of the rotary body **90** from this state, the user can replace the toner cartridge **70k**.

Replacement Operation of Toner Cartridge

A toner cartridge replacement operation will be described with reference to FIGS. 4A, 6A to 6C, 7A, and 7B. FIGS. 6A to 6C are each an exterior view of the apparatus body **1A**. FIGS. 7A and 7B are each a section view of the rotary body **90** and the surroundings thereof in toner cartridge replacement. To be noted, FIGS. 7A and 7B are each a section view of the apparatus taken along a virtual plane orthogonal to the rotational axis **90C** of the rotary body **90**.

FIG. 6A illustrates an external appearance of the apparatus body **1A** during the image forming operation and in a standby state. During the image forming operation is during a period in which a series of operations of the image forming apparatus **1** feeding a sheet **S**, forming an image on the sheet **S**, and then discharging the sheet **S** as a product are executed. The standby state is a state in which the image forming operation can be started if the image forming apparatus **1** receives an image forming instruction (printing instruction), and a state in which the image forming apparatus **1** is standing by for the image forming instruction from the user. As illustrated in FIG. 6A, the door **14** is closed during the image forming operation and in the standby state.

FIG. 6B illustrates the external appearance of the apparatus body **1A** at the time of toner cartridge replacement. At the time of toner cartridge replacement, the door **14** is opened, and the tray **80** and the toner cartridge **70** are moved to the outside of the apparatus body **1A**.

The toner cartridge **70** is movable to an attached position and a retracted position where the toner cartridge **70** is retracted from the attached position, with respect to the developing frame **53** of the developing unit **50**. In a state in which the toner cartridge **70** is at the attached position with respect to the developing frame **53**, the discharge opening

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71b and the inlet opening 53b face each other as illustrated in FIG. 3. As illustrated in FIGS. 4A and 4B, the rotary body 90 is configured to rotate about the rotational axis 90C to take the developing posture and the replacement posture in a state in which the toner cartridge 70 is at the attached position.

The toner cartridge replacement operation will be described. First, the user instructs the controller of the apparatus body 1A of a toner cartridge replacement operation. The instruction of toner cartridge replacement operation is given by, for example, input via an operation panel (operation portion) provided on the apparatus body 1A.

When the controller receives the instruction of the toner cartridge replacement operation, the rotary body 90 rotates to the replacement posture of the toner cartridge 70 serving as a replacement target (toner cartridge 70 whose toner has run out), and stops. That is, the controller rotates the rotary body 90 to the replacement posture of a toner cartridge specified in the instruction of the toner cartridge replacement (in FIG. 4B, the black replacement posture for replacing the black toner cartridge 70k). In the replacement posture, the tray 80 supporting the toner cartridge 70 whose replacement has been instructed faces the opening 16a of the frame 16 of the apparatus body 1A.

For example, the rotary body 90 of FIG. 4A is in the yellow developing posture in which the yellow developing roller 51y faces the photosensitive drum 2. At this time, the black toner cartridge 70k and the tray 80k do not have to face the opening 16a and the door 14. In other words, the toner cartridge 70 and the tray 80 do not have to face the opening 16a and the door 14 in the case where the rotary body 90 is in a developing posture or a replacement posture other than the replacement posture of the toner cartridge. Therefore, the opening 16a may have such a size that each of the toner cartridges 70 can individually pass therethrough. When the rotary body 90 rotates in the clockwise direction in the drawings by a predetermined angle from the yellow developing posture, the black toner cartridge 70k and the tray 80k face the opening 16a and the door 14 as illustrated in FIG. 4B.

Here, "the tray 80 facing the opening 16a" means that the tray 80 is positioned so as to be able to move to the outside of the apparatus body 1A through the opening 16a. That is, in the case where the tray 80 faces the opening 16a, a moving mechanism that will be described later moves the tray 80 outward in the rotational radius direction of the rotary body 90, and therefore the tray 80 and the toner cartridge 70 supported by the tray 80 are capable of projecting to the outside of the apparatus body 1A. In FIG. 4A, none of the trays 80y to 80k faces the opening 16a. In FIG. 4B, only the black tray 80k faces the opening 16a, and the other trays 80y to 80c do not face the opening 16a.

When the rotary body 90 is positioned in the replacement posture, the motor M2 moves the tray 80 supporting the toner cartridge 70 serving as a replacement target to the outside of the apparatus body 1A. As a result of this, the toner cartridge 70 serving as a replacement target moves from the attached position to the retracted position with respect to the rotary body 90. In addition, as illustrated in FIGS. 6B, 6C, 7A, and 7B, the tray 80 and the toner cartridge 70 serving as a replacement target supported by the tray 80 projects to the outside of the apparatus body 1A through the opening 16a.

More specifically, the tray 80 is movable to an accommodation position and a detachment position with respect to the rotary body 90. The accommodation position is a position where the tray 80 is accommodated in the rotary body

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90. The detachment position is a position (removal position or replaceable position) where the tray 80 projects to the outside of the rotary body 90 and the toner cartridge 70 can be detached from the tray 80. The positions of the trays 80y to 80k in FIGS. 4A and 4B serve as examples of the accommodation position. The position of the tray 80 in FIGS. 6B and 6C, the position of the tray 80k in FIG. 7A, and the position of the tray 80m in FIG. 7B serve as examples of the detachment position.

When the tray 80 is at the accommodation position, the toner cartridge 70 attached to the tray 80 is at the attached position. When the tray 80 is at the detachment position, the toner cartridge 70 attached to the tray 80 is at the retracted position.

Here, as illustrated in FIGS. 7A and 7B, the rotary body 90 has projection portions 95 for holding the tray 80 at the accommodation position and holding the toner cartridge 70 at the attached position. As illustrated in FIG. 8, the tray 80 has recess portions 87 configured to be fit on the projection portions 95. FIGS. 7A and 7B illustrate projection portions 95k and 95m corresponding to the trays 80k and 80m, and FIG. 8 illustrates recess portions 87y and 87m of the trays 80y and 80m. The projection portion 95 and the recess portion 87 are provided for each of the trays 80y to 80k. The projection portion 95 is preferably urged in such a direction as to engage with the recess portion 87.

The projection portion 95 fits in the recess portion 87 of the tray 80, and thus the tray 80 is locked with respect to the rotary frame 90f. As a result of this, the tray 80 stays in the accommodation position even when the rotary body 90 rotates, and thus movement of the toner cartridge 70 from the attached position can be suppressed. To be noted, in the case where the tray 80 is moved between the accommodation position and the detachment position by the moving device that will be described below, the projection portion 95 can be configured to be moved by the tray 80 and thus disengaged from the recess portion 87.

In the present embodiment, the door 14 is supported so as to be pivotable with respect to the apparatus body 1A. As illustrated in FIG. 7A, the door 14 is urged from the open position to the closed position by a spring 14s. The spring 14s is, for example, a tension spring, and urges the door 14 such that a moment in a counterclockwise direction in FIGS. 7A and 7B is generated about a support shaft 14c of the door 14.

The tray 80 pushes the door 14, and thus the door 14 takes the open state illustrated in FIG. 6B. This state can be referred to as a state in which the tray 80 is supported by the door 14. The door 14 supports at least part of the tray 80 projecting to the outside of the apparatus body 1A, and thus the toner cartridge 70 can be supported more stably. In other words, when the first toner cartridge (toner cartridge 70k) is at the first retracted position, the opening/closing member (door 14) at the open position supports the first support member (tray 80k). In addition, when the second toner cartridge (one of the toner cartridges 70y to 70c) is at the second retracted position, the opening/closing member (door 14) at the open position supports the second support member (one of the trays 80y to 80c).

To be noted, the door 14 is configured such that the door 14 at the open position comes in contact with part (for example, a lower edge 16c of the opening 16a) of the frame 16 of the apparatus body 1A and not to pivot downward beyond the open position. When the tray 80 is pulled back into the apparatus body 1A from the outside, the door 14 returns to the closed position by the urging force of the spring 14s.

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The toner cartridge 70 is detachably held by the tray 80. Therefore, as illustrated in FIG. 6C, the user can perform a work (replacement work) of detaching the toner cartridge 70 from the tray 80 and attaching a new toner cartridge 70. To be noted, in the case of replacing a plurality of toner cartridges 70, the replacement work can be performed by repeating the operation described above.

As described above, in the present embodiment, a configuration in which the opening/closing member (e.g., door 14) moves from the closed position to the open position in an interconnected manner with movement of the toner cartridge 70 from the attached position to the retracted position by the moving device 85 has been employed. As a result of this, in the present embodiment, the moving device 85 includes a support member (e.g., tray 80) supporting the toner cartridge 70, and the opening/closing member (e.g., door 14) moves from the closed position to the open position in an interconnected manner with the movement of the toner cartridge 70 from the attached position to the retracted position by the movement of the support member. As a result of this, the user does not have to open the door 14 manually, which makes it easier for the user to replace the toner cartridge.

In addition, in the present embodiment, since the opening/closing member (e.g., door 14) is urged toward the closed position by the spring 14s serving as an urging member, the opening/closing member moves from the open position to the closed position in an interconnected manner with the movement of the toner cartridge 70 from the retracted position to the attached position by the moving device 85. As a result of this, the user does not have to close the door 14 manually after replacing the toner cartridge, which makes it easier for the user to replace the toner cartridge.

In addition, in the present embodiment, in accordance with the movement of the toner cartridge 70 from the attached position to the retracted position, the support member (e.g., tray 80) pushes the opening/closing member (e.g., door 14), and thus the door 14 moves from the closed position to the open position. As a result of this, movement of the opening/closing member interconnected with the movement of the toner cartridge 70 can be realized without providing an additional member for moving the opening/closing member. To be noted, the opening/closing member can be also moved in an interconnected manner with the movement of the toner cartridge 70 by a configuration different from the configuration in which the support member pushes the opening/closing member as in a modification example that will be described later.

FIGS. 7A and 7B illustrate a cross-section of the rotary body 90 and the surroundings thereof at the time of toner cartridge replacement. FIG. 7A illustrates a state at the time of replacing the black toner cartridge 70k. FIG. 7B illustrates a state at the time of replacing the magenta toner cartridge 70m.

The image forming apparatus 1 includes the moving device 85 illustrated in FIG. 8 that moves the toner cartridge 70 from the attached position to the retracted position. In the present embodiment, it can be said that the moving device 85 includes the tray 80. A moving device 85k including the tray 80k can be referred to as an example of a first moving device including a first support member. A moving device 85m including the tray 80m can be referred to as an example of a second moving device including a second support member.

Even when the toner cartridge 70 is at the retracted position, the tray 80 is coupled to the rotary body 90 (supported by the rotary body 90). To easily detach the toner

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cartridge 70 from the rotary body 90, it is preferable that the length by which the toner cartridge 70 projects from the rotary body 90 at the retracted position is large. Since the toner cartridge 70 is configured to be attachable to and detachable from the rotary body 90 via the tray 80, the toner cartridge 70 can be stably supported by the tray 80 even in the case where the length by which the toner cartridge 70 projects from the rotary body 90 is large.

The movement direction of the toner cartridge 70 in which the toner cartridge 70 moves from the attached position to the retracted position will be referred to as a retraction direction. In the present embodiment, the retraction direction of the toner cartridge 70 is a direction intersecting with the direction of the rotational axis 90C (Y direction). Therefore, as illustrated in FIGS. 7A and 7B, as viewed in the direction of the rotational axis 90C (Y direction), the retraction direction of the toner cartridge 70 is a direction orthogonal to the direction of the rotational axis 90C (Y direction). In addition, the retraction direction of the toner cartridge 70 can be referred to as a direction outward in the rotational radius direction of the rotary body 90, that is, a direction away from the rotational axis 90C.

As illustrated in FIGS. 7A and 7B, since the user detaches the toner cartridge 70 from the rotary body 90, it is preferable that the at least part of the toner cartridge 70 projects from the rotary body 90 at the time of detaching the toner cartridge 70. In the present embodiment, the entirety of the toner cartridge 70 projects from the rotary body 90 when the toner cartridge 70 is at the retracted position.

It can be said that when the rotary body 90 rotates about the rotational axis 90C, the rotational trajectory of the rotary body 90 matches a circumscribed circle of the rotary body 90 centered on the rotational axis 90C, that is, a virtual circle 90V indicated by a broken line in FIGS. 7A and 7B. When the toner cartridge 70 is at the retracted position, half the length or more of the toner cartridge 70 in the retracted direction is preferably outside the rotational trajectory of the rotary body 90. That is, as viewed in the rotational axis direction of the rotary, half the total length or more of the toner cartridge is preferably positioned outside the rotational trajectory of the rotary in the movement direction of the toner cartridge from the attached position to the retracted position. This applies to the toner cartridges 70 including the toner cartridge 70k serving as an example of the first cartridge and the toner cartridge 70m serving as an example of the second cartridge. In addition, in the present embodiment, the entirety of the toner cartridge 70 is outside the rotational trajectory (virtual circle 90V) of the rotary body 90 when the toner cartridge 70 is at the retracted position as illustrated in FIGS. 7A and 7B.

Further, to make it easier for the user to grab the toner cartridge 70, at least part of the toner cartridge 70 is preferably outside the image forming apparatus 1, that is, outside the apparatus body 1A when the toner cartridge 70 is at the retracted position. The outside of the apparatus mentioned herein refers to a space that is outside the image forming apparatus 1 (outside the apparatus body 1A) when the image forming apparatus 1 is used for, for example, an image forming operation on the sheet S.

In the present embodiment, the exterior surface of the apparatus body 1A is constituted by the exterior surface of the frame 16. That is, outside the apparatus can be also referred to as outside the frame 16. Therefore, a state in which at least part of the toner cartridge 70 is outside the apparatus can be also referred to as a state in which at least

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part of the toner cartridge 70 projects to the outside of the frame 16 through the opening 16a of the frame 16 of the apparatus body 1A.

In the present embodiment, when the door 14 is at the closed position, the opening 16a of the frame 16 of the apparatus body 1A is covered by the door 14. Further, the exterior surface 14a of the door 14 at the closed position constitutes part of the exterior surface of the apparatus body 1A. In this case, outside of the apparatus refers to the outside of the exterior surface 14a of the door 14 at the closed position. That is, in the case where the position of the exterior surface 14a of the door 14 at the closed position is referred to as an exterior position, at least part of the toner cartridge 70 is positioned more outward than the exterior position with respect to the apparatus body 1A when the toner cartridge 70 is at the retracted position.

In other words, at least part of the toner cartridge 70 is positioned in a space that would be outside the apparatus body 1A if the door 14 were at the closed position. Further, at least part of the toner cartridge 70 is positioned downstream of the exterior position in the retraction direction of the toner cartridge 70.

In addition, in the case where the side surface 16b having the opening 16a is the front surface of the apparatus body 1A, it can be said at least part of the toner cartridge 70 projects to the front side more than the exterior surface on the front side of the apparatus body 1A when the toner cartridge 70 is at the retracted position. In this case, the user can easily access the toner cartridge 70 from the front side of the image forming apparatus and replace the toner cartridge 70.

To be noted, when the toner cartridge 70 is at the retracted position, half the length or more of the toner cartridge 70 in the retraction direction is preferably outside the apparatus. That is, as viewed in the rotational axis direction of the rotary, half the total length or more of the toner cartridge is preferably positioned outside the body frame in the movement direction of the toner cartridge from the attached position to the retracted position in a state in which the toner cartridge is at the retracted position. This applies to the toner cartridges 70 including the toner cartridge 70k serving as an example of the first toner cartridge and the toner cartridge 70m serving as an example of the second toner cartridge. In addition, the entirety of the toner cartridge 70 is preferably outside the apparatus when the toner cartridge 70 is at the retracted position. To be noted, although the exterior surface 14a of the door 14 and the side surface 16b constitute the exterior surface on the front side of the apparatus body 1A in the present embodiment, the configuration of the door 14 is not limited to this. For example, the size of the door 14 may be set so as to cover the entirety of the side surface 16b. In this case, the exterior surface 14a of the door 14 constitutes the exterior surface on the front side of the apparatus body 1A.

The tray 80 includes a cartridge holding portion 81 illustrated in FIGS. 3 and 6C that holds the toner cartridge 70. The cartridge holding portion 81 is an attached portion to which the toner cartridge 70 is attached. When the tray 80 is at the detachment position, the entirety of the cartridge holding portion 81 is preferably outside the rotational trajectory of the rotary body 90 in the retraction direction. When the tray 80 is at the detachment position, half the length or more of the cartridge holding portion 81 is preferably outside the apparatus in the retraction direction.

Here, as described above, the toner cartridge 70k and the tray 80k are larger in size than the other toner cartridges 70y to 70c and the other trays 80y to 80c. Therefore, as illustrated

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in FIGS. 7A and 7B, in the present embodiment, the movement amount of the tray 80 in toner cartridge replacement is changed in accordance with the size of the toner cartridge 70.

Specifically, as illustrated in FIG. 7A, the movement distance of the tray 80k (first support member) from the accommodation position (first accommodation position) to the detachment position (first detachment position) is L1. The movement distance of the tray 80m (second support member) from the accommodation position to the detachment position (third detachment position) is L2. Although a state in which the toner cartridge 70m and the tray 80m are moved is illustrated in FIG. 7B, the movement distance of the trays 80y and 80c from the accommodation position to the detachment position is also L2. In this case, L1 is larger than L2. In other words, it can be said that the movement distance of the first support member in the case where the first toner cartridge moves from the first attached position to the first retracted position is larger than the movement distance of the second support member in the case where the second toner cartridge moves from the second attached position to the second retracted position.

In addition, as illustrated in FIG. 7A, in a state in which the tray 80k is at the detachment position and the toner cartridge 70k is at the retracted position, the toner cartridge 70k projects from the exterior surface of the apparatus body 1A to the outside of the apparatus by a distance P1. In the present embodiment, the tray 80k also projects from the exterior surface of the apparatus body 1A to the outside of the apparatus by the distance P1.

In addition, as illustrated in FIG. 7B, in a state in which the tray 80m is at the detachment position and the toner cartridge 70m is at the retracted position, the toner cartridge 70m projects from the exterior surface of the apparatus body 1A to the outside of the apparatus by a distance P2. In the present embodiment, the tray 80m also projects from the exterior surface of the apparatus body 1A to the outside of the apparatus by the distance P2. To be noted, the toner cartridges 70y and 70c also project from the exterior surface of the apparatus body 1A to the outside of the apparatus by the distance P2.

The distance P1 described above is larger than the distance P2. That is, the length by which the first toner cartridge at the first retracted position projects through the opening 16a of the apparatus body 1A will be referred to as a first length (P1), and the length by which the second toner cartridge at the second retracted position projects through the opening 16a will be referred to as a second length (P2). In this case, it can be said that the first length is larger than the second length.

It is more preferable in terms of strength that the distance P2 by which the toner cartridges 70y to 70c smaller in size than the toner cartridge 70k project to the outside of the apparatus at the retracted position is smaller than the distance P1 by which the toner cartridge 70k projects to the outside at the retracted position. The reason for this is as follows. When the toner cartridge 70 is at the retracted position, at least part of the toner cartridge 70 projects to the outside of the apparatus from the outside of the rotational trajectory of the rotary body 90 or the exterior surface of the apparatus body 1A. At this time, the tray 80 supports the weight of the toner cartridge 70 in a state in which one side thereof is supported by the rotary body 90. Therefore, reducing the distance P2 by which the toner cartridges 70y to 70c project to the outside of the apparatus at the retracted position can reduce the load on the trays 80y to 80c or guide portions 97 of the rotary body 90 supporting the trays 80y to

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80k. In addition, since the toner cartridges 70y to 70c are smaller than the toner cartridge 70k, the operability in the cartridge replacement for the trays 80y to 80c can be maintained even if the distance P2 is set to be smaller than the distance P1.

Tray Arrangement in Rotary

The arrangement of the trays 80y to 80k in the rotary body 90 will be described with reference to FIGS. 8, 9, and 10. FIG. 8 is a perspective view illustrating the arrangement of the trays 80y to 80k in the rotary body 90. FIG. 9 is a section view illustrating the arrangement of the trays 80y to 80k in the rotary body 90. FIG. 10 is a diagram illustrating the member arrangement of the trays 80y to 80k on one end side in the Y direction. To be noted, FIG. 9 illustrates a cross-section of the rotary body 90 taken along a virtual plane orthogonal to the rotational axis 90C of the rotary body 90. In addition, the top half of FIG. 10 is a diagram of the rotary body 90 and the trays 80m and 80k of FIG. 8 as viewed from the upper-right side (+Z side) of FIG. 8, and the lower half of FIG. 10 is a diagram of the rotary body 90 and the trays 80c and 80y of FIG. 8 as viewed from the left side (-X) side of FIG. 8.

As illustrated in FIG. 8, the trays 80y to 80k are respectively provided with cartridge holding portions 81y to 81k and guided portions 82y to 82k.

The toner cartridges 70y to 70k are respectively attached to the cartridge holding portions 81y to 81k. The cartridge holding portions 81y to 81k respectively accommodate at least part of the toner cartridges 70y to 70k attached thereto.

The guided portions 82y to 82k are provided at respective end portions of the trays 80y to 80k such that the cartridge holding portions 81y to 81k are each interposed therebetween in the Y direction. The guided portions 82y to 82k are each a thin and long member extending in a direction orthogonal to the rotational axis of the rotary body 90.

In the present embodiment, a reinforcing rib 82k1 is formed on part of the guided portion 82k in a movement direction Dk of the tray 80k, and a reinforcing rib 82m1 is formed on part of the guided portion 82m in a movement direction Dm of the tray 80m as illustrated in FIGS. 11A and 11B. The reinforcing ribs 82k1 and 82m1 are thin and long rib shapes (ridges) projecting outward in the Y direction from the guided portions 82k and 82m of the respective end portions of the trays 80k and 80m in the Y direction and extending in the movement directions Dk and Dm of the trays 80k and 80m. The reinforcing ribs 82k1 and 82m1 improve the stiffness of the guided portions 82k and 82m.

To be noted, although the lengths of the reinforcing ribs 82m1 and 82k1 in the present embodiment are restricted so as to avoid the guided portions 82y and 82c, the reinforcing ribs 82m1 and 82k1 may be provided for the entire lengths of the guided portions 82m and 82k unless interference with the guided portions 82y and 82c occurs. Reinforcing ribs may be added to the guided portions 82y and 82c. In addition, in the case where the stiffness of the guided portions 82m and 82k is sufficient, a configuration in which the reinforcing ribs 82m1 and 82k1 are not provided may be employed.

The guided portions 82y to 82k have rack portions (rack gears) 83y to 83k formed therein. In addition, pinion gears 94y to 94k are rotatably held in the rotary body 90. The pinion gears 94y to 94k are respectively engaged with the rack portions 83y to 83k in a manner capable of transmitting a driving force.

The rack portions 83y to 83k and the pinion gears 94y to 94k are part of the moving devices 85y to 85k configured to move the toner cartridges 70y to 70k from the attached

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position to the retracted position. In addition, it can be said that the rack portions 83y to 83k and the pinion gears 94y to 94k are part of a driven device driven by the driving device 98 of the apparatus body 1A. It can be said that the pinion gears 94y to 94k are rotatable bodies (rotary members) that rotate to move the trays 80y to 80k with respect to the rotary body 90.

The pinion gears 94y to 94k and the rack portions 83y to 83k function as driven portions for the moving devices 85y to 85k of the rotary body 90 to receive the driving force from the driving device 98 of the apparatus body 1A. The pinion gear 94k and the rack portion 83k are examples of a first pinion gear and a first rack gear constituting at least part of a first driven portion included in a first moving device. The pinion gear 94m and the rack portion 83m are examples of a second pinion gear and a second rack gear constituting at least part of a second driven portion included in a second moving device.

The rotary body 90 includes guide portions 97 illustrated in FIGS. 7A and 7B that respectively engage with the guided portions 82y to 82k. FIG. 7A illustrates the guide portion 97 (97k) that engages with the guided portion 82k of the tray 80k, and FIG. 7B illustrates the guide portion 97 (97m) that engages with the guided portion 82m of the tray 80m. The rotary body 90 includes similar guide portions that respectively engage with the guided portions 82y and 82c of the trays 80y and 80c. In addition, although the guide portions 97 provided on one side (+Y side) of the rotary body 90 in the Y direction are illustrated in FIGS. 7A and 7B, similar guide portions 97 are also provided on the other side (-Y side) of the rotary body 90 in the Y direction.

When the tray 80 moves between the accommodation position and the detachment position, the guide portion 97 maintains the state of engaging with the guided portion 82 in at least part of the movement range, and guides the movement direction of the tray 80. In the present embodiment, the guide portion 97 maintains the state of engaging with the guided portion 82k in the entirety of the movement range of the tray 80k between the accommodation position and the detachment position. In addition, in the present embodiment, the guide portion 97 maintains the state of engaging with the guided portion 82m in the entirety of the movement range of the tray 80m between the accommodation position and the detachment position.

In addition, as illustrated in FIGS. 8 and 9, the four trays 80y to 80k are disposed in the rotary body 90 so as to overlap each other as will be described in detail below.

When the pinion gears 94y to 94k rotate, the rack portions 83y to 83k and the trays 80y to 80k move with respect to the rotary body 90. As illustrated in FIG. 9, the four trays 80y to 80k are arranged such that the movement directions thereof with respect to the rotary body 90 are rotated by 90° from each other. Therefore, the tray 80y and the tray 80c are held so as to be slidable in substantially the same direction (parallel directions) as each other, and the tray 80m and the tray 80k are held so as to be slidable in substantially the same direction (parallel directions) as each other. The movement directions of sliding of the trays 80y to 80k are regulated by the engagement between the guide portions 97 and the guided portions 82y to 82k described above.

To be noted, the trays 80y to 80k move to the outside of the apparatus through the opening 16a. When the trays 80y to 80k each move to the outside of the apparatus through the opening 16a, the movement directions of the trays are substantially the same (parallel).

As illustrated in FIG. 9, the trays are disposed such that with respect to the movement direction Dk of the tray 80k,

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the range in which the tray **80k** is disposed overlaps with the range in which the tray **80y** is disposed and the range in which the tray **80c**. In addition, with respect to the movement direction **Dk** of the tray **80k**, the range in which the tray **80k** is disposed overlaps with the rotational axis **90C** of the rotary body **90**. That is, it can be said that the toner cartridge **70k** held by the cartridge holding portion **81k** of the tray **80k** overlaps with the rotational axis **90C** of the rotary body **90** illustrated in FIG. 4B.

In contrast, the trays are disposed in a displaced manner such that with respect to the movement direction **Dm** of the tray **80m**, the range in which the tray **80m** is disposed does not overlap with the range in which the tray **80y** is disposed and the range in which the tray **80c** is disposed. Further, the trays are disposed in a displaced manner such that with respect to a movement direction **Dy** of the tray **80y**, the range in which the tray **80y** is disposed does not overlap with the range in which the tray **80m** is disposed and the range in which the tray **80k** is disposed. Similarly, the trays are disposed in a displaced manner such that with respect to a movement direction **Dc** of the tray **80c**, the range in which the tray **80c** is disposed does not overlap with the range in which the tray **80m** is disposed and the range in which the tray **80k** is disposed.

The positional relationship between the trays **80** can be also expressed as follows. As viewed in the movement direction **Dy** of the tray **80y**, the tray **80y** and the tray **80k** overlap with each other, but the tray **80y** and the tray **80m** do not overlap with each other. As viewed in the movement direction **Dm** of the tray **80m**, the tray **80m** and the tray **80k** overlap with each other, but the tray **80m** and the trays **80y** and **80c** do not overlap with each other. As viewed in the movement direction **Dc** of the tray **80c**, the tray **80c** and the tray **80k** overlap with each other, but the tray **80c** and the tray **80m** do not overlap with each other.

Here, two elements (such as members, parts, and units) overlapping as viewed in a specific direction refers to a projected region of one element and a projected region of the other element at least partially overlapping with each other in the case where the elements are orthogonally projected onto a virtual plane orthogonal to the specific direction.

As illustrated in FIGS. 8 and 10, with respect to the direction of the rotational axis **90C** (Y direction), a range in which the rack portion **83m** and the guided portion **82m** are disposed and a range in which the rack portion **83k** and the guided portion **82k** are disposed overlap with each other at least partially. That is, in the present embodiment, it can be said that with respect to the rotational axis direction of the rotary (Y direction), a range in which the first rack gear (rack portion **83k**) is disposed and a range in which the second rack gear (rack portion **83m**) is disposed overlap with each other at least partially. Therefore, as compared with a layout in which the rack portion **83m** and the guided portion **82m** do not overlap with the rack portion **83k** and the guided portion **82k**, the rack portions **83m** and **83k** and the guided portions **82m** and **82k** can be arranged in a smaller space in the Y direction.

With respect to the direction of the rotational axis **90C** (Y direction), a range in which the rack portion **83y** and the guided portion **82y** are disposed and a range in which the rack portion **83c** and the guided portion **82c** are disposed overlap with each other at least partially. That is, in the present embodiment, it can be said that with respect to the rotational axis direction of the rotary (Y direction), a range in which the third rack gear (rack portion **83y**) is disposed and a range in which the fourth rack gear (rack portion **83c**) is disposed overlap with each other at least partially. There-

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fore, as compared with a layout in which the rack portion **83y** and the guided portion **82y** do not overlap with the rack portion **83c** and the guided portion **82c**, the rack portions **83y** and **83c** and the guided portions **82y** and **82c** can be arranged in a smaller space in the Y direction.

Here, an engagement position between the rack portion **83** and the pinion gear **94** will be described with reference to FIG. 10. A top half portion of FIG. 10 illustrates an engagement position between the rack portion **83k** and the pinion gear **94k**. A lower half portion of FIG. 10 illustrates an engagement position between the rack portion **83y** and the pinion gear **94y**.

In a region **Y1** in the drawings in the direction of the rotational axis **90C** of the rotary body **90** (Y direction), the driving force transmitted from the motor **M2** illustrated in FIG. 2 serving as a drive source to a transmission device that will be described below is transmitted to the pinion gears **94y** to **94k**. In a region **Y2** in the drawings in the Y direction, the pinion gear **94k** is engaged with the rack portion **83k** in a manner capable of transmitting the driving force. In a region **Y3** in the drawings in the Y direction, the pinion gear **94y** is engaged with the rack portion **83y** in a manner capable of transmitting the driving force. To be noted, the rack portion **83m** is engaged with the pinion gear **94m** illustrated in FIG. 8 in the region **Y2** similarly to the rack portion **83k** in a manner capable of transmitting the driving force. The rack portion **83c** is engaged with the pinion gear **94c** illustrated in FIG. 8 in the region **Y3** similarly to the rack portion **83y** in a manner capable of transmitting the driving force.

Here, the regions **Y2** and **Y3** are at different positions in the Y direction, that is, are displaced from each other in the Y direction. In addition, the region **Y1** is at a position different from both the regions **Y2** and **Y3** in the Y direction. That is, the region **Y1** is displaced from the regions **Y2** and **Y3** in the Y direction.

Further, in a state in which the toner cartridges **70y** and **70c** are at the attached position, with respect to a movement direction of the rack portion **83y** (movement direction **Dy** of the tray **80y**), a range in which the rack portion **83y** is disposed and a range in which the rack portion **83c** is disposed overlap with each other at least partially. In the present embodiment, since the movement directions **Dy** and **Dc** of the trays **80y** and **80c** are substantially the same (parallel), also with respect to the movement direction **Dc** of the tray **80c**, a range in which the rack portion **83y** is disposed and a range in which the rack portion **83c** is disposed overlap with each other at least partially. Therefore, in a state in which the toner cartridges **70y** and **70c** are at the attached position, the tooth surface of the rack portion **83y** faces the tooth surface of the rack portion **83c** in a direction (left-right direction of FIG. 8) orthogonal to the movement directions **Dy** and **Dc** of the rack portions **83y** and **83c**.

Further, in a state in which the toner cartridges **70m** and **70k** are at the attached position, with respect to a movement direction of the rack portion **83m** (movement direction **Dm** of the tray **80m**), a range in which the rack portion **83m** is disposed and a range in which the rack portion **83k** is disposed overlap with each other at least partially. In the present embodiment, since the movement directions **Dm** and **Dk** of the trays **80m** and **80k** are substantially the same (parallel), also with respect to the movement direction **Dk** of the tray **80k**, a range in which the rack portion **83m** is disposed and a range in which the rack portion **83k** is disposed overlap with each other at least partially. Therefore, in a state in which the toner cartridges **70m** and **70k** are at the attached position, the tooth surface of the rack portion

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83m faces the tooth surface of the rack portion **83k** in a direction (up-down direction of FIG. 8) orthogonal to the movement directions **Dm** and **Dk** of the rack portions **83m** and **83k**.

In addition, as also illustrated in FIG. 12A that will be described later, the rack portion **83y** overlaps with the rack portions **83m** and **83k** as viewed in the direction (Y direction) of the rotational axis **90C**. As viewed in the direction (Y direction) of the rotational axis **90C**, the rack portion **83m** overlaps with the rack portions **83y** and **83c**. As viewed in the direction (Y direction) of the rotational axis **90C**, the rack portions **83c** and **83m** overlap with the rack portion **83k**. As viewed in the direction of the rotational axis **90C** (Y direction), the rack portions **83k** and **83y** overlap with the rack portion **83c**. In other words, it can be said that with respect to the rotational axis direction of the rotary (Y direction), a range in which the first rack gear (rack portion **83k**) is disposed does not overlap with a range in which the second rack gear (rack portion **83y**) is disposed. In addition, it can be said that, as viewed in the rotational axis direction of the rotary (Y direction), in a state in which the first toner cartridge **70k** is at the first attached position and the second toner cartridge **70y** is at the second attached position, the first rack gear (rack portion **83k**) and the second rack gear (rack portion **83y**) overlap with each other.

As described above, since the positions where the rack portions **83k** and **83m** are disposed and the positions where the rack portions **83y** and **83c** are disposed are different in the Y direction, the rack portions can be arranged such that the rack portions **83y** and **83c** overlap with the rack portions **83m** and **83k** as viewed in the Y direction.

As a result of this, the space for arranging the four trays in the rotary body **90** can be reduced, and thus the size of the rotary body **90** in the rotational radius direction can be reduced. That is, when it is attempted to arrange the rack portions **83** such that the movement distances of the trays **80y** to **80k** are equivalent to those of the present embodiment and the rack portions **83** do not overlap with each other as viewed in the Y direction, the area required for the arrangement of the four rack portions is large as viewed in the Y direction. As compared with such a configuration, as a result of arranging the plurality of rack portions **83** in a displaced manner in the Y direction such that the rack portions **83** overlap with each other as viewed in the Y direction, the installation area for the rack portions **83** as viewed in the Y direction can be reduced.

In addition, in the present embodiment, two pairs of two of the four rack portions **83y** to **83k** are arranged in a displaced manner in the Y direction. That is, it can be said that with respect to in the rotational axis direction of the rotary (Y direction), the range in which the first rack gear is disposed and the range in which the second rack gear is disposed overlap with each other, and the range in which the third rack gear is disposed and the range in which the fourth rack gear is disposed overlap with each other. In addition, it can be said that with respect to the Y direction, the range in which the first rack gear and the second rack gear are disposed does not overlap with the range in which the third rack gear and the fourth rack gear are disposed. As a result of this, the size of the rotary body **90** in the Y direction can be reduced as compared with a case where all the four rack portions **83y** to **83k** are arranged in a displaced manner in the Y direction.

Moving Elements of Tray

Elements related to the movement of the trays **80y** to **80k** disposed in the rotary body **90** will be described with reference to FIGS. 11A, 11B, 12A, and 12B. FIGS. 11A and

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11B are each a perspective view of elements related to the movement of the tray **80k**. FIGS. 12A and 12B are each a section view of elements related to the movement of the tray **80k**.

In the present embodiment, the trays **80y** to **80k** are respectively driven by receiving the driving force of the motor **M2** transmitted to the pinion gears **94y** to **94k** via the driving racks **15L** and **15R** serving as transmission devices. Here, the elements for moving the tray **80k** with respect to the rotary body **90** will be described, and since the elements for moving the trays **80y** to **80c** with respect to the rotary body **90** are substantially the same as the elements for moving the tray **80k**, description thereof will be omitted.

FIG. 11A illustrates a state in which the tray **80k** is inside the rotary body **90**, that is, in a state in which the toner cartridge **70k** is attached to the developing unit **50k**. That is, FIG. 11A illustrates a state in which the tray **80k** is at the accommodation position, and corresponds to a state in which the toner cartridge **70k** is at the attached position with respect to the developing frame **53k** illustrated in FIG. 4A. FIG. 11B illustrates a state in which the tray **80k** has slid to the outside of the rotary body **90**. That is, FIG. 11B illustrates a state in which the tray **80k** is at the attached position, and corresponds to a state in which the toner cartridge **70k** is at the retracted position with respect to the developing frame **53k** illustrated in FIG. 4B.

The apparatus body **1A** of the present embodiment includes the driving racks **15L** and **15R** serving as driving gears that drive the pinion gears **94**. The driving racks **15** are each driven by the motor **M2** via the unillustrated drive transmission mechanism.

As described above, two rack portions **83k** are formed at respective end portions of the tray **80k** in the Y direction. Two pinion gears **94k** and the driving racks **15L** and **15R** are disposed at positions respectively corresponding to the rack portions **83k** at the respective end portions. That is, the apparatus body **1A** of the present embodiment includes the driving racks **15L** and **15R** serving as the first driving gear and the second driving gear. It can be said that the driving rack **15L** is an example of a first driving gear, and the driving rack **15R** is an example of a second driving gear. However, this numbering is merely used for the sake of convenience of description, and can be appropriately interchanged in principle. In the case where there is no need to distinguish the driving racks **15L** and **15R** from each other, the driving racks will be each described as a "driving rack **15**".

The rack portions **83** of the present embodiment are configured as a pair of rack gears, and the pinion gears **94** of the present embodiment are configured as a pair of pinion gears. The pair of rack gears and the pair of pinion gears are disposed on the two end sides (i.e., one end side and the other end side) of the support member (tray **80**) in the Y direction in the present embodiment, but the pair of rack gears and the pair of pinion gears may be disposed at other positions. The rack portions **83k** and the pinion gears **94k** of the moving device **85k** corresponding to the tray **80k** can be respectively referred to as examples of a first pair of rack gears and a first pair of pinion gears. The rack portions **83y** to **83c** and the pinion gears **94y** to **94c** of the moving devices **85y** to **85c** respectively corresponding to the other trays **80y** to **80c** can be respectively referred to as examples of second pairs of rack gears and second pairs of pinion gear.

One of the pair of rack gears engages with one of the pair of pinion gears, and the other of the pair of rack gears engages with the other of the pair of pinion gears. At least one of the pair of pinion gears is driven by the driving rack **15L** serving as a first driving rack. In the present embodi-

ment, two of the pair of pinion gears are simultaneously driven by the driving racks **15L** and **15R** serving as a first driving rack and a second driving rack. As a result of this, rotation of the tray **80** is less likely to occur, and thus stable movement of the toner cartridge **70** is made possible.

To be noted, a configuration in which the tray **80** includes one rack portion **83** and is moved by one driving rack **15** via one pinion gear **94** may be employed.

The tray **80k** is held to be slidable in a direction parallel to the guided portion **82k** (that is, the movement direction **Dk**) with respect to the rotary body **90**. The driving rack **15** is held to be slidable in a direction intersecting with the movement direction **Dk** of the tray **80k** with respect to the apparatus body **1A**. The driving rack **15** is configured to slide (reciprocate) in a first direction (vertically upward direction in the present embodiment) and a second direction (vertically downward direction in the present embodiment) opposite to the first direction, with respect to the apparatus body **1A**. That is, the movement direction of the driving rack **15** of the present embodiment is a direction intersecting with (preferably orthogonal to) both the movement direction **Dk** of the tray **80k** and the direction of the rotational axis **90C** of the rotary body **90** (**Y** direction).

A tray moving operation of sliding the tray **80k** between the accommodation position and the detachment position will be described with reference to FIGS. **11A** and **11B**. The tray moving operation of the tray **80k** is performed by the motor **M2** illustrated in FIG. **2**, an unillustrated drive transmission mechanism, the driving rack **15**, the pinion gear **94k**, and the rack portion **83k**.

First, a tray moving operation (tray pull-out operation) of detaching the toner cartridge **70k** from the rotary body **90** will be described. In a state before the tray pull-out operation is started, the driving rack **15** illustrated in FIG. **11A** is positioned below a position where the driving rack **15** engages with the pinion gear **94k**. In addition, as described above, in the replacement operation of the toner cartridge **70k**, the rotary body **90** takes the replacement posture illustrated in FIG. **4B** for the toner cartridge **70k**.

When the tray pull-out operation is started, the driving racks **15** are slid upward with respect to the apparatus body **1A** by the driving force of the motor **M2**. In the course of movement of the driving racks **15**, the driving racks **15** engage with the pinion gears **94k**, and the pinion gears **94k** are rotationally driven.

As illustrated in FIG. **11B**, the pinion gears **94k** are rotationally driven in an arrow direction in FIG. **11B**, and thus the driving force is input to the rack portions **83k** engaged with the pinion gears **94k**. As a result of this, the tray **80k** is pushed to the outside of the apparatus, and moves from the accommodation position to the detachment position with respect to the rotary body **90**. The movement direction of the tray **80k** in this movement is guided to the predetermined movement direction **Dk** by the engagement between the guided portion **82k** and the guide portion **97k** of the rotary body **90** illustrated in FIG. **7A**. As a result of the tray **80k** moving from the accommodation position to the detachment position, the toner cartridge **70k** is moved from the attached position to the retracted position with respect to the developing unit **50k**.

In a state in which the tray **80k** is positioned at the detachment position and the toner cartridge **70k** is positioned at the retracted position, the user can attach and detach the toner cartridge **70k** to and from the tray **80k**.

The tray moving operation (tray pull-in operation or tray insertion operation) at the time of attaching the toner cartridge **70** to the rotary body **90** is performed in a process

reversed from the tray pull-out operation. In the state before the tray pull-in operation is started, the driving racks **15** are positioned above positions where the driving racks **15** engage with the pinion gears **94k**. For example, the tray pull-in operation is started in response to the user operating a predetermined operation portion. When the tray pull-in operation is started, the driving racks **15** are slid downward with respect to the apparatus body **1A** by the driving force of the motor **M2**. Here, the rotational direction of the motor **M2** in the tray pull-in operation is opposite to that in the tray pull-out operation. In the course of movement of the driving racks **15**, the driving racks **15** engage with the pinion gears **94k**, and thus the pinion gears **94k** are rotationally driven.

As a result of the pinion gears **94k** being rotationally driven in a direction opposite to the arrow of FIG. **11B**, the driving force is input to the rack portions **83k** engaged with the pinion gears **94k**. As a result of this, the tray **80k** is pulled into the apparatus, and moves from the detachment position to the accommodation position with respect to the rotary body **90**. The movement direction of the tray **80k** is guided to the movement direction **Dk** opposite to the arrow of FIG. **11B** by the engagement between the guided portion **82k** and the guide portion **97k** of the rotary body **90**. As a result of the tray **80k** moving from the detachment position to the accommodation position, the toner cartridge **70k** is moved from the retracted position to the attached position with respect to the developing unit **50k**.

Movement of the tray **80k** and the toner cartridge **70k** for black has been described above, and movement of the other trays **80y** to **80c** and toner cartridges **70y** to **70c** is also performed by a similar mechanism. That is, the driving racks **15** transmit drive to the pinion gears **94y** to **94c** in respective replacement postures of the toner cartridges.

The motor **M2** provided in the apparatus body **1A** and the transmission device including the driving racks **15** (**15L** and **15R**) and the drive transmission mechanism constitute the driving device **98** for driving the moving devices **85** provided in the rotary body **90**.

As described above, in the present embodiment, a plurality of moving devices **85k** to **85y** corresponding to the plurality of toner cartridges **70k** to **70y** are disposed in the rotary body **90**. The driving device **98** of the apparatus body **1A** is a common driving device that drives the plurality of moving devices **85k** to **85y** (plurality of driven devices) of the rotary body **90**.

In addition, in the present embodiment, the driving target of the driving device **98** switches in accordance with the rotation of the rotary body **90**. In other words, the driving device of the present embodiment includes the driving racks **15** each serving as a transmission member that transmits the driving force of the drive source. The driving device can take a state in which the transmission member engages with a first driven portion (e.g., pinion gear **94k**) in a manner capable of transmitting the driving force and a state in which the transmission member engages with a second driven portion (e.g., pinion gear **94m**) in a manner capable of transmitting the driving force. In addition, the driving device can take a state in which the transmission member is disengaged from the first driven portion and the second driven portion.

As described above, the pinion gears **94y** to **94k** are held by the rotary body **90**. Therefore, when the rotary body **90** rotates, it is preferable that the engagement between the driving racks **15** and the pinion gears **94y** to **94k** is released.

FIG. **12A** illustrates a state in which the tray **80k** is inside the rotary body **90**, that is, a state in which the tray **80k** is at the accommodation position. FIG. **12B** illustrates a state in

which the tray **80k** has moved to the outside of the rotary body **90**, that is, a state in which the tray **80k** has moved to the detachment position.

As illustrated in FIG. 12A, when the tray **80k** is inside the rotary body **90**, the driving racks **15** are positioned in a lower portion in the apparatus body **1A**. At this time, the driving racks **15** are retracted from the pinion gears **94k**. Therefore, the rotary body **90** can be rotated without being interrupted by the driving racks **15**. More specifically, the driving racks **15** can be retracted to the outside of the rotational trajectory of the rotary body **90** indicated by dotted lines in FIGS. 12A and 12B.

As described above, by rotationally driving the motor **M2** in a normal direction and a reverse direction, the tray **80** attached to the rotary body **90** can be moved from the accommodation position to the detachment position and from the detachment position to the accommodation position with respect to the rotary body **90**. That is, the driving device of the present embodiment can drive the moving devices such that not only the toner cartridges move from the attached position to the retracted position but also the toner cartridges move from the retracted position to the attached position.

Here, as described above, in the present embodiment, the movement amount of the tray **80** in toner cartridge replacement is changed in accordance with the size of the toner cartridge **70**. Specifically, the movement distance **L1** of the movement of the black tray **80k** from the accommodation position to the detachment position is larger than the movement distance **L2** of the movement of the other trays **80y** to **80c** from the accommodation position to the detachment position as illustrated in FIGS. 7A and 7B.

Therefore, in the present embodiment, when moving the toner cartridges **70y** to **70k** from the attached position to the retracted position, a value obtained by dividing the speed of the rack portions **83k** by the speed of the driving racks **15** is larger than a value obtained by dividing the speed of the rack portions **83y** to **83c** by the speed of the driving racks **15**.

For example, as illustrated in FIG. 10, the pinion gear **94y** is formed as a stepped gear including a large-diameter gear **941** that engages with the driving rack **15** and a small-diameter gear **942** that has a smaller pitch radius than the large-diameter gear **941** and engages with the rack portion **83y**. The pinion gears **94m** and **94c** are formed as similar stepped gears. In contrast, the pinion gear **94k** is formed such that a portion thereof that engages with the driving rack **15** and a portion thereof that engages with the rack portion **83k** have the same pitch radius. In this case, the pitch radius of the pinion gear **94k** can be equal to the pitch radius of the large-diameter gear **941** of the pinion gears **94y** to **94c**. According to this configuration, the movement distance of the rack portions **83k** can be made larger than the movement distance of the other rack portions **83y** to **83c** even in the case where the movement distance of the driving racks **15** is the same. That is, the movement distance **L1** of the movement of the black tray **80k** from the accommodation position to the detachment position can be made larger than the movement distance **L2** of the movement of the other trays **80y** to **80c** from the accommodation position to the detachment position.

In addition, by forming the pinion gears **94y** to **94c** as stepped gears, the movement distance **L1** of the tray **80k** can be made larger than the movement distance **L2** of the other trays **80y** to **80c** even in the configuration in which the pinion gears **94y** to **94k** receive the driving force from the same driving racks **15**.

To be noted, the pinion gear **94k** may be formed as a stepped gear instead of (or in combination with) the configuration in which the pinion gears **94y** to **94c** are formed as stepped gears. In this case, a portion of the pinion gear **94k** that engages with the driving rack **15** may be formed as a small-diameter gear, and a portion of the pinion gear **94k** that engages with the rack portion **83k** may be formed as a large-diameter gear having a larger pitch radius than the small-diameter gear. In addition, the stepped gear is an example of a speed reduction mechanism, and may be replaced by a known speed reduction mechanism that makes the movement amount of a member on the input side (drive source side) smaller than a movement amount of a member on the output side (tray **80** side).

In addition, the movement amount of the driving rack **15** when the toner cartridge **70k** moves from the attached position to the retracted position may be made larger than the movement amount of the driving rack **15** when the toner cartridges **70y** to **70c** move from the attached position to the retracted position.

Incidentally, in the case where the distance of the movement of the toner cartridge **70** from the attached position to the retracted position is smaller, the movement time of the toner cartridge **70** can be made shorter, and the time in which the user waits for the toner cartridge **70** to move can be made shorter. If a configuration in which the movement amount of the driving racks **15** for the toner cartridge **70k** is larger than the movement amount of the driving racks **15** for the toner cartridges **70y** to **70c** is employed as described above, the time in which the user waits for the toner cartridges **70y** to **70c** to move can be made shorter.

According to the configurations described above, the movement distance **L1** can be made larger than the movement distance **L2**. These configurations may be employed in combination.

Modification Example

Although a configuration in which the driven portion includes the pinion gears **94** that engage with both the driving racks **15** and the rack portions **83** has been described, the driven portion may include a gear that engages with the driving rack **15** and a gear that engages with the rack portion **83**.

In addition, the configuration of the moving device **85** that move the tray **80** is not limited to a so-called rack-and-pinion configuration. For example, the member corresponding to the pinion gear **94** may be replaced by a roller that receives the drive of the motor **M2** to rotate, and the tray **80** may be moved by friction between the roller and the tray **80**.

In addition, in the case of using the roller that receives the drive of the motor **M2** to rotate, the roller may be brought into contact with the toner cartridge **70**. In this case, the toner cartridges **70y** to **70k** can be made directly attachable to and detachable from the rotary body **90** without using the trays **80y** to **80k**. In this case, the moving device **85** is constituted by a roller.

A moving device **85'** serving as a modification example will be described with reference to FIGS. 20A and 20B. FIGS. 20A and 20B are diagrams illustrating the moving device **85'** according to the present modification example. The moving device **85'** includes a rotary member **494a** that receives the driving force of the motor **M2** to rotate.

In the present modification example, the direction of the rotational axis of the rotary member **494a** is parallel to the direction of the rotational axis **90C** of the rotary body **90**. The rotary member **494a** abuts the toner cartridge **70** and

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rotates, and thus the toner cartridge 70 can reciprocate between the attached position indicated by a solid line in FIG. 20B and the retracted position indicated by a dot line in FIG. 20B.

To be noted, the rotary member 494a may be a roller that moves the toner cartridge 70 by friction by abutting the toner cartridge 70 and rotating. In addition, the toner cartridge 70 may be moved by a configuration in which the rotary member 494a is a gear engaging with a gear shape (rack shape) formed on the toner cartridge 70.

The moving device 85' may include a plurality of rotary members 494a. The plurality of rotary members 494a may be arranged arbitrarily. For example, as illustrated in FIG. 20A, the moving device 85' may include a rotary member 494a that abuts one end portion of the toner cartridge 70 and a rotary member 494a that abuts the other end portion of the toner cartridge 70 in the longitudinal direction of the toner cartridge 70 parallel to the rotational axis 90C. In addition, the moving device 85' may include a rotary member 494a that abuts the center of the toner cartridge 70.

In addition, the moving device 85' may include only one rotary member 494a. In this case, the moving device 85' may be disposed at an arbitrary position. For example, the moving device 85' may include a rotary member 494a that abuts the center of the toner cartridge 70.

Further, the rotary member 494a may be urged toward the toner cartridge 70. In addition, the moving device 85' may include a driven roller 494b as illustrated in FIG. 20B. The toner cartridge 70 is interposed between the rotary member 494a and the driven roller 494b. To be noted, with respect to the longitudinal direction of the toner cartridge 70, the position of the rotary member 494a and the position of the driven roller 494b may overlap with each other, or may be different from each other. In addition, at least one of the rotary member 494a and the driven roller 494b may be urged toward the toner cartridge 70.

In addition, the rotary member 494a and the driven roller 494b may be provided in the rotary body 90.

In the first embodiment, as an embodiment in which the door 14 moves from the closed position to the open position in an interconnected manner with the movement of the toner cartridge 70 from the attached position to the retracted position, a configuration in which the tray 80 pushes the door 14 has been described. Instead of this configuration, for example, a configuration in which part of the toner cartridge 70 pushes the door 14 when the toner cartridge 70 moves from the attached position to the retracted position, and thus the door 14 moves from the closed position to the open position, may be employed. In addition, the door 14 may be moved by transmitting a driving force from a member of the moving device 85 different from the tray 80 (support member) or part of the driving device 98. For example, an input gear may be attached to the door 14, and rotation may be transmitted from the pinion gear 94 of the moving device 85 illustrated in FIG. 11A or the driving rack 15 of the driving device 98 to this input gear via a gear train.

Further, the embodiment in which the door 14 moves from the closed position to the open position in an interconnected manner with the movement of the toner cartridge 70 from the attached position to the retracted position is not limited to mechanical interconnection, and the movement of the toner cartridge 70 may be interconnected with the movement of the door 14 by electrical control. For example, the door 14 is configured to be driven by a drive source (for example, a solenoid) different from the driving device 98 for moving the toner cartridge 70. Further, control of the driving device 98 to move the toner cartridge 70 from the attached position

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to the retracted position (control of the motor M2) and control of the other drive source described above to move the door 14 from the closed position to the open position are executed in the same sequence process.

The sequence process is a series of processing performed by the controller described above to enable replacement of the toner cartridge 70 by the user and then restoring the image forming apparatus 1 to a state in which the image forming operation can be executed again in the case where an instruction to perform the toner cartridge replacement operation is received from the user. A period in which the driving device 98 is driven to move the toner cartridge 70 from the attached position to the retracted position and a period in which the other drive source is driven to move the door 14 from the closed position to the open position may overlap (progress in parallel) with each other at least partially.

Second Embodiment

A second embodiment will be described with reference to FIGS. 13 to 14C. Although the pinion gear 94 is driven by the driving rack 15 in the first embodiment, the pinion gear 94 may be driven by a different mechanism. In the present embodiment, a configuration employing a swing gear will be described as an example. In the description below, it is assumed that elements denoted by the same reference signs as in the first embodiment have substantially the same configurations and functions as those described in the first embodiment unless otherwise described, and parts different from the first embodiment will be mainly described.

FIGS. 13 to 14C are schematic diagrams for describing a configuration for moving the tray 80 in the second embodiment. An apparatus body 101A of an image forming apparatus 101 according to the second embodiment includes, as a driving device, the motor M2 serving as a drive source, and a swing gear unit 115 serving as a transmission device.

A moving device (driven device) that is driven by the driving device to move the tray 80 is disposed in the rotary body 90. That is, the rotary unit 90U includes the rotary body 90 and the moving device including the tray 80.

In the second embodiment, a plurality of moving devices are disposed in correspondence with the plurality of toner cartridges 70y to 70k. The moving devices each include a driven portion 194 (one of driven portions 194y to 194k), and a rack portion 83 (one of rack portions 83y to 83k). The swing gear unit 115 transmits the driving force of the motor M2 to the driven portion 194, and moves the tray 80 via the rack portion 83.

Similarly to the first embodiment in which the pinion gears 94y to 94k are driven by the driving racks 15, the first to fourth driven portions 194y to 194k are each driven by the swing gear unit 115. The first to fourth driven portions 194y to 194k are respectively engaged with the rack portions 83y to 83k, and respectively move the trays 80y to 80k.

The configuration in which each of the first to fourth driven portions 194y to 194k is driven by the swing gear unit 115 is substantially the same. In addition, the configuration for moving each of the trays 80y to 80k via each of the first to fourth driven portions 194y to 194k and each of the rack portions 83y to 83k is the same. Therefore, in the present embodiment, the first to fourth driven portions 194y to 194k, the rack portions 83y to 83k, and the trays 80y to 80k will be respectively referred to as driven portions 194, rack portions 83, and trays 80 without distinguishing each other.

As illustrated in FIG. 14A, the swing gear unit 115 includes an input gear 115a, an output gear 115b, and an arm

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115c. The input gear 115a is rotationally driven by the motor M2. The arm 115c rotatably supports the output gear 115b (such that the output gear 115b can rotate) and allows the output gear 115b to move (swing or revolve) around the input gear 115a.

Further, the swing gear unit 115 includes an arm restriction member 115d that restricts the arm 115c. The arm restriction member 115d includes a restriction portion 115d2 and a restriction rack portion 115d1. The restriction rack portion 115d1 is engaged with the input gear 115a.

In addition, the driven portion 194 includes a driven gear 194a and a pinion gear 194b. The pinion gear 194b is engaged with the rack portion 83 of the tray 80.

In the case of moving the tray 80 from the accommodation position to the detachment position, that is, in the case of moving the toner cartridge 70 from the attached position to the retracted position, the motor M2 rotationally drives the input gear 115a in a clockwise direction in FIGS. 14A to 14C.

When the input gear 115a rotates in the clockwise direction in the drawing, the arm 115c swings in the clockwise direction in the drawing similarly to the input gear 115a by a frictional force received from the input gear 115a. The output gear 115b supported by the arm 115c rotates in a clockwise direction about the input gear 115a together with the arm 115c, and engages with the driven gear 194a as illustrated in FIG. 14B. At this time, the reaction force that the output gear 115b receives from the driven gear 194a preferably acts in such a direction as to move the output gear 115b and the arm 115c in a clockwise direction about the input gear 115a. That is, the direction of the reaction force that the output gear 115b receives from the driven gear 194a when rotating in a counterclockwise direction in FIG. 14B while pressing the driven gear 194a is preferably such a direction that a moment in a clockwise direction in the drawings with respect to the rotational axis of the input gear 115a is generated. As a result of this, the engagement between the output gear 115b and the driven gear 194a is more likely to be maintained.

As illustrated in FIG. 14C, when the input gear 115a rotates in the clockwise direction in a state in which the output gear 115b is engaged with the driven gear 194a, the driven gear 194a drives the pinion gear 194b. As a result of the rack portion 83 being moved by the rotation of the pinion gear 194b, the tray 80 moves from the accommodation position to the detachment position. Further, the toner cartridge 70 moves from the attached position to the retracted position.

To be noted, the positional relationship between the toner cartridge 70 and the frame 16 of the apparatus body 101A when the toner cartridge 70 is at the retracted position and the positional relationship between the toner cartridge 70 and the rotary body 90 are the same as those described in the first embodiment, and therefore description thereof will be omitted.

Meanwhile, in the course of clockwise rotation of the input gear 115a described above, the arm restriction member 115d including the restriction rack portion 115d1 that engages with the input gear 115a moves down. Then, the arm restriction member 115d is positioned at such a position that the restriction portion 115d2 restricts the arm 115c. The restriction portion 115d2 restricts such movement of the arm 115c that the output gear 115b disengages from the driven gear 194a.

In the case of moving the tray 80 from the detachment position to the accommodation position, that is, in the case of moving the toner cartridge 70 from the retracted position

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to the attached position, the motor M2 rotationally drives the input gear 115a in a counterclockwise direction in FIGS. 14A to 14C.

When the input gear 115a rotates in the counterclockwise direction, the output gear 115b supported by the arm 115c receives a force in such a direction as to move in the counterclockwise direction about the input gear 115a together with the arm 115c. However, since the movement of the arm 115c is restricted by the arm restriction member 115d, the engagement between the output gear 115b and the driven gear 194a is maintained.

When the input gear 115a rotates in the counterclockwise direction in a state in which the output gear 115b is engaged with the driven gear 194a, the pinion gear 194b is driven by the driven gear 194a, and the tray 80 moves from the detachment position to the accommodation position. Then, the toner cartridge 70 moves from the retracted position to the attached position.

After the tray 80 has moved to the accommodation position, the restriction of the arm 115c by the restriction portion 115d2 of the arm restriction member 115d is cancelled. As a result, the rotation of the input gear 115a in the counterclockwise direction moves the output gear 115b in the counterclockwise direction about the input gear 115a together with the arm 115c, and thus the engagement between the output gear 115b and the driven gear 194a is cancelled.

According to the configuration described above, the tray 80 can be moved from the accommodation position to the detachment position. In addition, the toner cartridge 70 can be moved from the attached position to the retracted position. Further, the tray 80 can be moved from the detachment position to the accommodation position. In addition, the toner cartridge 70 can be moved from the retracted position to the attached position.

Modification Example

Although the driving force of the motor M2 is transmitted to the tray 80 via the driven gear 194a and the pinion gear 194b in the present embodiment, one gear may be configured to engage with the output gear 115b and the rack portion 83.

In addition, similarly to the first embodiment, the configuration for moving the tray 80 is not limited to a so-called rack-and-pinion configuration.

As yet another modification example, a partially-toothed gear may be used as a drive transmission mechanism from the motor M2 to the pinion gear engaged with the rack portion 83 instead of the configuration of the first and second embodiments. For example, a partially-toothed gear that engages with a pinion gear of the rotary body 90 may be provided in the apparatus body 1A, and the partially-toothed gear may be rotationally driven by the motor M2. In this case, the partially-toothed gear engages with a pinion gear that engages with the rack portion 83 of the tray 80 facing the opening 16a when the rotary body 90 takes the replacement posture illustrated in FIG. 4B. In the black replacement posture illustrated in FIG. 12A, the partially-toothed gear engages with the pinion gear 94k. By switching the rotation direction of the partially-toothed gear, the movement direction of the tray 80 can be switched. In addition, it is preferable that the engagement between the partially-toothed gear and the pinion gear 94 is cancelled in a period other than when the tray 80 is moved with respect to the rotary body 90.

A third embodiment will be described with reference to FIGS. 15 to 16B. Although the toner cartridge 70 is moved from the attached position to the retracted position by the driving force of the motor M2 of the apparatus body 1A in the first and second embodiments, the configuration for moving the toner cartridge 70 is not limited to this. In the present embodiment, a configuration in which the toner cartridge 70 is moved by an elastic force (urging force) of a spring (elastic member) will be described as an example. In the description below, it is assumed that elements denoted by the same reference signs as in the first embodiment have substantially the same configurations and functions as those described in the first embodiment unless otherwise described, and parts different from the first embodiment will be mainly described.

As illustrated in FIG. 15, in the image forming apparatus according to the present embodiment, the toner cartridge 70 is moved from the attached position to the retracted position by using an urging member 294 serving as an example of a moving device. More specifically, the rotary body 90 of the image forming apparatus includes an urging member 294k corresponding to the toner cartridge 70k and the tray 80k, and urging members 294y to 294c corresponding to the toner cartridges 70y to 70c and the trays 80y to 80c. That is, the rotary of the present embodiment includes first to fourth urging members corresponding to the first to fourth toner cartridges. The urging member 294k is an example of the first urging member. The urging members 294y to 294c are each an example of a second urging member. To be noted, this numbering is merely used for the sake of convenience of description, and can be interchanged appropriately in principle.

The urging member 294 of the present embodiment includes an elastic member, and is attached to the rotary body 90. In the present embodiment, a compression spring can be used as the elastic member.

In the present embodiment, the urging member 294 urges the tray 80, and thus the tray 80 moves from the accommodation position to the detachment position and the toner cartridge 70 moves from the attached position to the retracted position. Although the urging member 294 urges the toner cartridge 70 via the tray 80 in the present embodiment, the urging member 294 may directly urge the toner cartridge 70.

That is, although the urging member 294 of the present embodiment is configured to be into contact with the tray 80, the urging member 294 may be in contact with the toner cartridge 70. In this case, the tray 80 may be omitted.

To be noted, also in the present embodiment, the toner cartridge 70k and the tray 80k are larger in size than the other toner cartridges 70y to 70c and the trays 80y to 80c. Therefore, the distance by which the toner cartridge 70k moves from the attached position to the retracted position is larger than the distance by which the toner cartridges 70y to 70c move from the attached position to the retracted position. In addition, the distance by which the tray 80k moves from the accommodation position to the detachment position is larger than the distance by which the trays 80y to 80c move from the accommodation position to the detachment position.

In a state in which the tray 80k is at the accommodation position and the toner cartridge 70k is at the attached position, the elastic member of the urging member 294k is deformed. Similarly, in a state in which the trays 80y to 80c are at the accommodation position and the toner cartridges

70y to 70c are at the attached position, the elastic members of the urging members 294y to 294c are deformed.

In this state, the urging force of the elastic member of the urging member 294k may be set to be larger than the urging force of the elastic members of the urging members 294y to 294c. In addition, the deformation amount of the elastic member of the urging member 294k may be set to be larger than the deformation amounts of the elastic members of the urging members 294y to 294c. As a result of this, the distance by which the toner cartridge 70k moves from the attached position to the retracted position can be made larger than the distance by which the toner cartridges 70y to 70c move from the attached position to the retracted position. In addition, the distance by which the tray 80k moves from the accommodation position to the detachment position can be made larger than the distance by which the trays 80y to 80c move from the accommodation position to the detachment position.

To be noted, when the rotary body 90 rotates, it is preferable that the movement of the toner cartridge 70 to the retracted position and the movement of the tray 80 to the detachment position are restricted.

Therefore, as illustrated in FIG. 16A, the movement of the tray 80 to the detachment position and the movement of the toner cartridge 70 to the retracted position may be restricted by a restriction member 295. The restriction member 295 is movable to a restricting position where the restriction member 295 restricts movement of the tray 80 to the detachment position and movement of the toner cartridge 70 to the retracted position, and to a releasing position where the restriction member 295 allows movement of the tray 80 to the detachment position and movement of the toner cartridge 70 to the retracted position.

In the present embodiment, the restriction member 295 is urged toward the restricting position. In a state in which the rotary body 90 is in the replacement posture, when the restriction member 295 is moved to the releasing position, the urging member 294 moves the tray 80 to the detachment position, and thus the toner cartridge 70 moves to the retracted position. The movement of the restriction member 295 to the releasing position may be manually performed by the user, and the apparatus body 1A may include a releasing means or a releasing mechanism that moves the restriction member 295. As the releasing means (releasing mechanism), a drive source such as a solenoid or a motor can be used. To be noted, one of the motors M1, M2, and M3 may be used as the drive source.

In addition, the urging member 294 may be configured to switch the urging state each time the toner cartridge 70 or the tray 80 is pushed in.

For example, as illustrated in FIG. 16B, the image forming apparatus may include a received portion 296 interlocked with the urging member 294, and a receiving portion 297 that receives the received portion 296. For example, the received portion 296 is provided on the tray 80, and the receiving portion 297 is provided on the rotary body 90. The received portion 296 is urged rightward in the drawing by the urging force of the urging member 294.

In a state in which the received portion 296 is at a position PP, the urging force of the urging member 294 is received by a first recess portion 297d of the receiving portion 297, and the received portion 296 stays in the position PP. In this state, the tray 80 is positioned at the detachment position, and the toner cartridge 70 is positioned at the retracted position.

When the tray 80 is pushed in toward the accommodation position or the toner cartridge 70 is pushed in toward the attached position against the urging force of the urging

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member 294 in this state, the received portion 296 moves to a position Pt1 along a first guiding surface 297a of the receiving portion 297. Then, when the force to push in the tray 80 or the toner cartridge 70 is no longer present, the urging force of the urging member 294 moves the received portion 296 to a position PR along the receiving portion 297. At this time, the urging force of the urging member 294 is received by a second recess portion 297e of the receiving portion 297. As a result, the tray 80 is positioned at the accommodation position, and the toner cartridge 70 is positioned at the attached position.

When the tray 80 or the toner cartridge 70 is pushed in again against the urging force of the urging member 294 in this state, the received portion 296 moves to a position Pt2 along a second guiding surface 297b of the receiving portion 297. Then, when the force to push in the tray 80 or the toner cartridge 70 is no longer present, the urging force of the urging member 294 moves the received portion 296 to the position PP along a third guiding surface 297c of the receiving portion 297. As a result, the tray 80 is positioned at the detachment position, and the toner cartridge 70 is positioned at the retracted position.

To be noted, in the case where the receiving portion 297 is attached to the rotary body 90, the urging member 294 may include the elastic member and the received portion 296. In this case, in a state in which the received portion 296 is positioned at the position PR, the urging force of the urging member 294 is received by the second recess portion 297e of the receiving portion 297, and does not act on the tray 80 or the toner cartridge 70. In addition, when the received portion 296 moves from the position Pt2 to the position PP, the urging force of the urging member 294 acts on the tray 80 and the toner cartridge 70, the tray 80 is positioned at the detachment position, and the toner cartridge 70 is positioned at the retracted position. To be noted, a member corresponding to the received portion 296 may be provided in the rotary body 90, and a member corresponding to the receiving portion 297 may be provided in the urging member 294 or the tray 80.

To be noted, the image forming apparatus may include a locking portion 298 (tray holding member) illustrated in FIG. 16C. Although the locking portion 298 engages with the tray 80 in the present embodiment, the locking portion 298 may engage with the toner cartridge 70. The locking portion 298 holds the tray 80 at the accommodation position or holds the toner cartridge 70 at the attached position in a state in which the urging member 294 is not urging the tray 80 or the toner cartridge 70. For example, also in a state in which the direction of the gravity acting on the toner cartridge 70 has come to match the retraction direction or has become close to the retraction direction of the toner cartridge 70 due to the rotation of the rotary body 90, the toner cartridge 70 is held at the attached position by the locking portion 298. When the urging force of the urging member 294 acts on the tray 80 or the toner cartridge 70, the locking portion 298 is moved by the tray 80 or the toner cartridge 70. To be noted, the position where the locking portion 298 engages with a locked portion of the tray 80 or the toner cartridge 70 is not limited to the position of FIG. 16C.

As the locking portion 298, for example, the same part as the projection portion 95 provided in the rotary body 90 of the first embodiment illustrated in FIGS. 7A and 7B can be used.

Modification Example

Although the elastic member is a compression spring in the present embodiment, the elastic member may be a

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torsion coil spring, a plate spring, or a tension spring each provided at an appropriate position.

In addition, for example, magnets may be used as the urging member. For example, a first magnet may be attached to the rotary body 90, and the tray 80 may be urged toward the retracted position by a repulsive force between the first magnet and a second magnet attached to the toner cartridge 70 or the tray 80.

Fourth Embodiment

A fourth embodiment will be described with reference to FIGS. 17A and 17B. Although the toner cartridge 70 is moved from the attached position to the retracted position by the driving force of the motor M2 of the apparatus body 1A in the first and second embodiments and by the urging member 294 in the third embodiment, the configuration for moving the toner cartridge 70 is not limited to these. In the present embodiment, a configuration in which the toner cartridge 70 is moved by using a solenoid will be described as an example. In the description below, it is assumed that elements denoted by the same reference signs as in the first embodiment have substantially the same configurations and functions as those described in the first embodiment unless otherwise described, and parts different from the first embodiment will be mainly described.

FIGS. 17A and 17B are each a diagram for describing a configuration for moving the tray 80 in the fourth embodiment. In the image forming apparatus according to the present embodiment, the toner cartridge 70 is moved from the attached position to the retracted position by using a solenoid unit 394 serving as an example of a moving device. More specifically, the rotary body 90 of the present embodiment includes a solenoid unit 394k corresponding to the toner cartridge 70k and the tray 80k. In addition, the rotary body 90 of the present embodiment includes solenoid units 394y to 394c corresponding to the toner cartridges 70y to 70c and the trays 80y to 80c.

The solenoid unit 394 includes a coil and a movable member, and the movable member is moved by a magnetic field induced in the coil. The movable member is capable of moving such that the tray 80 moves from the accommodation position to the detachment position or the toner cartridge 70 moves from the attached position to the retracted position.

In addition, the movable member is capable of moving such that the tray 80 moves from the detachment position to the accommodation position or the toner cartridge 70 moves from the retracted position to the attached position.

To be noted, as illustrated in FIG. 17B, the image forming apparatus may include a locking portion 397 (tray holding member). Although the locking portion 397 engages with the tray 80 in the present embodiment, the locking portion 397 may engage with the toner cartridge 70. The locking portion 397 holds the tray 80 at the accommodation position or holds the toner cartridge 70 at the attached position in a state in which the movable member of the solenoid unit 394 is not urging the tray 80 or the toner cartridge 70. That is, the locking portion 397 restricts movement of the tray 80 and the toner cartridge 70 in a state in which the movable member of the solenoid unit 394 is not urging the tray 80 and the toner cartridge 70. In contrast, when the movable member of the solenoid unit 394 acts on the tray 80 or the toner cartridge 70, the locking portion 397 is moved by the tray 80 or the toner cartridge 70. To be noted, the position where the

locking portion **397** engages with a locked portion of the tray **80** or the toner cartridge **70** is not limited to the position of FIG. **17B**.

As the locking portion **397**, for example, the same part as the projection portion **95** provided in the rotary body **90** of the first embodiment illustrated in FIGS. **7A** and **7B** can be used.

Fifth Embodiment

An image forming apparatus **201** according to a fifth embodiment will be described with reference to FIG. **18**. In the present embodiment, a second surface of a body frame is provided at a more inward position than a first surface of the body frame constituting the exterior surface of the image forming apparatus **201**, and the opening/closing member is disposed on the second surface. In the description below, it is assumed that elements denoted by the same reference signs as in the first embodiment have substantially the same configurations and functions as those described in the first embodiment unless otherwise described, and parts different from the first embodiment will be mainly described.

As illustrated in FIG. **18**, a frame (body frame) **215** of the image forming apparatus **201** of the present embodiment includes an outer frame **217** and an inner frame **216** accommodated inside the outer frame **217**. The outer frame **217** constitutes the exterior surface (appearance surface) of the image forming apparatus **201**.

A door **214** serving as an opening/closing member of the present embodiment is disposed on a side surface **216b** on the +X side of the inner frame **216**. The door **214** is movable between a closed position where the door **214** covers an opening **216a** provided in the side surface **216b** and an open position where the door **214** allows attachment and detachment of the toner cartridge **70** to and from the rotary body **90** via the opening **216a**, with respect to the inner frame **216**.

The side surface **216b** of the inner frame **216** is positioned more on the inside than a side surface **217a** of the outer frame **217** with respect to the image forming apparatus **201**. The side surface **217a** of the outer frame **217** constitutes at least part of the exterior surface on the +X side of the image forming apparatus **201**. That is, the side surface **217a** is an example of a first surface constituting part of the exterior surface of the image forming apparatus, and the side surface **216b** of the inner frame **216** is an example of a second surface provided at a more inward position than the first surface with respect to the image forming apparatus.

At least part of the side surface **217a** of the outer frame **217** is configured as an openable portion **217b** capable of pivoting (or opening and closing) with respect to the other part of the outer frame **217**. Further, the user is allowed to replace the toner cartridge **70** by accessing a space between the outer frame **217** and the inner frame **216** by, for example, manually opening the openable portion **217b**.

In the present embodiment, for example, after the user has opened the openable portion **217b** of the outer frame **217**, the door **214** moves from the closed position to the open position in an interconnected manner with the movement of the toner cartridge **70** from the attached position to the retracted position. Also according to such a configuration, the user does not have to open the door **214** manually, which makes it easier for the user to replace the toner cartridge.

To be noted, the side surface **216b** of the inner frame **216** and the side surface **217a** of the outer frame **217** are each a surface extending in a direction intersecting with the horizontal direction (that is, a direction intersecting with the X direction) as viewed in the rotational axis direction of the

rotary body **90** (Y direction). The side surface **217a** of the outer frame **217** constitutes part of the exterior surface on the downstream side of the image forming apparatus **201** in a discharge direction in which the sheet S on which an image has been formed is discharged through a discharge port of the apparatus body **1A**. In addition, it can be said that the side surface **217a** of the outer frame **217** constitutes part of the exterior surface on the front side of the image forming apparatus **201**.

Here, in the first embodiment, a configuration in which the sheet S stored in the sheet storage portion **300** projects to the outside of the image forming apparatus **1** with respect to the side surface **16b** of the frame **16** in which the opening **16a** and the door **14** are provided as illustrated in FIG. **1** has been described. In the present embodiment, the outer frame **217** is formed in accordance with the size of a cassette **305** serving as a sheet storage portion that stores the sheet S. Specifically, a side surface **305a** on the downstream side of the cassette **305** in a draw-out direction Dp of the cassette **305** constitutes the exterior surface of the image forming apparatus **201** together with the side surface **217a** of the outer frame **217**. As a result of this, the aesthetic appearance of the apparatus can be improved by not projecting the sheet S from the body frame **215**, and thus, for example, a possibility that the user erroneously staining the sheet S can be reduced.

To be noted, the draw-out direction Dp is a movement direction in which the cassette **305** is drawn out from a stored position where the cassette **305** is stored in the body frame **215** to a drawn-out position where at least part of the cassette **305** projects to the outside of the body frame **215**. A configuration in which the side surface **305a** of the cassette **305** and the side surface **217a** of the outer frame **217** are formed along the same virtual plane, that is, are flush may be employed.

Sixth Embodiment

An image forming apparatus **301** according to a sixth embodiment will be described with reference to FIGS. **19A** and **19B**. In the present embodiment, a second opening/closing member is movable to a position where the second opening/closing member covers the opening/closing member and a position where the second opening/closing member exposes the opening/closing member. In the description below, it is assumed that elements denoted by the same reference signs as in the first embodiment have substantially the same configurations and functions as those described in the first embodiment unless otherwise described, and parts different from the first embodiment will be mainly described.

As illustrated in FIGS. **19A** and **19B**, the image forming apparatus **301** according to the present embodiment includes an outer door **302**. The outer door **302** is movable between a position illustrated in FIG. **19A** where the outer door **302** covers the door **14** (inner door) at the closed position and a position illustrated in FIG. **19B** where the outer door **302** exposes the door **14**, with respect to the frame **16**. In the state of FIG. **19B**, movement of the door **14** from the closed position to the open position, and attachment and detachment of the toner cartridge **70** to and from the rotary body **90** via the opening **16a** are possible. The outer door **302** is an example of a second opening/closing member movable between a second closed position where the second opening/closing member covers the opening/closing member (door **14**, first opening/closing member) at the closed position and a second open position where the second opening/closing

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member exposes the opening/closing member so as to allow movement of the opening/closing member from the closed position to the open position, with respect to the body frame.

The outer door 302 is pivotably supported by the frame 16 about a support shaft 302a provided at a lower end portion of the outer door 302. As a result of this, the outer door 302 is pivotable with respect to the frame 16 about an axis substantially parallel to the rotational axis direction of the rotary body 90 (Y direction). To be noted, the configuration for making the outer door 302 movable is not limited to this, and for example, a configuration in which the outer door 302 pivots about the rotational axis extending in a direction (Z direction) intersecting with the Y direction as a result of a support shaft provided at a side end portion of the outer door 302 in the Y direction may be employed.

In the present embodiment, for example, after the user has opened the outer door 302 manually, the door 14 moves from the closed position to the open position in an interconnected manner with the movement of the toner cartridge 70 from the attached position to the retracted position. Also according to such a configuration, the user does not have to open the door 14 manually, which makes it easier for the user to replace the toner cartridge.

Modification Example

To be noted, in the fifth and sixth embodiments, the door 14 may be omitted. In this case, the openable portion 217b of the outer frame 217 of the fifth embodiment or the outer door 302 of the sixth embodiment may be opened in an interconnected manner with the movement of the toner cartridge 70 from the attached position to the retracted position. In this case, the user does not have to open the openable portion 217b or the outer door 302 manually, which makes it easier for the user to replace the toner cartridge similarly to the other embodiments.

Seventh Embodiment

A seventh embodiment will be described with reference to FIG. 21. In the first to fourth embodiments, a configuration in which the rotary body 90 includes four developing units 50j to 50k and a color image can be formed by using toners of four colors has been described. In the present embodiment, a configuration in which a toner cartridge is moved with respect to a developing device capable of forming a monochromatic image will be described. In the description below, it is assumed that elements denoted by the same reference signs as in the first to fourth embodiments have substantially the same configurations and functions as those described in the first to fourth embodiments unless otherwise described, and parts different from the first to fourth embodiments will be mainly described.

As illustrated in FIG. 21, an image forming apparatus 501 includes a toner cartridge 570 attachable to and detachable from the apparatus body 1A. In addition, the apparatus body 1A includes a developing device (developing unit) 590.

The developing device 590 is an example of a developing means or a developing portion that develops (visualizes) an electrostatic latent image formed on the photosensitive drum 2 into a toner image by using toner. The developing device 590 of the present embodiment develops the electrostatic latent image formed on the photosensitive drum 2 by using black toner.

The developing device 590 includes a developing roller 51, a supply roller 52, and a developing blade. The toner cartridge 570 is attached to the developing device 590. The

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toner cartridge 570 stores therein black toner to be supplied to the developing device 590.

The toner cartridge 570 includes a toner frame 571. The toner frame 571 includes a toner storage portion 571a that stores toner, and a discharge opening 571b communicating with the toner storage portion 571a.

The developing device 590 includes a developing frame (storage frame) 553 including a developing-side storage portion 553a that stores toner. In addition, the developing frame 553 has an inlet opening 553b communicating with the developing-side storage portion (toner supply chamber) 553a.

The toner cartridge 570 is attachable to and detachable from the developing device 590 through the opening 16a provided in the frame 16 of the apparatus body 1A. More specifically, the toner cartridge 570 is movable to an attached position and a retracted position where the toner cartridge 570 is retracted from the attached position, through the opening 16a with respect to the developing frame 553.

In a state in which the toner cartridge 570 is at the attached position with respect to the developing frame 553, the discharge opening 571b faces the inlet opening 553b. That is, the toner storage portion 571a of the toner cartridge 570 and the developing-side storage portion 553a of the developing device 590 communicate with each other via the discharge opening 571b and the inlet opening 553b. When toner is supplied from the toner cartridge 570 to the developing device 590, at least part of the inlet opening 553b is positioned below at least part of the discharge opening 571b.

Then, toner stored in the toner storage portion 571a is discharged through the discharge opening 571b, and the toner discharged through the discharge opening 571b is stored in the developing-side storage portion 553a through the inlet opening 553b. The toner stored in the developing-side storage portion 553a is supplied to the developing roller 51 by the supply roller 52. To be noted, the developing-side storage portion 553a may include a toner conveyance member that conveys toner toward the supply roller 52.

The function of the toner cartridge 570 is substantially the same as the functions of the toner cartridge 70 of the first to fourth embodiments. In addition, the function of the developing device 590 is substantially the same as the function of one of the developing units 50j, 50m, 50c, and 50k of the first to fourth embodiments.

Meanwhile, the apparatus body 1A includes a transfer roller 512. The transfer roller 512 is an example of a transfer means or a transfer unit that transfers an image from the photosensitive drum 2 onto the sheet S. A conveyance roller pair 320 conveys the sheet S to a transfer portion that is a nip portion between the photosensitive drum 2 and the transfer roller 512. The image on the photosensitive drum 2 is transferred onto the surface of the sheet S that has been conveyed thereto.

The apparatus body 1A includes a moving device configured to move the toner cartridge 570 from the attached position to the retracted position with respect to the developing device 590, more specifically with respect to the developing frame 553 of the developing device 590. As this moving device, a moving device described in the first to fourth embodiments and modification examples thereof can be used.

In this case, among the moving devices of the first to fourth embodiments, part provided in the rotary body 90 may be provided in the developing device 590. In addition, the replacement posture and the developing posture of the developing device 590 may be the same or different. For example, the developing device 590 may be movable to a

contact position where the developing roller **51** is in contact with the photosensitive drum **2** and a separation position where the developing roller **51** is separated from the photosensitive drum **2**, and the developing device **590** may be at the separation position in a state in which the developing device **590** is in the replacement posture.

For example, the developing device **590** may have the tray **80** and an element (moving device) for moving the tray **80**. As the element for moving the tray **80**, those described in the first to fourth embodiments and modification examples thereof can be used. In addition, the developing device **590** may include the rotary member **494a** and the driven roller **494b** that have been described in the modification example of the first embodiment.

That is, the apparatus body **1A** of the present embodiment may include a moving device including a support member (tray **80**) that supports the toner cartridge **570**. In this case, the opening/closing member (door **14**) moves from the closed position to the open position in an interconnected manner with the movement of the toner cartridge **570** from the attached position to the retracted position by the movement of the support member. As a result of this, the user does not have to open the door **14** manually, which makes it easier for the user to replace the toner cartridge.

Also in the present embodiment, when the toner cartridge **570** is at the retracted position, at least part of the toner cartridge **570** is preferably outside the apparatus of the image forming apparatus **501** (outside the apparatus body **1A**). That is, when the toner cartridge **570** is at the retracted position, at least part of the toner cartridge **570** is positioned more outward than the exterior position with respect to the apparatus body **1A** when the toner cartridge **570** is at the retracted position. In other words, at least part of the toner cartridge **570** is positioned in a space outside the apparatus body **1A** in the case where the door **14** is at the closed position. Further, at least part of the toner cartridge **570** is positioned downstream of the exterior position in the retraction direction of the toner cartridge **570**.

In addition, in the case where the side surface **16b** having the opening **16a** is the front surface of the apparatus body **1A**, it can be said at least part of the toner cartridge **570** projects to the front side more than the exterior surface on the front side of the apparatus body **1A** when the toner cartridge **570** is at the retracted position.

Also in the present embodiment, when the toner cartridge **570** is at the retracted position, half the length or more of the toner cartridge **570** in the retraction direction is preferably outside the apparatus.

As described above, although the toner cartridges **70** are detachably attached to the rotary body **90** in the first to fourth embodiments and modification examples thereof, the toner cartridge **570** is detachably attached to the developing device **590** in the present embodiment.

Other Embodiments

In the embodiments described above, a configuration in which the tray **80** moves from the accommodation position to the detachment position by a driving device, an urging member, or the like of the apparatus body **1A** without the user operating the tray **80** has been described. Instead of this, a configuration in which the user grips the tray **80** and pulls out the tray **80** from the accommodation position to the detachment position in a state in which the rotary body **90** is at the replacement posture, that is, a configuration in which the support member is manually moved may be employed. Also in this case, since the user is only required

to attach or detach the toner cartridge **70** to or from the tray **80** pulled out of the apparatus body, the operability is improved as compared with a case where the toner cartridge is directly inserted into or removed from the apparatus body.

In addition, in the first to fourth embodiments described above, a configuration in which the rotary body **90** includes the four developing units **50y** to **50k** and a color image can be formed by using toners of four colors has been described. However, the number of developing units that the rotary body **90** includes may be 3 or less or 5 or more. In these cases, the number and arrangement of the trays and toner cartridges can be appropriately changed in accordance with the number of developing units. For example, in the first to fourth embodiments described above, a configuration in which the four toner cartridges **70y** to **70k** are attachable to and detachable from the rotary body **90** has been described as an example. However, a configuration in which the rotary body **90** includes only one developing unit **50k** and only one toner cartridge **70k** is attached to the rotary body **90** may be employed. In this case, the rotary body **90** can rotate in a clockwise direction in FIG. **1** about the rotational axis **90C** to alternately take the black replacement posture and the black developing posture.

In addition, in the first to fourth embodiments described above, a configuration in which the rotary body **90** includes the four developing units **50y** to **50k** and a color image can be formed by using toners of four colors has been described. However, the rotary body **90** may include a plurality of developing units capable of executing image formation by using toner of the same color. For example, a configuration in which the rotary body **90** includes four black developing units **50k** and four toner cartridges **70k** are attached to the rotary body **90** may be employed.

In addition, although a configuration in which the door **14** serving as an opening/closing member pivots about an axis substantially parallel to the rotational axis direction of the rotary body **90** (Y direction) has been described in the embodiments described above, the mode of movement of the door **14** is not limited to this. For example, a double door-like configuration in which the door **14** is split into two at the center thereof in the Y direction and the two parts of the door **14** respectively pivot about support shafts provided at respective end portions of the door **14** in the Y direction may be employed. In addition, for example, a configuration in which the door **14** is provided so as to be slidable in the Z direction with respect to the frame **16**, and the door **14** slides downward by being pushed by an inclined surface provided on the tray **80** in the case where the toner cartridge **70** moves in the retraction direction may be employed.

According to the present disclosure, a new form of an image forming apparatus that advances conventional technology can be provided.

While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

This application claims the benefit of Japanese Patent Application Applications No. 2022-178517, filed on Nov. 8, 2022, and No. 2023-165975, filed on Sep. 27, 2023, which are hereby incorporated by reference herein in their entirety.

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What is claimed is:

1. An image forming apparatus comprising:
 - a rotary configured to rotate and including a developing roller and a storage frame including a storage portion configured to store toner to be supplied to the developing roller;
 - a body frame with an opening, the body frame being configured to accommodate the rotary;
 - a toner cartridge which is configured to store the toner, which is attachable to and detachable from the rotary through the opening, which is movable to an attached position and a retracted position where the toner cartridge is retracted from the attached position, with respect to the storage frame, and which is configured to supply the toner to the storage portion in a state in which the toner cartridge is at the attached position;
 - a moving device configured to move the toner cartridge from the attached position to the retracted position; and
 - an opening/closing member provided to be, with respect to the body frame, movable between a closed position where the opening/closing member covers the opening and an open position where the opening/closing member opens the opening to allow attachment and detachment of the toner cartridge to and from the rotary through the opening,
 wherein the opening/closing member is configured to move from the closed position to the open position in an interconnected manner with movement of the toner cartridge from the attached position to the retracted position by the moving device.
2. The image forming apparatus according to claim 1, wherein the moving device includes a support member which is configured to support the toner cartridge, to which the toner cartridge is detachably attached, and which is attached to the rotary, and
- wherein the support member is movable with respect to the storage frame such that the toner cartridge moves from the attached position to the retracted position.
3. The image forming apparatus according to claim 2, wherein in a case where the moving device moves the toner cartridge from the attached position to the retracted position, the support member is configured to push the opening/closing member such that the opening/closing member moves from the closed position to the open position.
4. The image forming apparatus according to claim 2, further comprising:
 - an urging member configured to urge the opening/closing member toward the closed position,
 - wherein the opening/closing member is configured to be moved from the open position to the closed position by an urging force of the urging member in an interconnected manner with movement of the toner cartridge from the retracted position to the attached position by the moving device.
5. The image forming apparatus according to claim 4, wherein the opening/closing member is pivotable to the open position and the closed position about a rotational axis extending in a direction intersecting with a direction in which the support member moves.
6. The image forming apparatus according to claim 1, wherein the body frame includes an exterior surface constituting an exterior of the image forming apparatus, wherein the opening is provided in the exterior surface, and

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- wherein in a case where the opening/closing member is at the closed position, the opening/closing member constitutes part of the exterior surface.
7. The image forming apparatus according to claim 6, wherein the toner cartridge is configured such that, in a state in which the toner cartridge is at the retracted position, at least part of the toner cartridge projects outside the body frame with respect to a position of the exterior surface of a case where the opening/closing member is at the closed position.
8. The image forming apparatus according to claim 1, wherein the body frame has a first surface constituting part of an exterior surface of the image forming apparatus and a second surface provided at a more inward position of the image forming apparatus than the first surface, and
- wherein the opening is provided in the second surface.
9. The image forming apparatus according to claim 8, further comprising:
 - a sheet storage portion which is configured to store a sheet and which is movable, with respect to the body frame, between a stored position where the sheet storage portion is stored in the body frame and a drawn-out position where the sheet storage portion is drawn out from the stored position in a draw-out direction such that at least part of the sheet storage portion projects outside the body frame,
 - wherein in a state in which the sheet storage portion is at the stored position, a surface of the sheet storage portion on a downstream side in the draw-out direction, together with the first surface, constitutes an exterior surface of the image forming apparatus on the downstream side in the draw-out direction.
10. The image forming apparatus according to claim 1, further comprising:
 - a second opening/closing member movable to a position where the second opening/closing member covers the opening/closing member at the closed position and a position where the second opening/closing member exposes the opening/closing member so as to allow movement of the opening/closing member from the closed position to the open position.
11. The image forming apparatus according to claim 1, wherein the opening/closing member is configured such that in a state in which the toner cartridge is at the retracted position and the opening/closing member is at the open position, at least part of a weight of the toner cartridge is supported by the opening/closing member.
12. The image forming apparatus according to claim 1, wherein a movement direction of the toner cartridge from the attached position toward the retracted position is a direction intersecting with a rotational axis direction of the rotary.
13. The image forming apparatus according to claim 1, further comprising:
 - a photosensitive drum,
 - wherein the rotary is rotatable to a developing posture in which the developing roller faces the photosensitive drum and a replacement posture where detachment of the toner cartridge from the rotary is allowed.
14. The image forming apparatus according to claim 1, wherein in a case where the moving device moves the toner cartridge from the attached position to the retracted position, the toner cartridge is configured to push the opening/closing member such that the opening/closing member moves from the closed position to the open position.

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15. The image forming apparatus according to claim 14, wherein the moving device includes a support member which is configured to support the toner cartridge, to which the toner cartridge is detachably attached, and which is attached to the rotary, and
- wherein the support member is movable with respect to the storage frame such that the toner cartridge moves from the attached position to the retracted position.
16. The image forming apparatus according to claim 15, further comprising:
- an urging member configured to urge the opening/closing member toward the closed position,
 - wherein the opening/closing member is configured to be moved from the open position to the closed position by an urging force of the urging member in an interconnected manner with movement of the toner cartridge from the retracted position to the attached position by the moving device.
17. The image forming apparatus according to claim 16, wherein the opening/closing member is pivotable to the open position and the closed position about a rotational axis extending in a direction intersecting with a direction in which the support member moves.
18. The image forming apparatus according to claim 14, wherein the body frame has a first surface constituting part of an exterior surface of the image forming apparatus and a second surface provided at a more inward position of the image forming apparatus than the first surface, and
- wherein the opening is provided in the second surface.
19. The image forming apparatus according to claim 18, further comprising:
- a sheet storage portion which is configured to store a sheet and which is movable, with respect to the body frame, between a stored position where the sheet storage portion is stored in the body frame and a drawn-out position where the sheet storage portion is drawn out from the stored position in a draw-out direction such that at least part of the sheet storage portion projects outside the body frame,
 - wherein in a state in which the sheet storage portion is at the stored position, a surface of the sheet storage portion on a downstream side in the draw-out direction, together with the first surface, constitutes an exterior surface of the image forming apparatus on the downstream side in the draw-out direction.
20. An image forming apparatus comprising:
- a rotary including a first developing roller, a second developing roller, a first storage frame, and a second storage frame, the first storage frame including a first storage portion configured to store first toner to be supplied to the first developing roller, the second storage frame including a second storage portion configured to store second toner to be supplied to the second developing roller;
 - a body frame configured to accommodate the rotary and having an opening;
 - a first toner cartridge which is configured to store the first toner, which is attachable to and detachable from the rotary through the opening, which is movable to a first attached position and a first retracted position where the first toner cartridge is retracted from the first attached position, with respect to the first storage frame, and which is configured to supply the first toner to the first storage portion in a state in which the first toner cartridge is at the first attached position;

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- a second toner cartridge which is configured to store the second toner, which is attachable to and detachable from the rotary through the opening, which is movable to a second attached position and a second retracted position where the second toner cartridge is retracted from the second attached position, with respect to the second storage frame, and which is configured to supply the second toner to the second storage portion in a state in which the second toner cartridge is at the second attached position;
 - a first moving device configured to move the first toner cartridge from the first attached position to the first retracted position; and
 - a second moving device configured to move the second toner cartridge from the second attached position to the second retracted position; and
 - an opening/closing member provided to be, with respect to the body frame, movable between a closed position where the opening/closing member covers the opening and an open position where the opening/closing member opens the opening to allow attachment and detachment of the first toner cartridge and the second toner cartridge to and from the rotary through the opening, wherein the opening/closing member is configured to move from the closed position to the open position in an interconnected manner with movement of the first toner cartridge from the first attached position to the first retracted position by the first moving device, and wherein the opening/closing member is configured to move from the closed position to the open position in an interconnected manner with movement of the second toner cartridge from the second attached position to the second retracted position by the second moving device.
21. The image forming apparatus according to claim 20, wherein the first moving device includes a first support member which is configured to support the first toner cartridge, to which the first toner cartridge is detachably attached, and which is attached to the rotary,
- wherein the second moving device includes a second support member which is configured to support the second toner cartridge, to which the second toner cartridge is detachably attached, and which is attached to the rotary,
- wherein the first support member is movable with respect to the first storage frame such that the first toner cartridge moves from the first attached position to the first retracted position, and
- wherein the second support member is movable with respect to the second storage frame such that the second toner cartridge moves from the second attached position to the second retracted position.
22. The image forming apparatus according to claim 21, wherein in a case where the first moving device moves the first toner cartridge from the first attached position to the first retracted position, the first support member is configured to push the opening/closing member such that the opening/closing member moves from the closed position to the open position,
- wherein in a case where the second moving device moves the second toner cartridge from the second attached position to the second retracted position, the second support member is configured to push the opening/closing member such that the opening/closing member moves from the closed position to the open position.

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23. The image forming apparatus according to claim 20, wherein the opening/closing member is configured such that in a state in which the first toner cartridge is at the first retracted position and the opening/closing member is at the open position, at least part of a weight of the first toner cartridge is supported by the opening/closing member, and
- wherein the opening/closing member is configured such that in a state in which the second toner cartridge is at the second retracted position and the opening/closing member is at the open position, at least part of a weight of the second toner cartridge is supported by the opening/closing member.
24. The image forming apparatus according to claim 20, wherein in a case where a length by which the first toner cartridge at the first retracted position projects from the opening is a first length and a length by which the second toner cartridge at the second retracted position projects from the opening is a second length, the first length is larger than the second length.
25. The image forming apparatus according to claim 20, wherein a movement amount of the first toner cartridge from the first attached position to the first retracted position is larger than a movement amount of the second toner cartridge from the second attached position to the second retracted position.

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26. The image forming apparatus according to claim 20, wherein the first toner cartridge is configured to store a first amount of the first toner, wherein the second toner cartridge is configured to store a second amount of the second toner, and wherein the first amount is larger than the second amount.
27. The image forming apparatus according to claim 20, wherein a movement direction of the first toner cartridge in which the first toner cartridge moves from the first attached position to the first retracted position intersects with a rotational axis direction of the rotary, and wherein a movement direction of the second toner cartridge in which the second toner cartridge moves from the second attached position to the second retracted position intersects with the rotational axis direction of the rotary.
28. The image forming apparatus according to claim 20, further comprising:
a photosensitive drum,
wherein the rotary is rotatable to a first developing posture in which the first developing roller faces the photosensitive drum, a first replacement posture in which detachment of the first toner cartridge from the rotary is allowed, a second developing posture in which the second developing roller faces the photosensitive drum, and a second replacement posture in which detachment of the second toner cartridge from the rotary is allowed.

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